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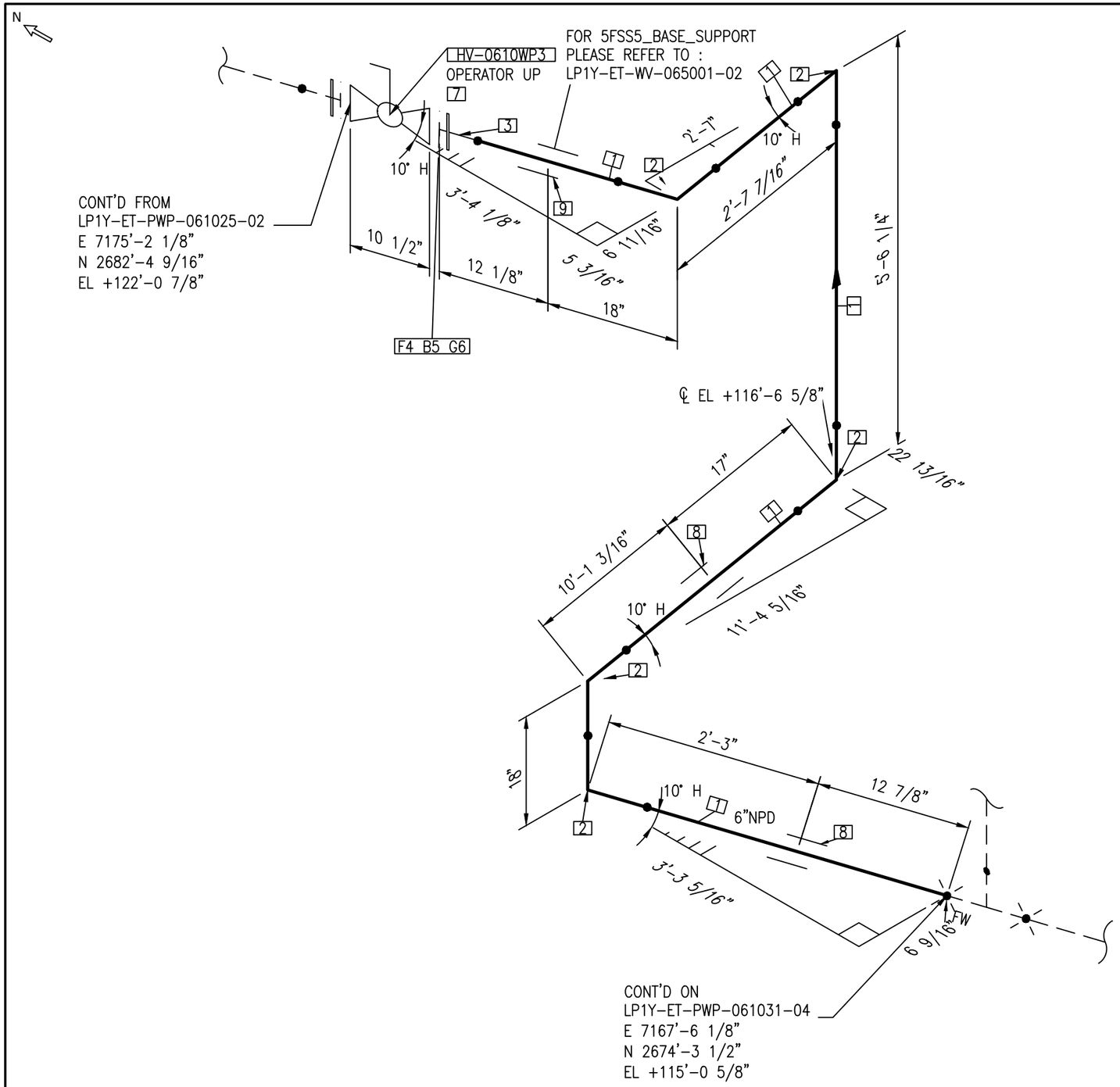

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	19'-3"	6"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	5	6"	ELL 90 DEG LR SCH 10S DUPLEX STL S32205-W	5460743
3	1	6"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
4	1	6"	FLG LAP JOINT 150# 304 SS	5599576L
5	8	3/4"X3 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
6	1	6"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
7	1	6"	BALL CL 150 RF DI, PFA LINED FP GR OP	5300941L4
8	2	6"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-06-01
9	1	6"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-06-01

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By dac54131 at Mar 02, 2026

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0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW
REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	PWP
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 6" DUP PWP (344) (108) 061025

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-ET-PWP-061025-01

REV
0

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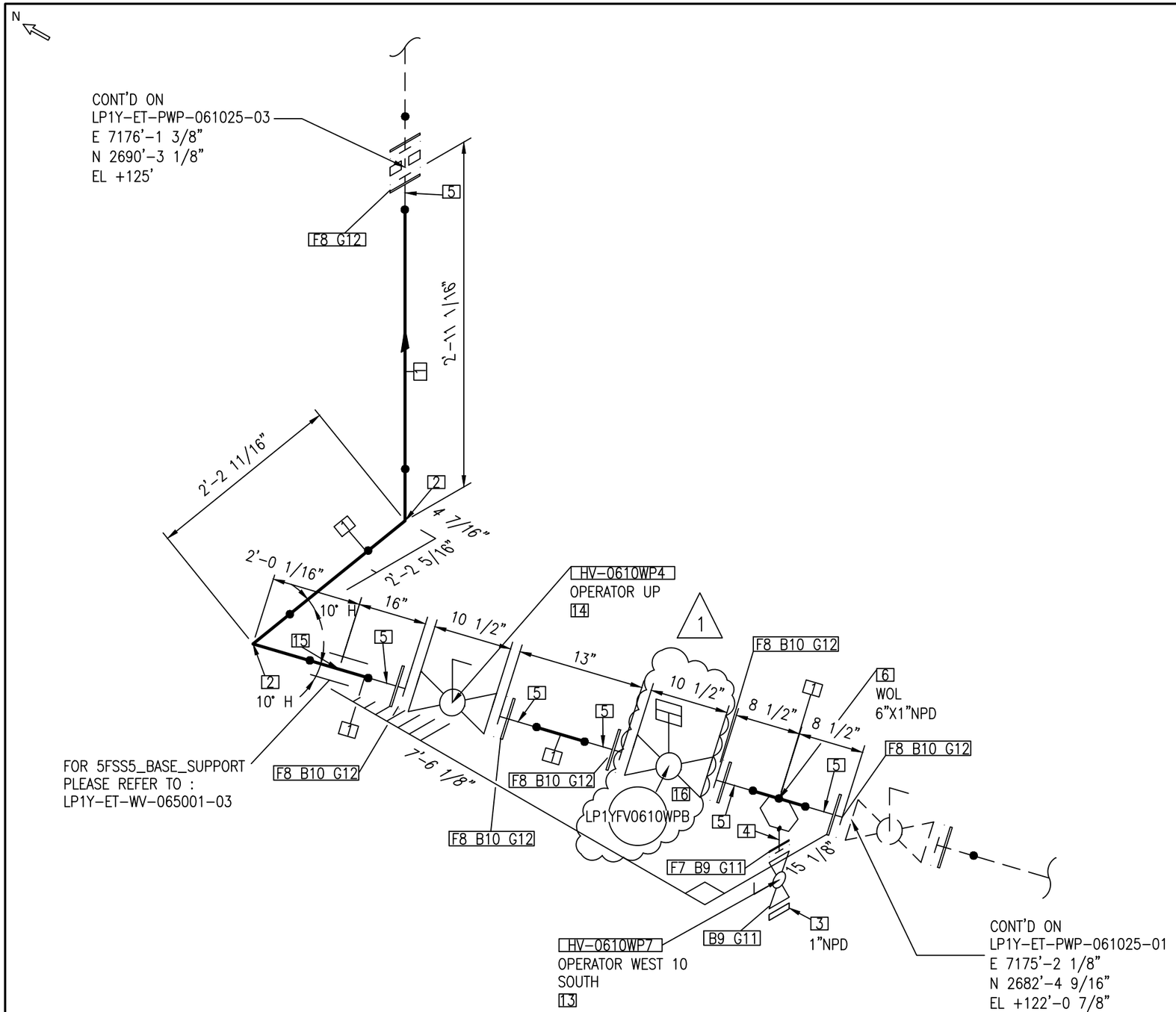

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	6'-3"	6"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	2	6"	ELL 90 DEG LR SCH 10S DUPLEX STL S32205-W	5460743
3	1	1"	FLG BLIND 150# RF DUPLEX STL F60	5616811
4	1	1"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
5	6	6"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
6	1	6"X1"	PIPET SCH 10S DUPLEX STL F60	5512022
7	1	1"	FLG LAP JOINT 150# 304 SS	5599576L
8	6	6"	FLG LAP JOINT 150# 304 SS	5599576L
9	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
10	40	3/4"X3 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
11	2	1"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
12	6	6"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
13	1	1"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
14	1	6"	BALL CL 150 RF DI, PFA LINED FP GR OP	5300941L4
15	1	6"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-06-01
16	1	6"	KNA BALL VALVE ASME CL. 150# RF	LP1YFV0610WPB

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	JHN / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	PWP
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 6" DUP PWP (344) (108) 061025

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-ET-PWP-061025-02

REV
1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026

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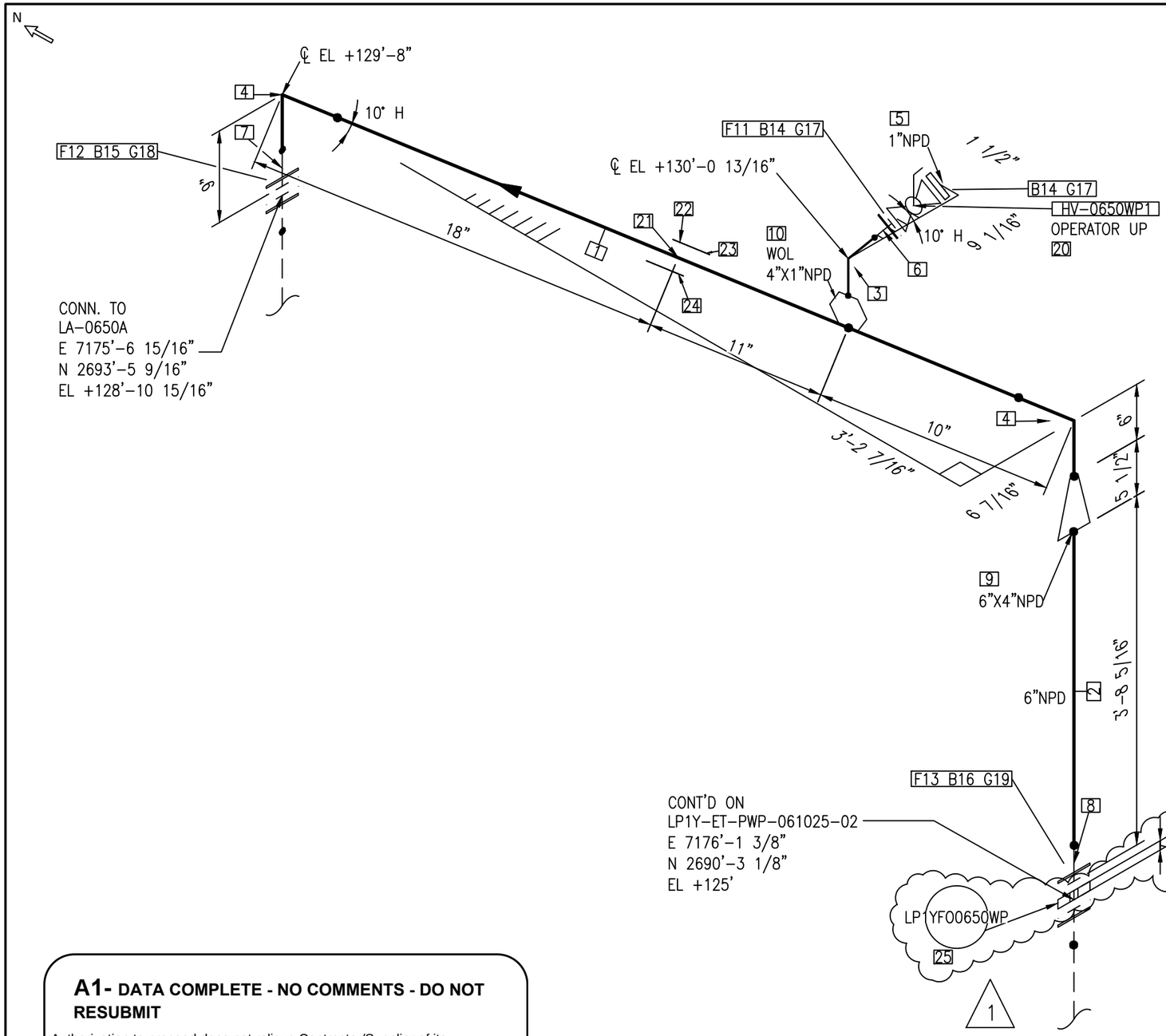

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	2'-3"	4"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	3'-5"	6"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
3	1	1"	ELL 90 DEG LR SCH 10S DUPLEX STL S32205-W	5460743
4	2	4"	ELL 90 DEG LR SCH 10S DUPLEX STL S32205-W	5460743
5	1	1"	FLG BLIND 150# RF DUPLEX STL F60	5616811
6	1	1"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
7	1	4"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
8	1	6"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
9	1	6"X4"	REDUCER CONC SCH 10S DUPLEX STL S32205-W	5460754
10	1	4"X1"	PIPET SCH 10S DUPLEX STL F60	5512022
11	1	1"	FLG LAP JOINT 150# 304 SS	5599576L
12	1	4"	FLG LAP JOINT 150# 304 SS	5599576L
13	1	6"	FLG LAP JOINT 150# 304 SS	5599576L
14	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
15	8	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
16	8	3/4"X3 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
17	2	1"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
18	1	4"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
19	1	6"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
20	1	1"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
21	1	4"	5S1D4S (SHOE)	5S1D4S
22	1	4"	5G2 (GUIDE FOR WELDED SHOE)	5G2
23	1	4"	5DA1 (DIRECTIONAL ANCHOR)	5DA1-D
24	1	4"	5FS1 (FIELD CANTILEVER SUPPORT)	5FS1S2
25	1	6"	RESTRICTION ORIFICE	LP1YF00650WP

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	PWP
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 6" DUP PWP (344) (108) 061025

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-PWP-061025-03	1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026

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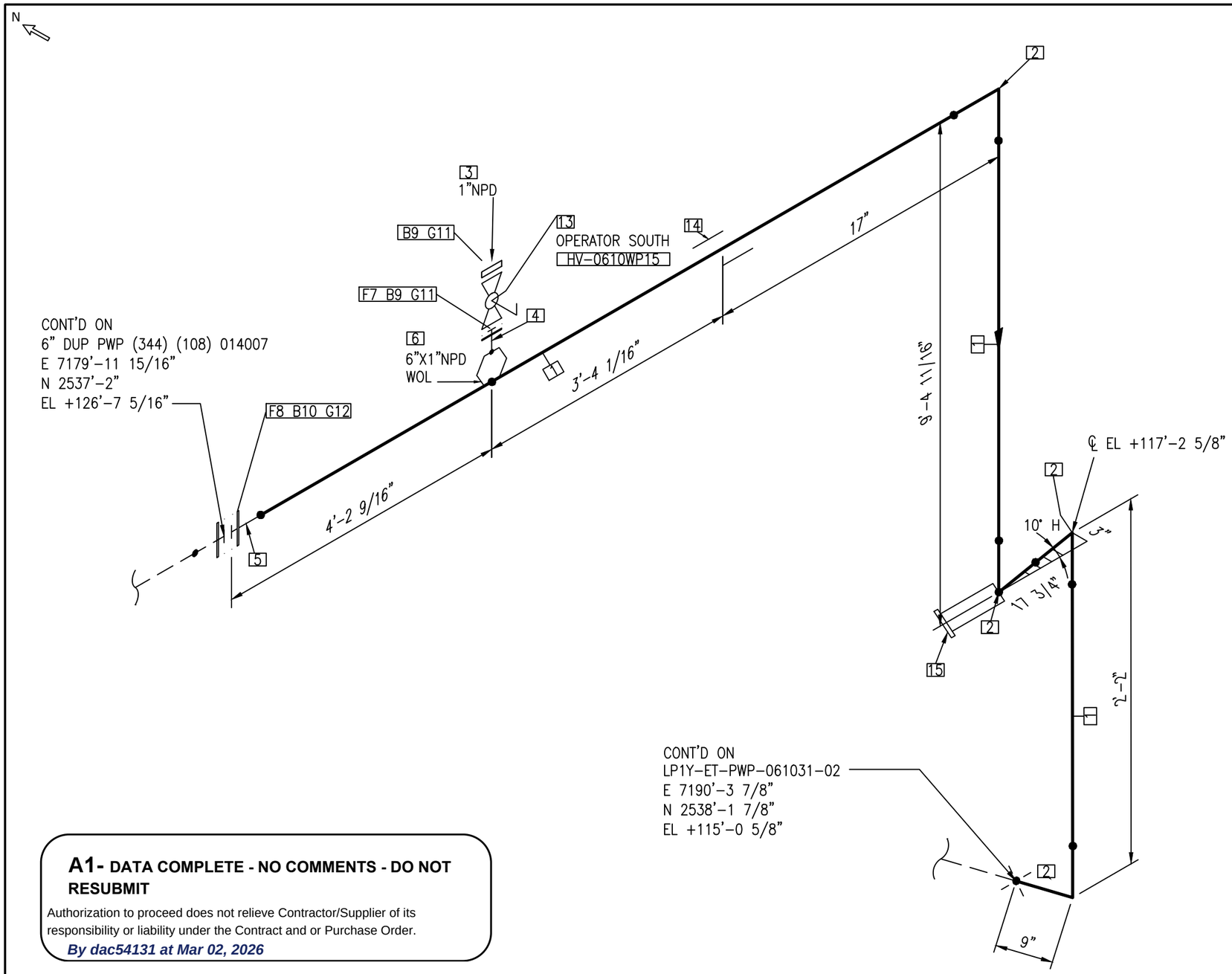

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	16'-6"	6"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
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3	1	1"	FLG BLIND 150# RF DUPLEX STL F60	5616811
4	1	1"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
5	1	6"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
6	1	6"X1"	PIPET SCH 10S DUPLEX STL F60	5512022
7	1	1"	FLG LAP JOINT 150# 304 SS	5599576L
8	1	6"	FLG LAP JOINT 150# 304 SS	5599576L
9	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
10	8	3/4"X3 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
11	2	1"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
12	1	6"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
13	1	1"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
14	1	6"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-06-01
15	1	6"	5DS1 (DUMMY SUPPORT) L=1'-5"	5DS1-06

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By dac54131 at Mar 02, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	PWP
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 6" DUP PWP (344) (108) 061031

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-PWP-061031-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 23-FEB-2026

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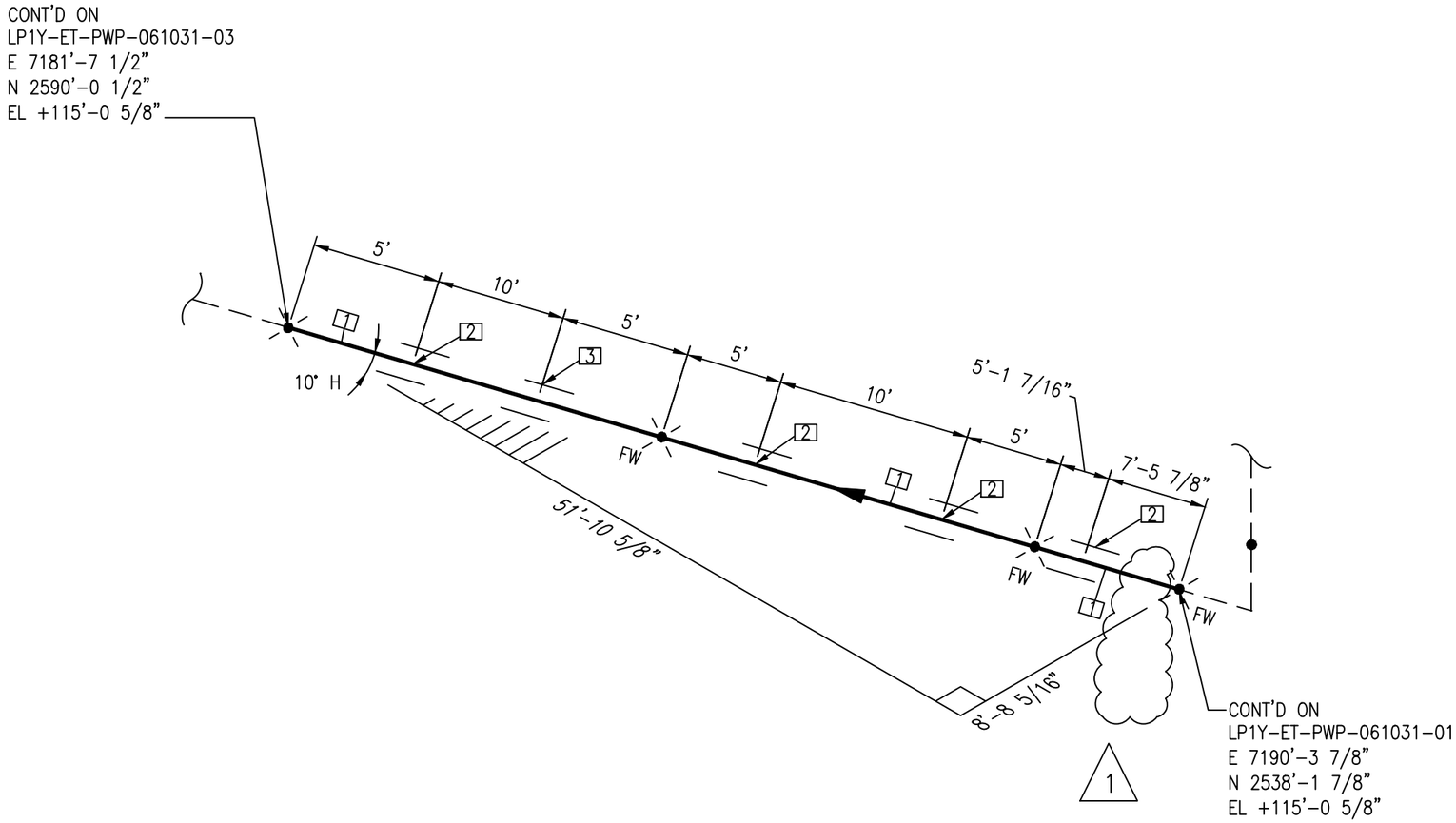
N

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	52'-8"	6"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	4	6"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-06-01
3	1	6"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-06-01



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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	PWP
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

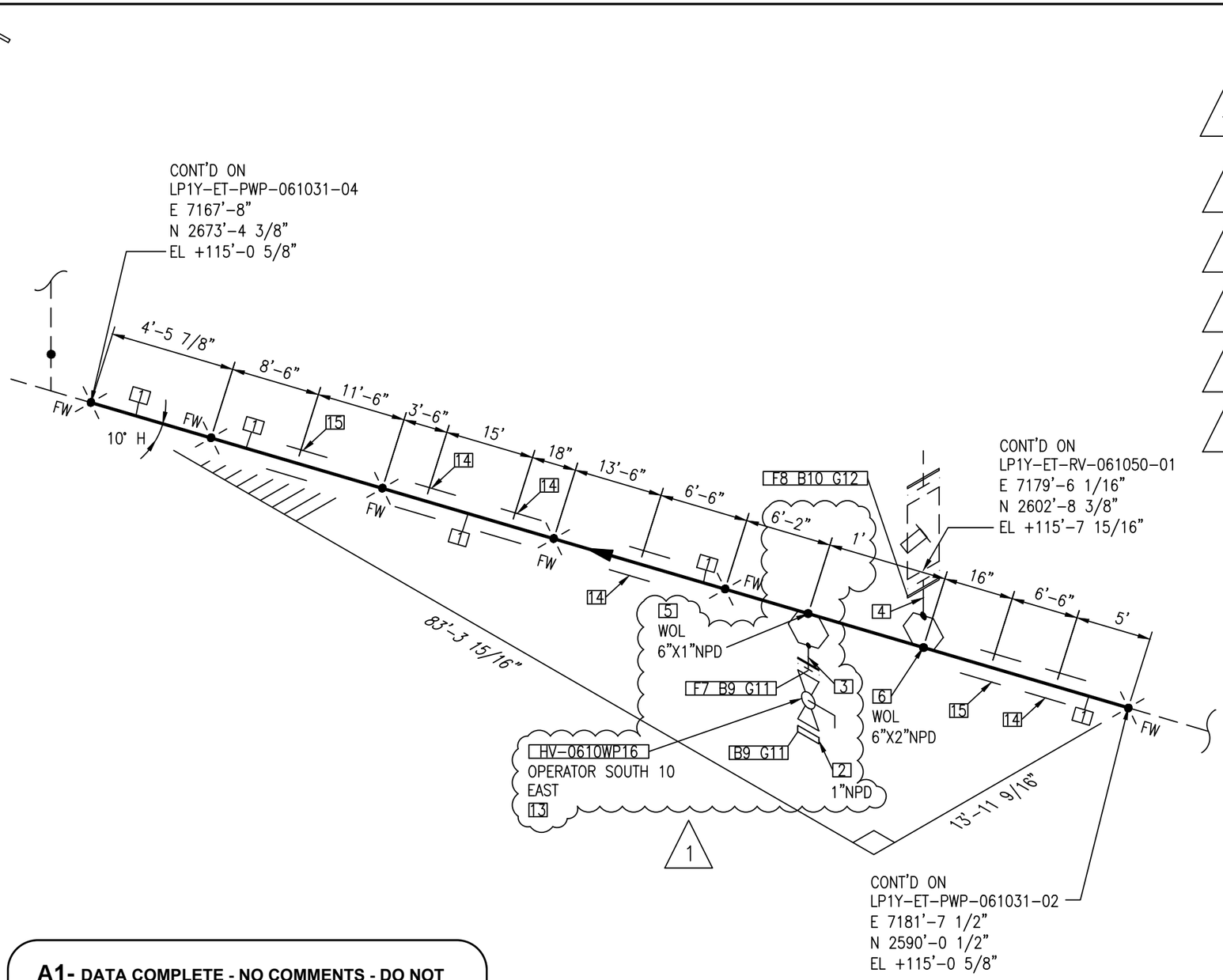
HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 6" DUP PWP (344) (108) 061031

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-PWP-061031-02	1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	84'-6"	6"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	1	1"	FLG BLIND 150# RF DUPLEX STL F60	5616811
3	1	1"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
4	1	2"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
5	1	6"X1"	PIPET SCH 10S DUPLEX STL F60	5512022
6	1	6"X2"	PIPET SCH 10S DUPLEX STL F60	5512022
7	1	1"	FLG LAP JOINT 150# 304 SS	5599576L
8	1	2"	FLG LAP JOINT 150# 304 SS	5599576L
9	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
10	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	2	1"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
12	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	1	1"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
14	4	6"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-06-01
15	2	6"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-06-01

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

GENERAL NOTE:
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3. WELDS AND/OR COUPLINGS DEFINING RANDOM PIPE SPOOL LENGTHS ARE FOR REFERENCE ONLY. FINAL WELD OR COUPLING LOCATION IS TO BE DETERMINED BY THE CONTRACTOR.
4. THE MECHANICAL CONTRACTOR TO FIELD VERIFY INTERFACE POINTS BETWEEN EQUIPMENT & PLUMBING FIXTURES IN 3rd FIX (3F SERIES CWP) AND YS YARD STICK BUILT (YS SERIES CWP EQUIPMENT CONNECTIONS) ISOMETRICS PRIOR TO FABRICATION.
5. THE MECHANICAL CONTRACTOR IS TO VERIFY ALL DIMENSIONAL DATA WITH ACTUAL FIELD CONDITIONS PRIOR TO INSTALLATION OF THE SYSTEM.
6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	PWP
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 6" DUP PWP (344) (108) 061031

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-PWP-061031-03	1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026

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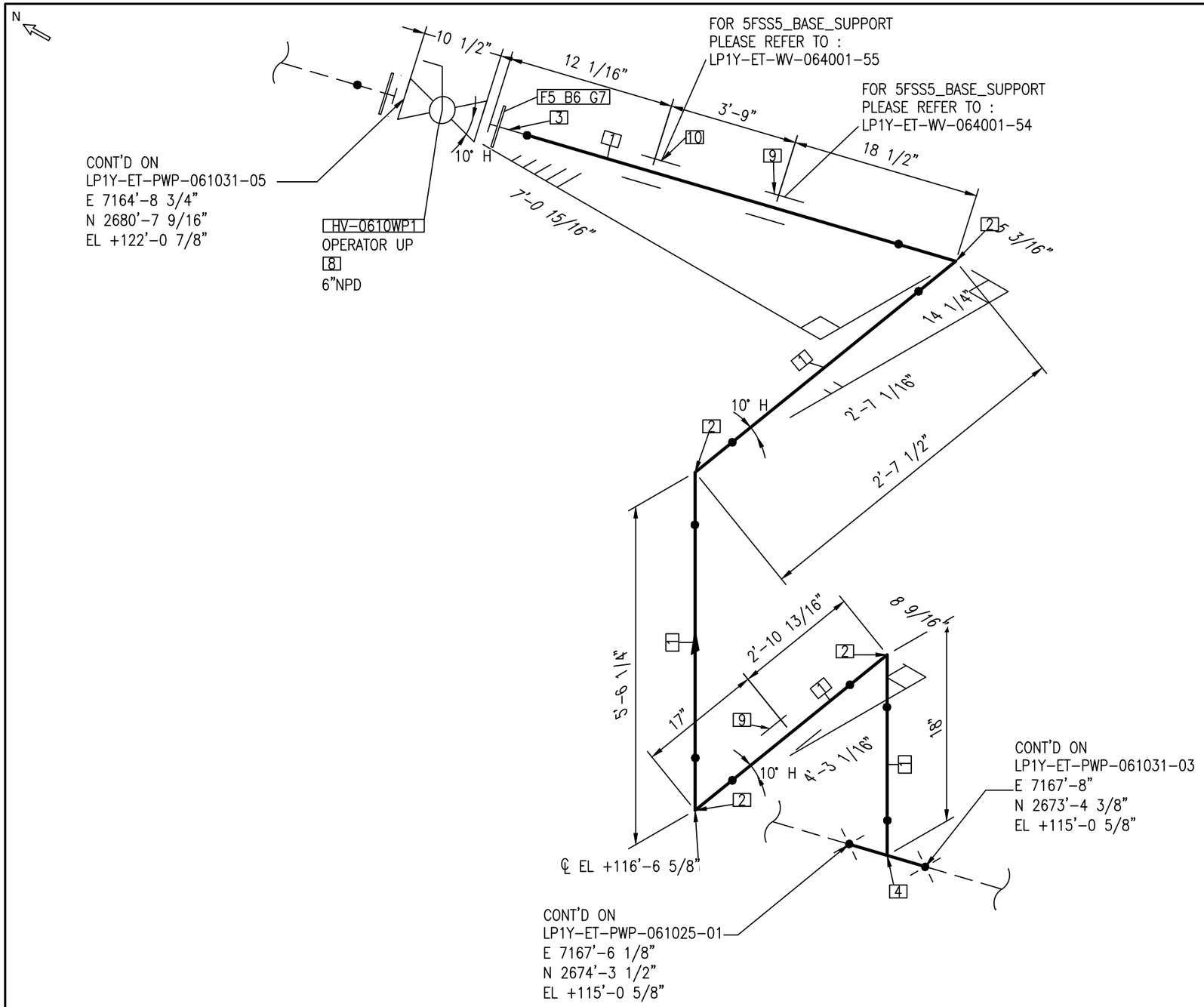

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	13'-6"	6"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	4	6"	ELL 90 DEG LR SCH 10S DUPLEX STL S32205-W	5460743
3	1	6"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
4	1	6"	TEE SCH 10S DUPLEX STL S32205-W	5460752
5	1	6"	FLG LAP JOINT 150# 304 SS	5599576L
6	8	3/4"X3 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
7	1	6"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
8	1	6"	BALL CL 150 RF DI, PFA LINED FP GR OP	5300941L4
9	2	6"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-06-01
10	1	6"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-06-01

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By dac54131 at Mar 02, 2026

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7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	PWP
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
						/	
						/	
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 6" DUP PWP (344) (108) 061031

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-PWP-061031-04	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 23-FEB-2026

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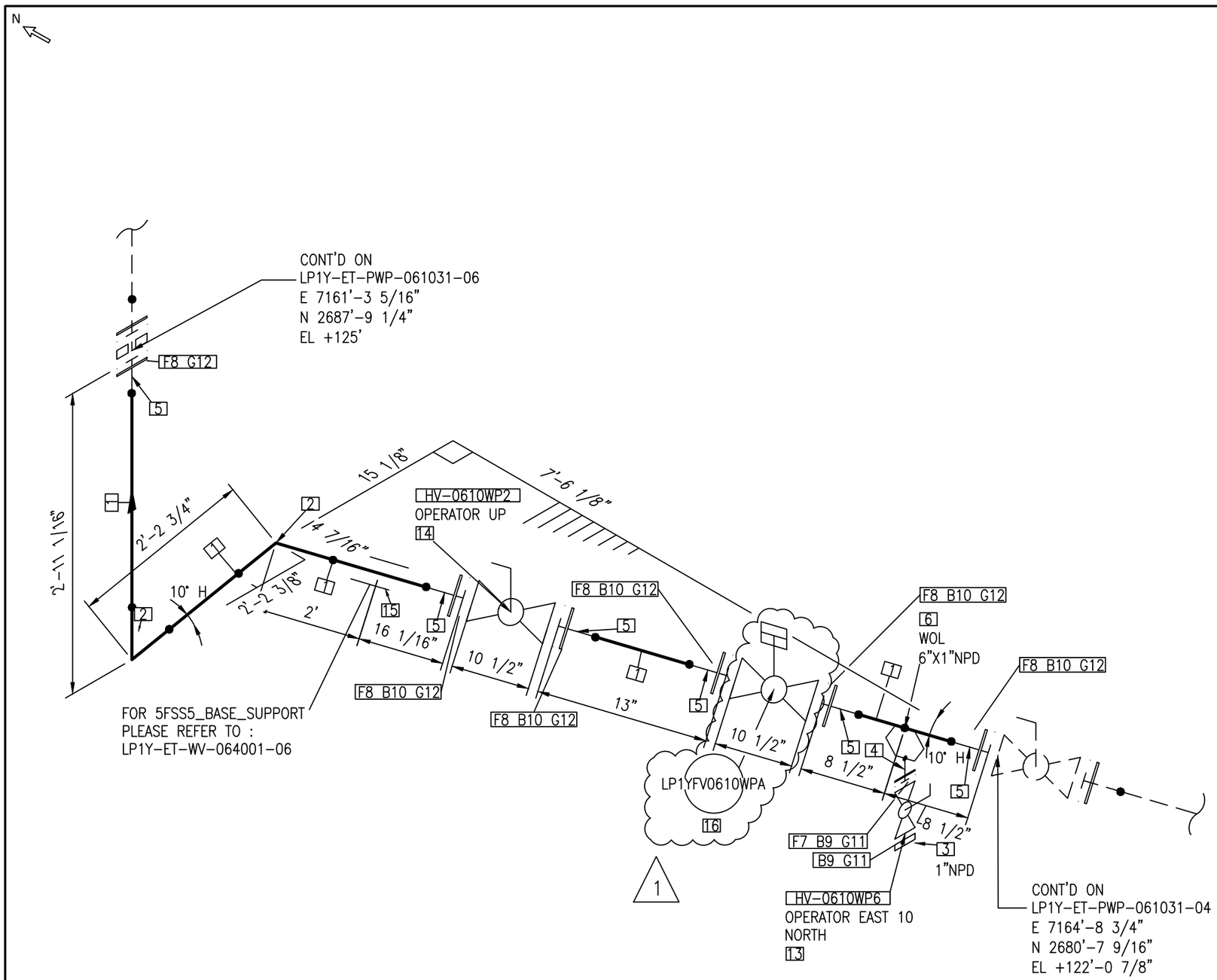

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	6'-3"	6"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	2	6"	ELL 90 DEG LR SCH 10S DUPLEX STL S32205-W	5460743
3	1	1"	FLG BLIND 150# RF DUPLEX STL F60	5616811
4	1	1"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
5	6	6"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
6	1	6"X1"	PIPET SCH 10S DUPLEX STL F60	5512022
7	1	1"	FLG LAP JOINT 150# 304 SS	5599576L
8	6	6"	FLG LAP JOINT 150# 304 SS	5599576L
9	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
10	40	3/4"X3 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
11	2	1"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
12	6	6"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
13	1	1"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
14	1	6"	BALL CL 150 RF DI, PFA LINED FP GR OP	5300941L4
15	1	6"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-06-01
16	1	6"	KNA BALL VALVE ASME CL. 150# RF	LP1YFV0610WPA

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	PWP
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

LINE NUMBER
LP1Y 6" DUP PWP (344) (108) 061031

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-PWP-061031-05	1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026

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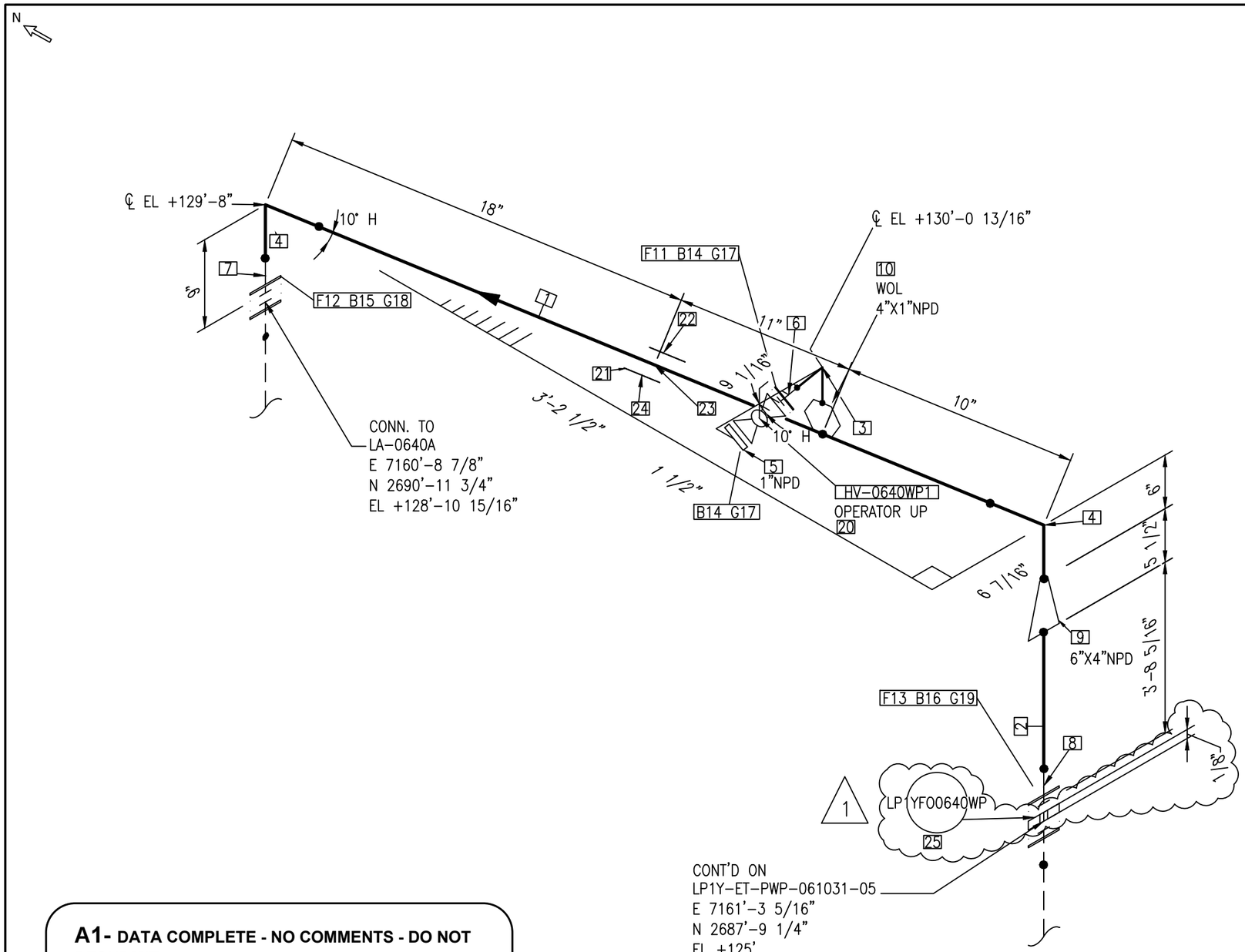

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	2'-4"	4"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	3'-5"	6"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
3	1	1"	ELL 90 DEG LR SCH 10S DUPLEX STL S32205-W	5460743
4	2	4"	ELL 90 DEG LR SCH 10S DUPLEX STL S32205-W	5460743
5	1	1"	FLG BLIND 150# RF DUPLEX STL F60	5616811
6	1	1"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
7	1	4"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
8	1	6"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
9	1	6"X4"	REDUCER CONC SCH 10S DUPLEX STL S32205-W	5460754
10	1	4"X1"	PIPET SCH 10S DUPLEX STL F60	5512022
11	1	1"	FLG LAP JOINT 150# 304 SS	5599576L
12	1	4"	FLG LAP JOINT 150# 304 SS	5599576L
13	1	6"	FLG LAP JOINT 150# 304 SS	5599576L
14	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
15	8	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
16	8	3/4"X3 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
17	2	1"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
18	1	4"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
19	1	6"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
20	1	1"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
21	1	4"	5S1D4S (SHOE)	5S1D4S
22	1	4"	5G2 (GUIDE FOR WELDED SHOE)	5G2
23	1	4"	5DA1 (DIRECTIONAL ANCHOR)	5DA1-D
24	1	4"	5FS1 (FIELD CANTILEVER SUPPORT)	5FS1S2
25	1	6"	RESTRICTION ORIFICE	LP1YFO0640WP

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	JHN / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	PWP
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 6" DUP PWP (344) (108) 061031

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-PWP-061031-06	1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026

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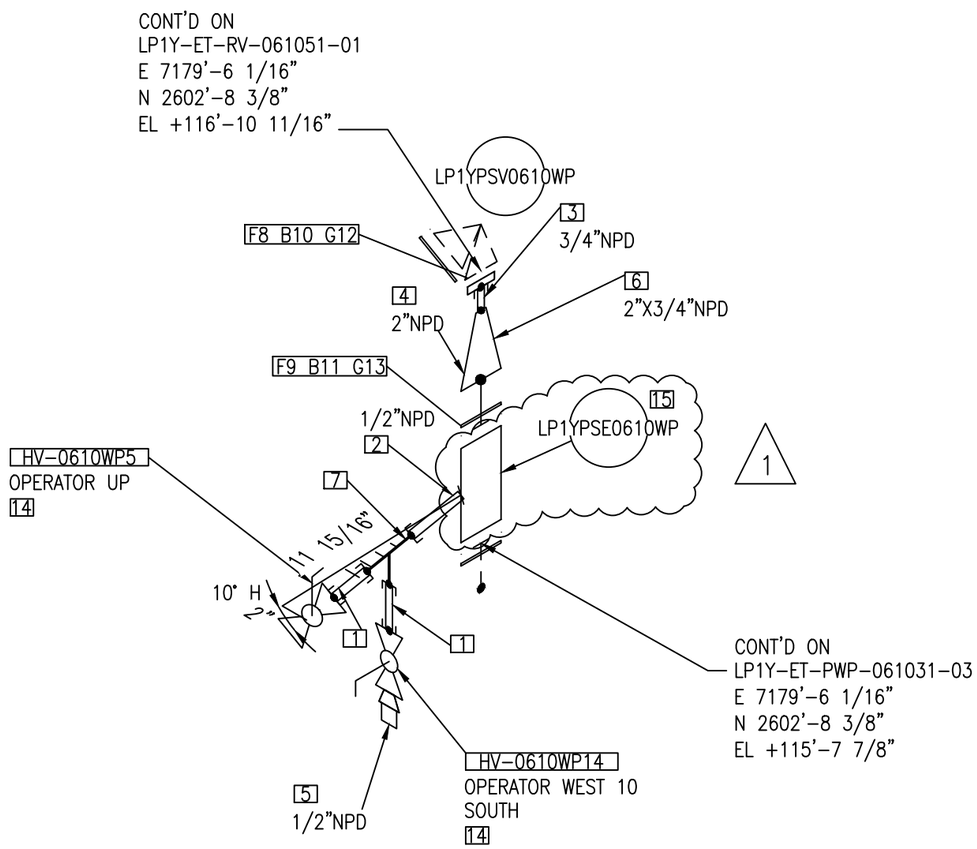
N



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	2	1/2"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
2	1	1/2"	NIPPLE SCH 80S 316/316L SS TOE 6" LONG	5540563
3	1	3/4"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
4	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	1/2"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
6	1	2"X3/4"	SWAGE CONC SCH 10S 316/316L SS BBE	5546561
7	1	1/2"	TEE 3000# SW 316/316L SS	5480140
8	1	3/4"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
9	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
10	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
11	4	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
12	1	3/4"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
13	1	2"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
14	2	1/2"	BALL 1000# CWP SW X FNPT 316SS TFE FP HNDL OP	5352192L6
15	1	2"	RUPTURE DISK AND HOLDER ASSEMBLY	LP1YPSE0610WP

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By dac54131 at Apr 10, 2026

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1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	JHN / N/A	MDW
REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	RV
PIPE SCHEDULE:	315
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS RV (315) (108) 061050

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-ET-RV-061050-01

REV
1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026

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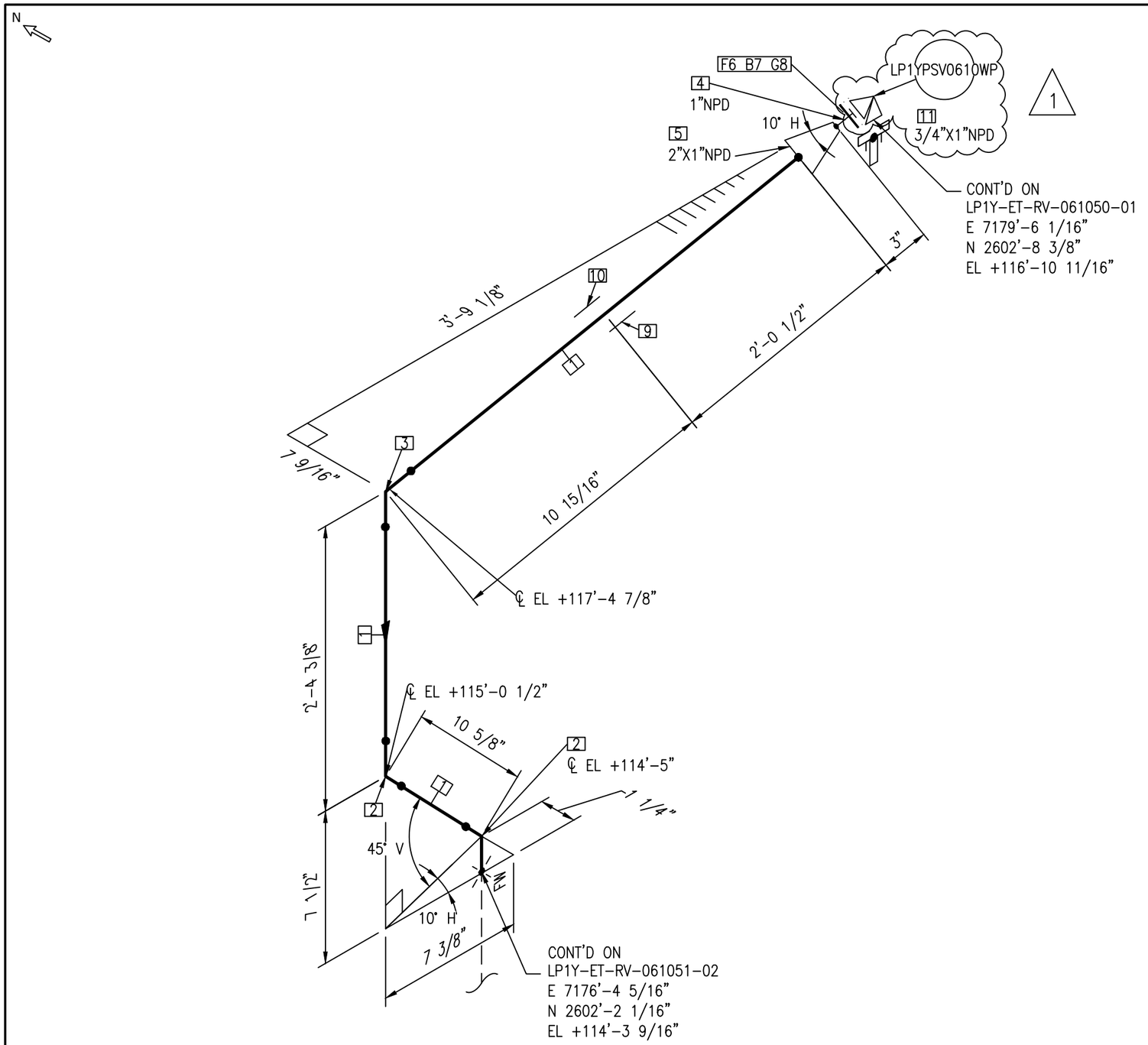

BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	5'-5"	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	ELL 45 SCH 10S 316/316L SS A403-W	5450712
3	1	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
4	1	1"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
5	1	2"X1"	REDUCER CONC SCH 10S 316/316L SS A403-W	5450717
6	1	1"	FLG LAP JOINT 150# 316 SS	5599578L
7	4	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS	5676576L
8	1	1"	GASKET 150# EPDM RUBBER RING 1/8" THK	5664315
9	1	2"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-02-01
10	1	2"	5FS1 (FIELD CANTILEVER SUPPORT)	5FS1S1
11	1	3/4"X1"	PRESSURE SAFETY VALVE	LP1YPSV0610WP

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	RV
PIPE SCHEDULE:	315
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS RV (315) (108) 061051

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-RV-061051-01	1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026

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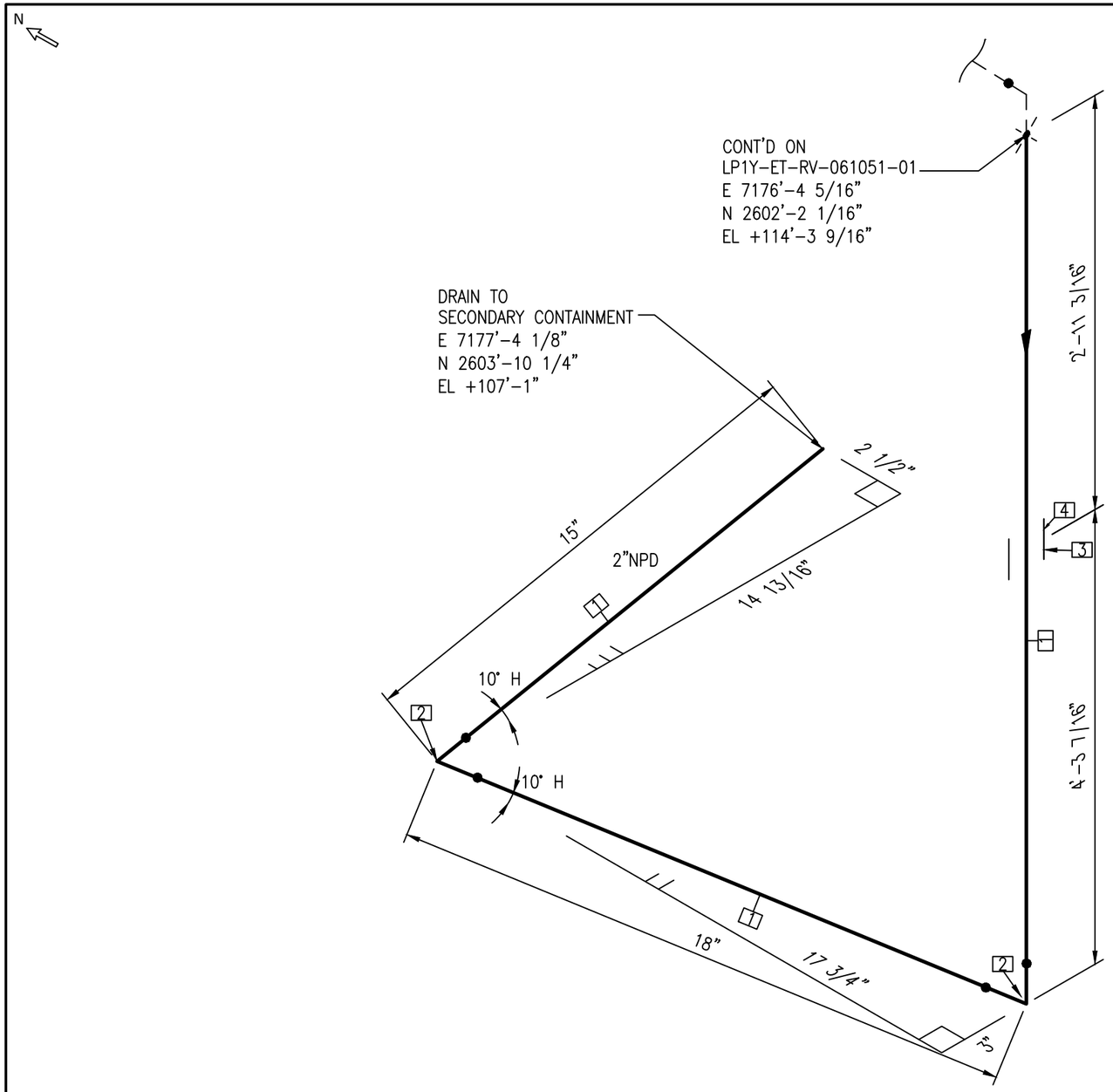

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	9'	2"	PIPE SCH 10S ERW 316/316L SS A312	5368751
2	2	2"	ELL 90 LR SCH 10S 316/316L SS A403-W	5450706
3	1	2"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-02-01
4	1	2"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-02-01

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By dac54131 at Mar 02, 2026

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0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW
REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	RV
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS RV (315) (108) 061051

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-RV-061051-02	0

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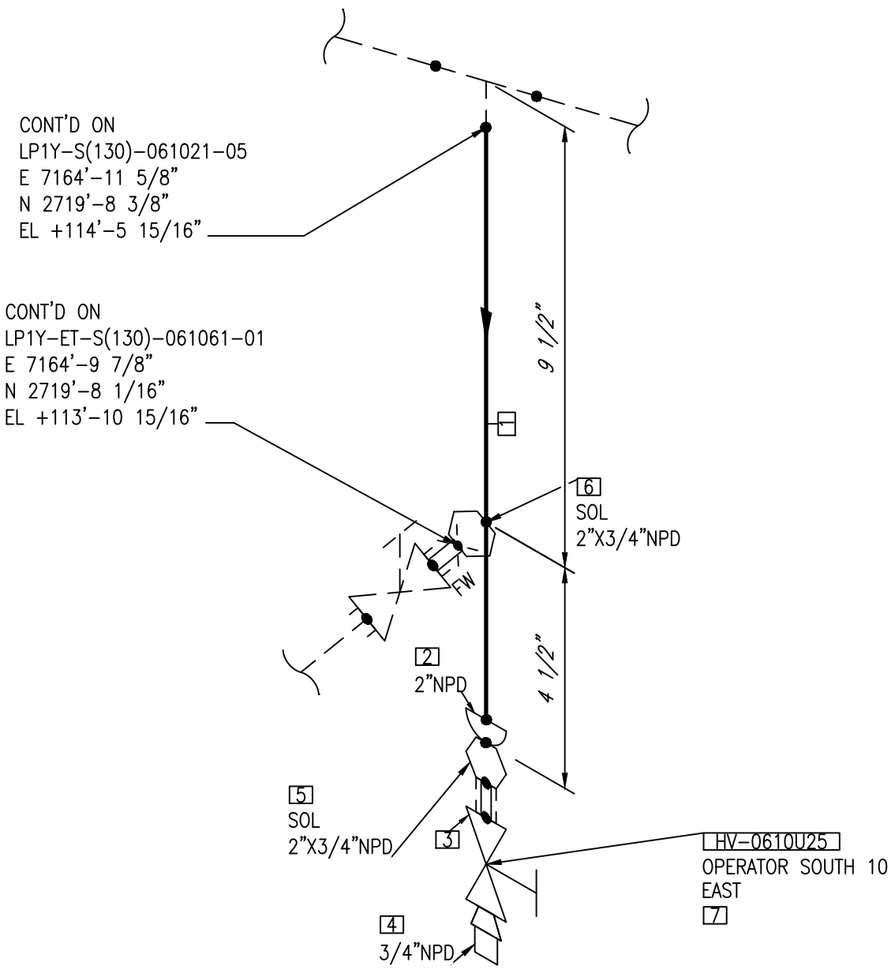
N



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	10"	2"	PIPE SCH 40 ERW STL A53-B	5356648
2	1	2"	CAP SCH 40 STL A234 WPB	5374070
3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
4	1	3/4"	PLUG ROUND HEAD SCRD STL A105	5490665
5	1	2"X3/4"	FLAT SOCKOLET 3000# STL A105	5532672
6	1	2"X3/4"	SOCKOLET 3000# STL A105	5532671
7	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273

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By dac54131 at Mar 02, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061021

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-S(130)-061021-06	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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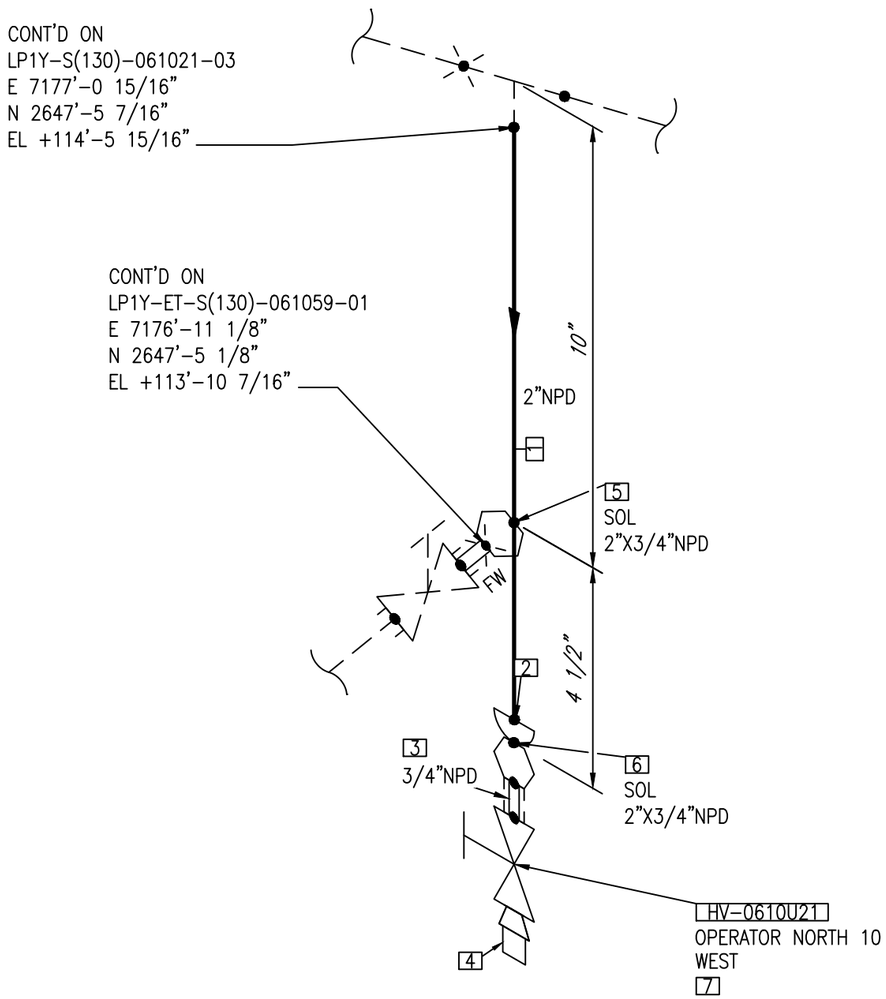

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	11"	2"	PIPE SCH 40 ERW STL A53-B	5356648
2	1	2"	CAP SCH 40 STL A234 WPB	5374070
3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
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5	1	2"X3/4"	SOCKOLET 3000# STL A105	5532671
6	1	2"X3/4"	FLAT SOCKOLET 3000# STL A105	5532672
7	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273

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By dac54131 at Mar 02, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061021

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-S(130)-061021-07	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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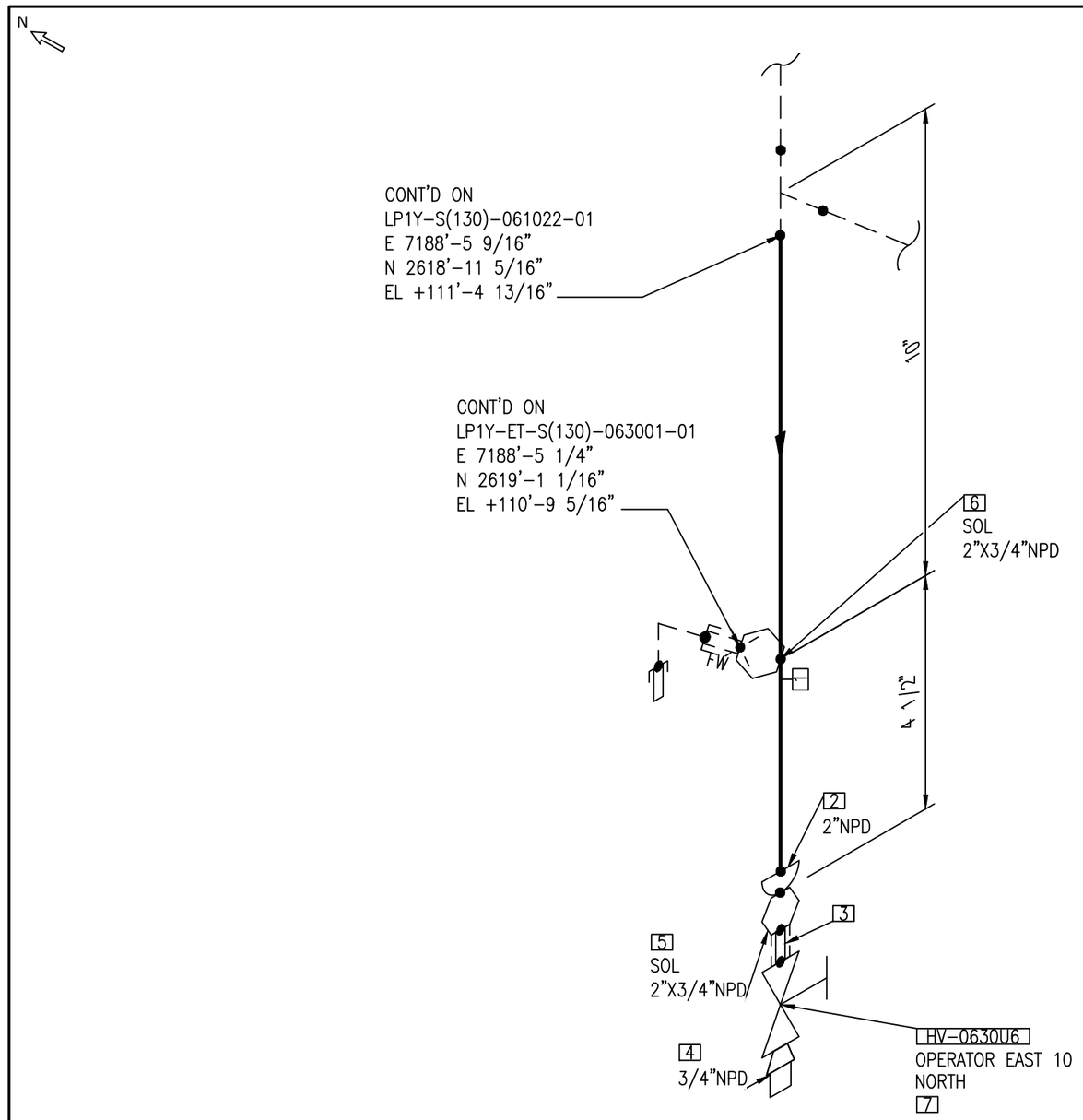

BILL OF MATERIALS				
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By dac54131 at Mar 02, 2026

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REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061022

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-ET-S(130)-061022-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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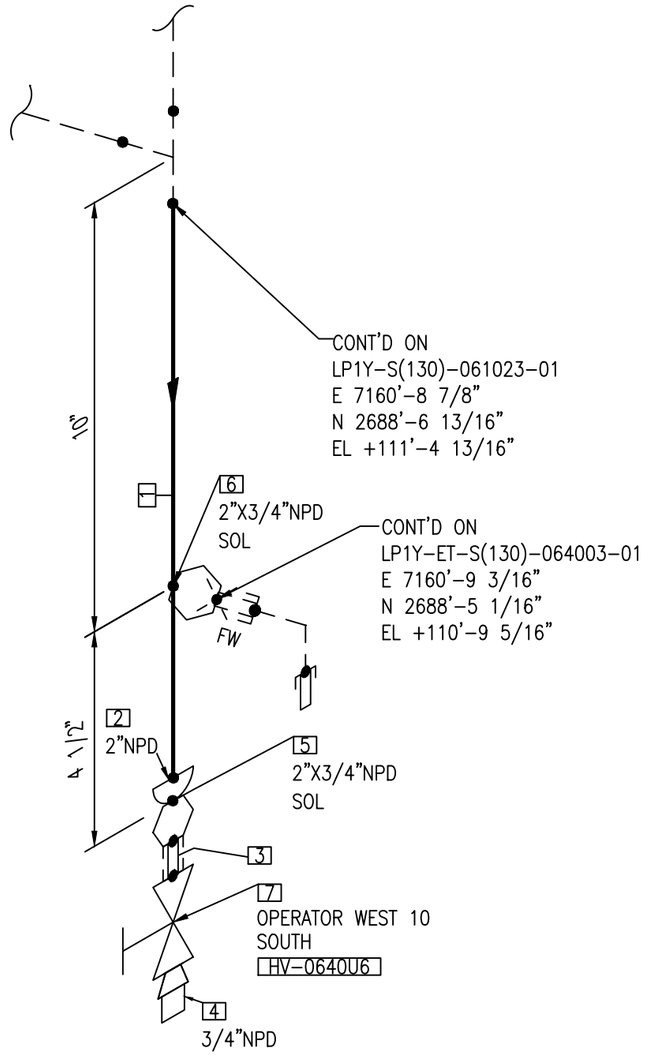
N



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	11"	2"	PIPE SCH 40 ERW STL A53-B	5356648
2	1	2"	CAP SCH 40 STL A234 WPB	5374070
3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
4	1	3/4"	PLUG ROUND HEAD SCRD STL A105	5490665
5	1	2"X3/4"	FLAT SOCKOLET 3000# STL A105	5532672
6	1	2"X3/4"	SOCKOLET 3000# STL A105	5532671
7	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Mar 02, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

GENERAL NOTE:
1. REFER TO THE PIPING LINE LIST FOR PROCESS DATA/CONDITIONS AND OTHER FABRICATION DETAILS SUCH AS TESTING, CLEANING, AND PAINTING REQUIREMENTS.
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3. WELDS AND/OR COUPLINGS DEFINING RANDOM PIPE SPOOL LENGTHS ARE FOR REFERENCE ONLY. FINAL WELD OR COUPLING LOCATION IS TO BE DETERMINED BY THE CONTRACTOR.
4. THE MECHANICAL CONTRACTOR TO FIELD VERIFY INTERFACE POINTS BETWEEN EQUIPMENT & PLUMBING FIXTURES IN 3rd FIX (3F SERIES CWP) AND YS YARD STICK BUILT (YS SERIES CWP EQUIPMENT CONNECTIONS) ISOMETRICS PRIOR TO FABRICATION.
5. THE MECHANICAL CONTRACTOR IS TO VERIFY ALL DIMENSIONAL DATA WITH ACTUAL FIELD CONDITIONS PRIOR TO INSTALLATION OF THE SYSTEM.
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7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061023

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000064	LP1Y-ET-S(130)-061023-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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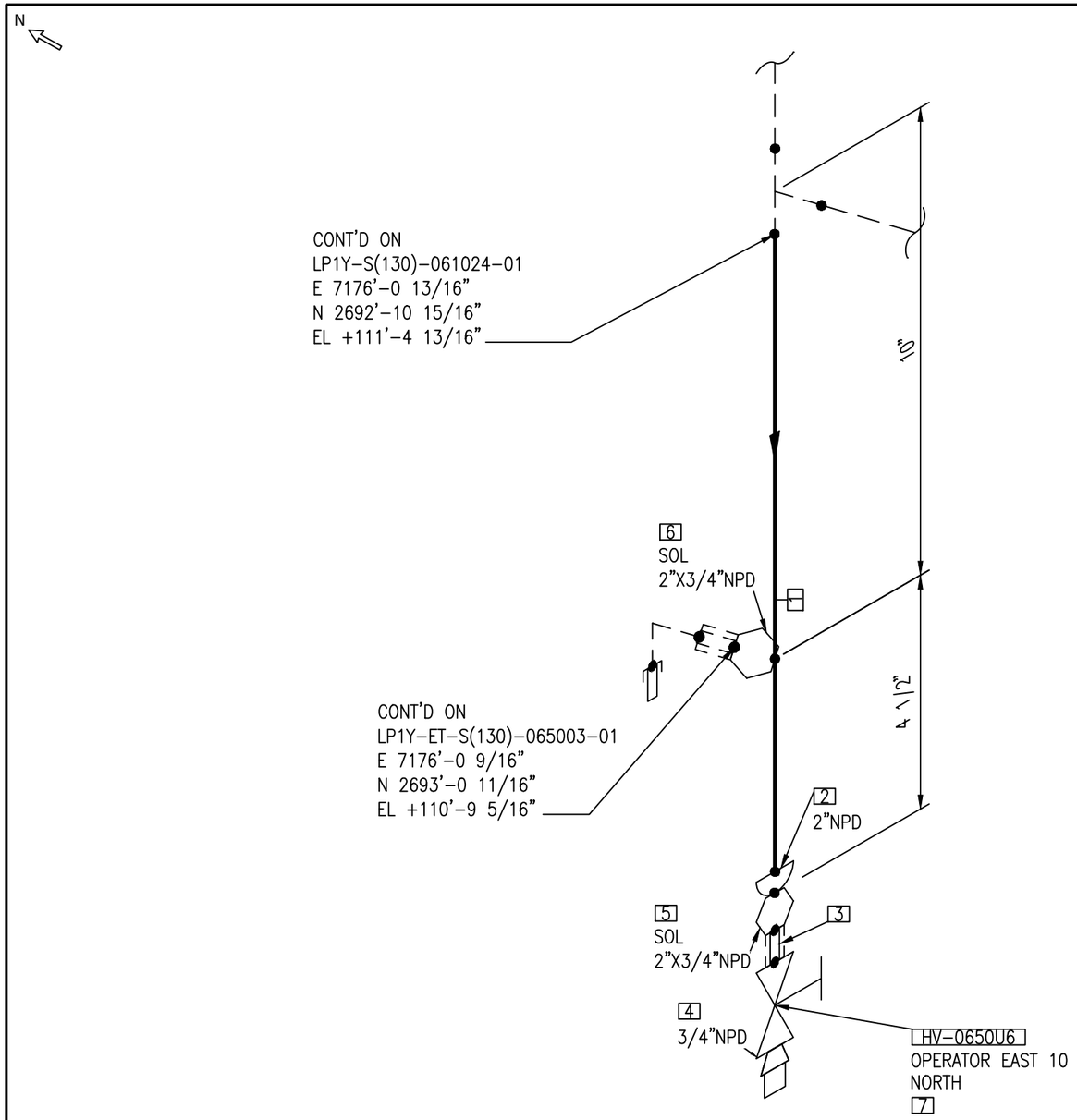

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	11"	2"	PIPE SCH 40 ERW STL A53-B	5356648
2	1	2"	CAP SCH 40 STL A234 WPB	5374070
3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
4	1	3/4"	PLUG ROUND HEAD SCRD STL A105	5490665
5	1	2"X3/4"	FLAT SOCKOLET 3000# STL A105	5532672
6	1	2"X3/4"	SOCKOLET 3000# STL A105	5532671
7	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Mar 02, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061024

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000065	LP1Y-ET-S(130)-061024-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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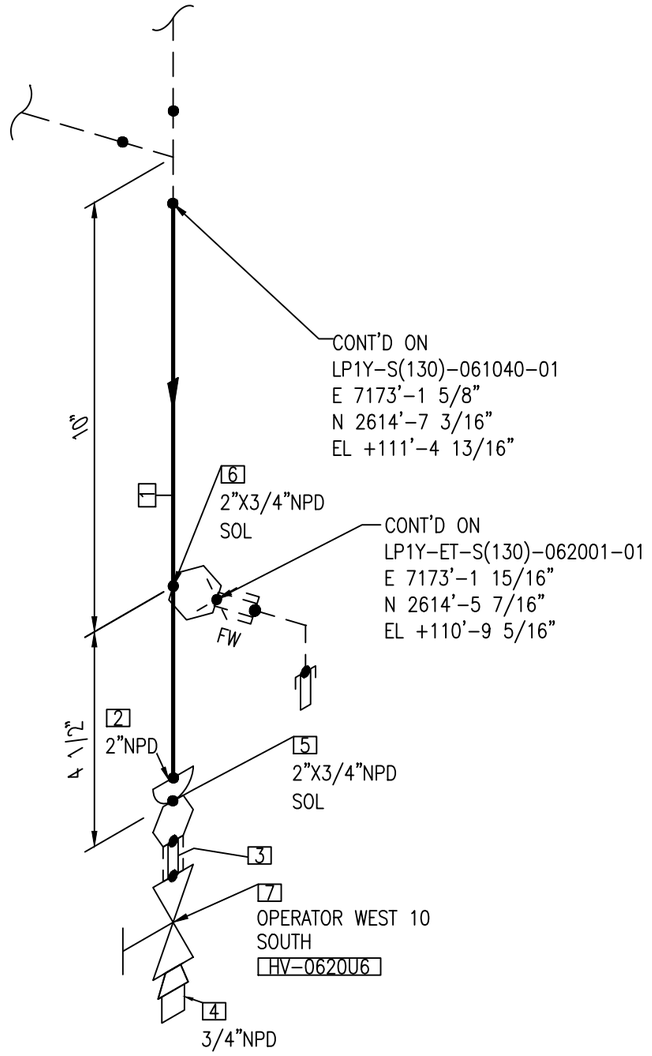
N



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	11"	2"	PIPE SCH 40 ERW STL A53-B	5356648
2	1	2"	CAP SCH 40 STL A234 WPB	5374070
3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
4	1	3/4"	PLUG ROUND HEAD SCRD STL A105	5490665
5	1	2"X3/4"	FLAT SOCKOLET 3000# STL A105	5532672
6	1	2"X3/4"	SOCKOLET 3000# STL A105	5532671
7	1	3/4"	GATE 800# SW/SCRD STL, F6/STEL SEATS	5115273

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By dac54131 at Mar 02, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" CS S(130) (2) (103) 061040

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-ET-S(130)-061040-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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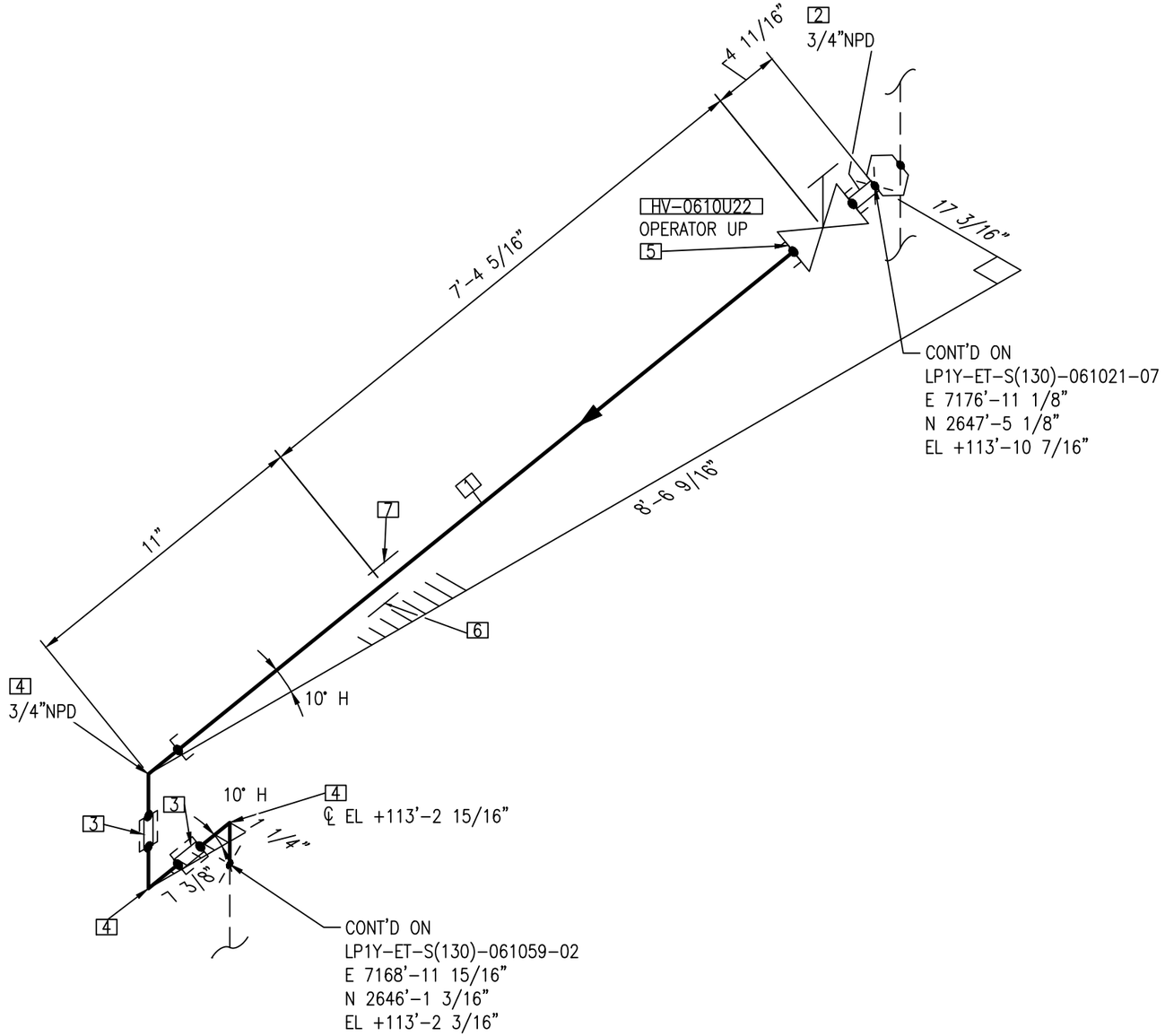

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	8'-1"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
2	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
3	2	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
4	3	3/4"	ELL 90 DEG 3000# SW STL A105	5478105
5	1	3/4"	GATE 800# SW STL, F6/STEL SEATS	5061780L0
6	1	3/4"	5C (INSULATION CRADLE)	5C-75-15
7	1	3/4"	5FS1 (FIELD SUPPORT)	5FS1S1

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS S(130) (2) (103) 061059

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-S(130)-061059-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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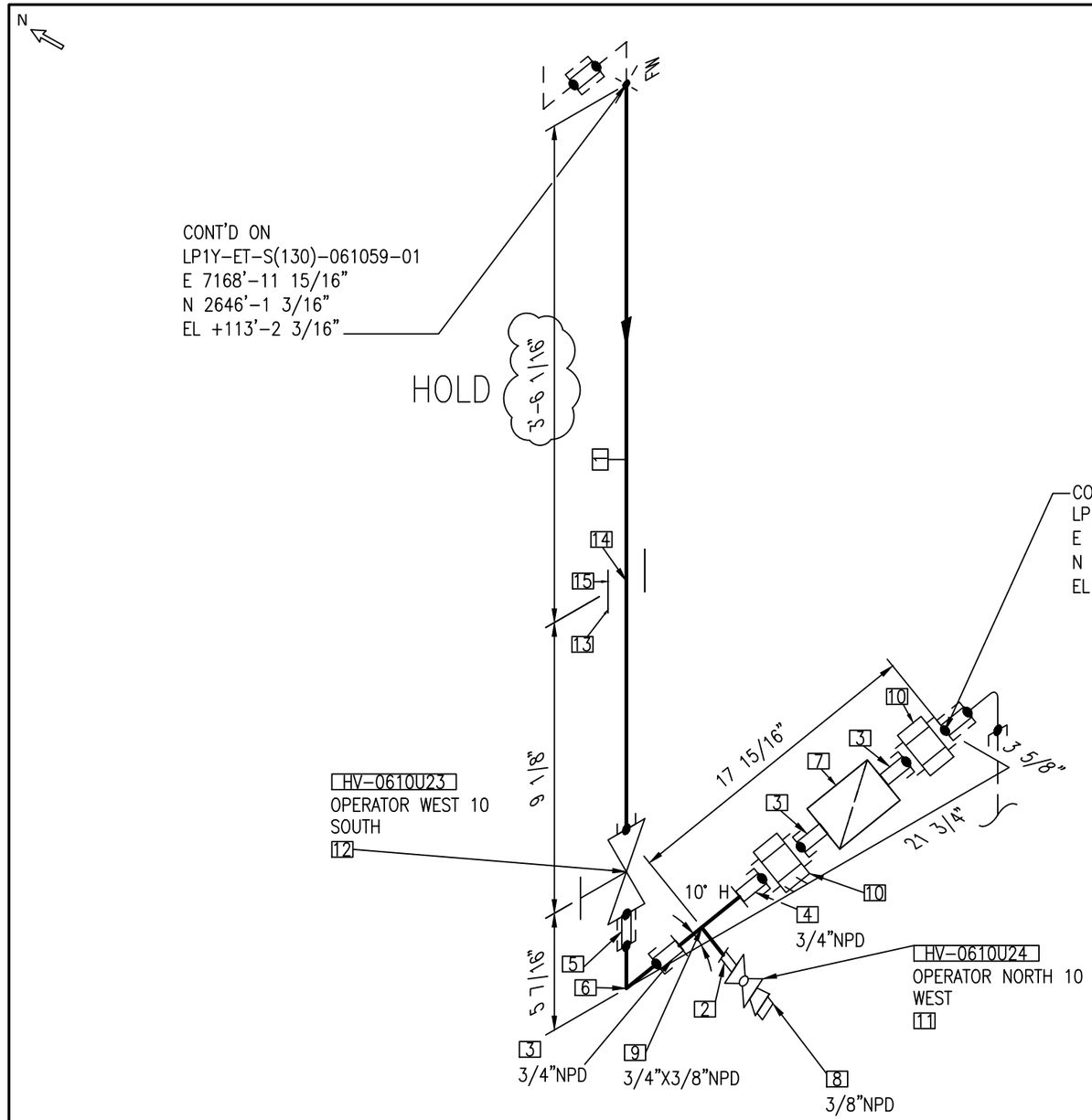

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	4'-2"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
2	1	3/8"	NIPPLE XXS STL A106-B TBE 3" LONG	5539171L
3	3	3/4"	NIPPLE SCH 160 STL A106-B TOE 3" LONG	5539141
4	1	3/4"	NIPPLE SCH 160 STL A106-B TOE 6" LONG	5539143
5	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
6	1	3/4"	ELL 90 DEG 3000# SW STL A105	5478105
7	1	3/4"	ARMSTRONG 2010 NPT INVERTED BUCKET STEAM TRAP	LP1YSTT0610U3
8	1	3/8"	PLUG ROUND HEAD SCRD STL A105	5490665
9	1	3/4"X3/8"	Y-STRAINER, KECKLEY STYLE SB7, CL 600 FNPT	LP1YSTR0610U3
10	2	3/4"	UNION 3000# SW STL A105	5478111
11	1	3/8"	BALL 2000# SCRD STL UNIBODY	5058791L5
12	1	3/4"	GATE 800# SW STL, F6/STEL SEATS	5061780L0
13	1	3/4"	5C (INSULATION CRADLE)	5C-75-15
14	1	3/4"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-75-15
15	1	3/4"	5FS1 (FIELD SUPPORT)	5FS1S1

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS S(130) (2) (103) 061059

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-S(130)-061059-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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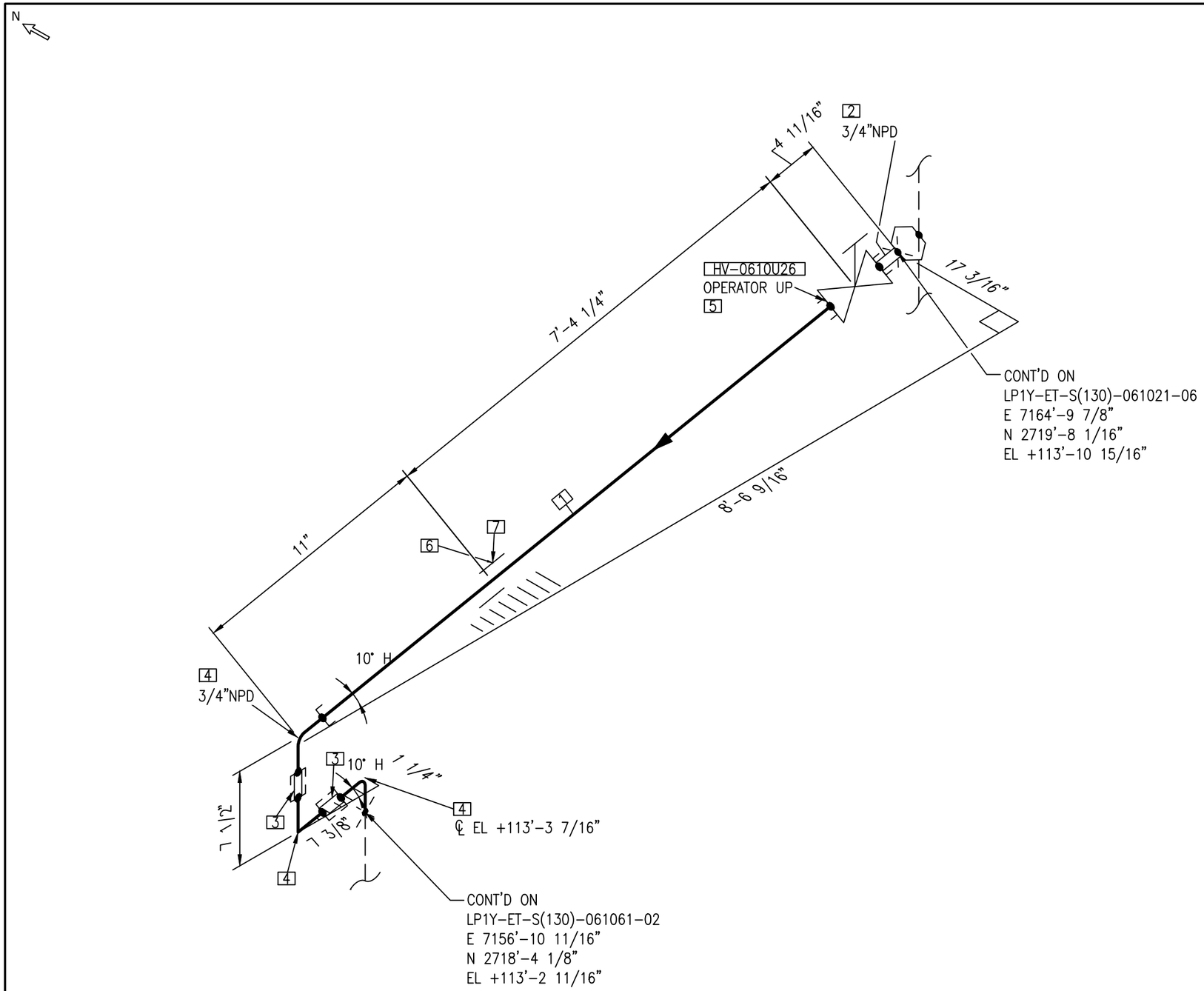

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	8'-1"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
2	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
3	2	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
4	3	3/4"	ELL 90 DEG 3000# SW STL A105	5478105
5	1	3/4"	GATE 800# SW STL, F6/STEL SEATS	5061780L0
6	1	3/4"	5C (INSULATION CRADLE)	5C-75-15
7	1	3/4"	5FS1 (FIELD SUPPORT)	5FS1S1

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By dac54131 at Mar 03, 2026

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6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
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8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS,
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW
REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS S(130) (2) (103) 061061

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-S(130)-061061-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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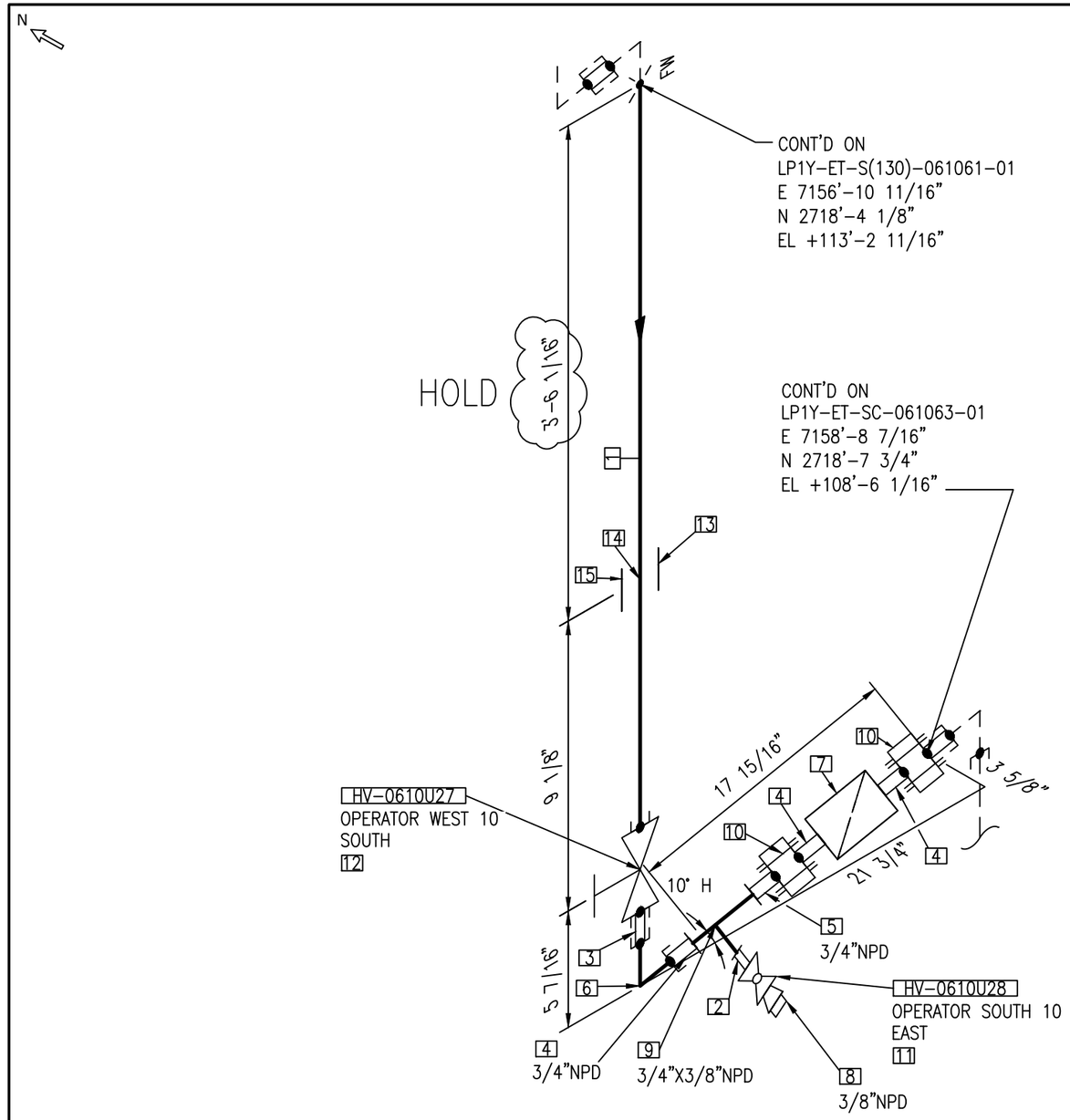

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	4'-2"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
2	1	3/8"	NIPPLE XXS STL A106-B TBE 3" LONG	5539171L
3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
4	3	3/4"	NIPPLE SCH 160 STL A106-B TOE 3" LONG	5539141
5	1	3/4"	NIPPLE SCH 160 STL A106-B TOE 6" LONG	5539143
6	1	3/4"	ELL 90 DEG 3000# SW STL A105	5478105
7	1	3/4"	ARMSTRONG 2010 NPT INVERTED BUCKET STEAM TRAP	LP1YSTT0610U4
8	1	3/8"	PLUG ROUND HEAD SCRD STL A105	5490665
9	1	3/4"X3/8"	Y-STRAINER, KECKLEY STYLE SB7, CL 600 FNPT	LP1YSTR0610U4
10	2	3/4"	UNION 3000# SW STL A105	5478111
11	1	3/8"	BALL 2000# SCRD STL UNIBODY	5058791L5
12	1	3/4"	GATE 800# SW STL, F6/STEL SEATS	5061780L0
13	1	3/4"	5C (INSULATION CRADLE)	5C-75-15
14	1	3/4"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-75-15
15	1	3/4"	5FS1 (FIELD SUPPORT)	5FS1S1

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS S(130) (2) (103) 061061

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-S(130)-061061-02	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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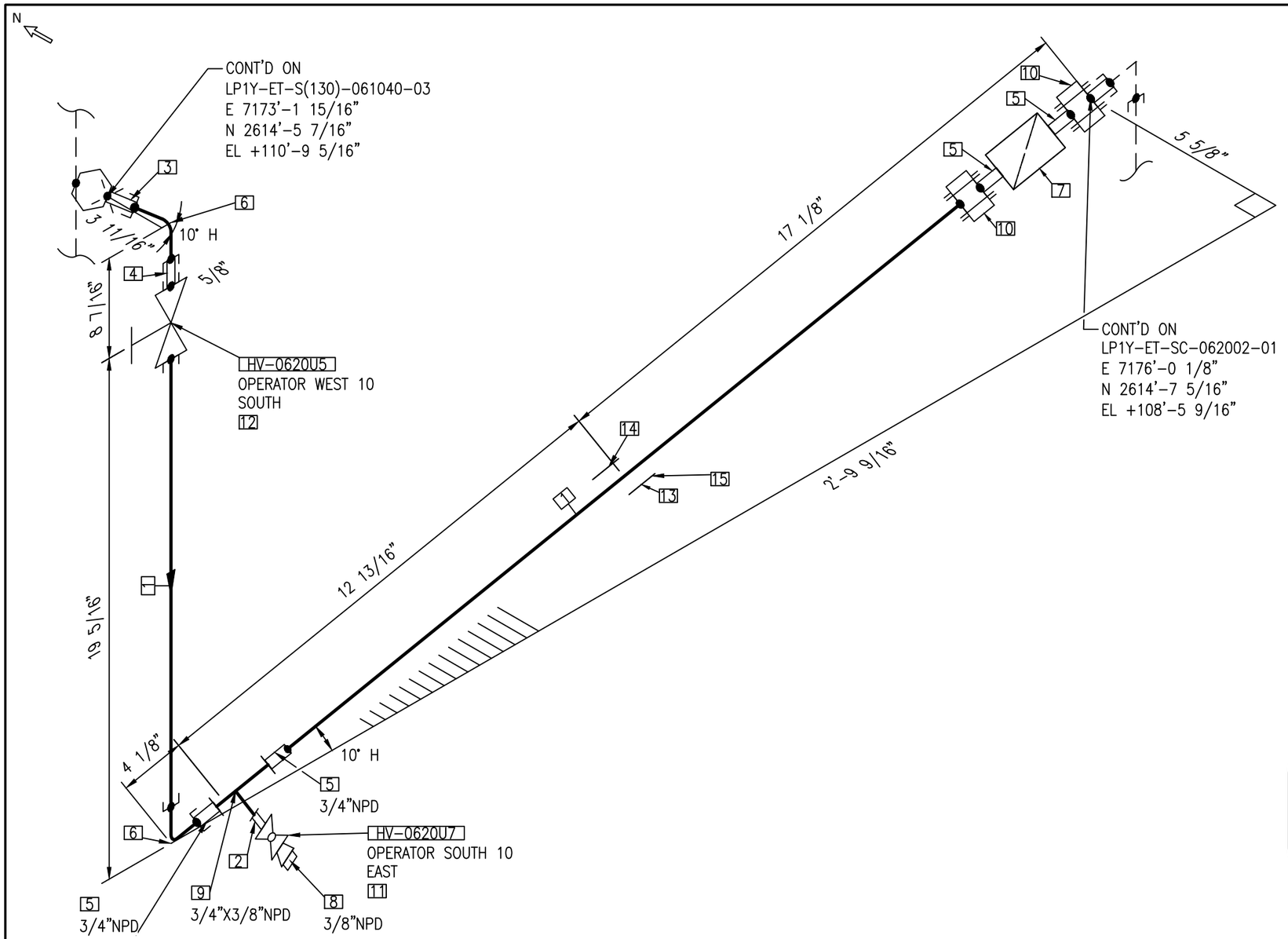

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	2'-8"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
2	1	3/8"	NIPPLE XXS STL A106-B TBE 3" LONG	5539171L
3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
4	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
5	4	3/4"	NIPPLE SCH 160 STL A106-B TOE 3" LONG	5539141
6	2	3/4"	ELL 90 DEG 3000# SW STL A105	5478105
7	1	3/4"	ARMSTRONG 2010 NPT INVERTED BUCKET STEAM TRAP	LP1YSTT0620U1
8	1	3/8"	PLUG ROUND HEAD SCRD STL A105	5490665
9	1	3/4"X3/8"	Y-STRAINER, KECKLEY STYLE SB7, CL 600 FNPT	LP1YSTR0620U2
10	2	3/4"	UNION 3000# SW STL A105	5478111
11	1	3/8"	BALL 2000# SCRD STL UNIBODY	5058791L5
12	1	3/4"	GATE 800# SW STL, F6/STEL SEATS	5061780L0
13	1	3/4"	5C (INSULATION CRADLE)	5C-75-15
14	1	3/4"	5FS1 (FIELD SUPPORT)	5FS1S1
15	1	2"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-02-15

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS S(130) (2) (103) 062001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-ET-S(130)-062001-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

B1LG-00-K215-00123-1_LP1Y-ET-S130-063001-01_A1.PDF

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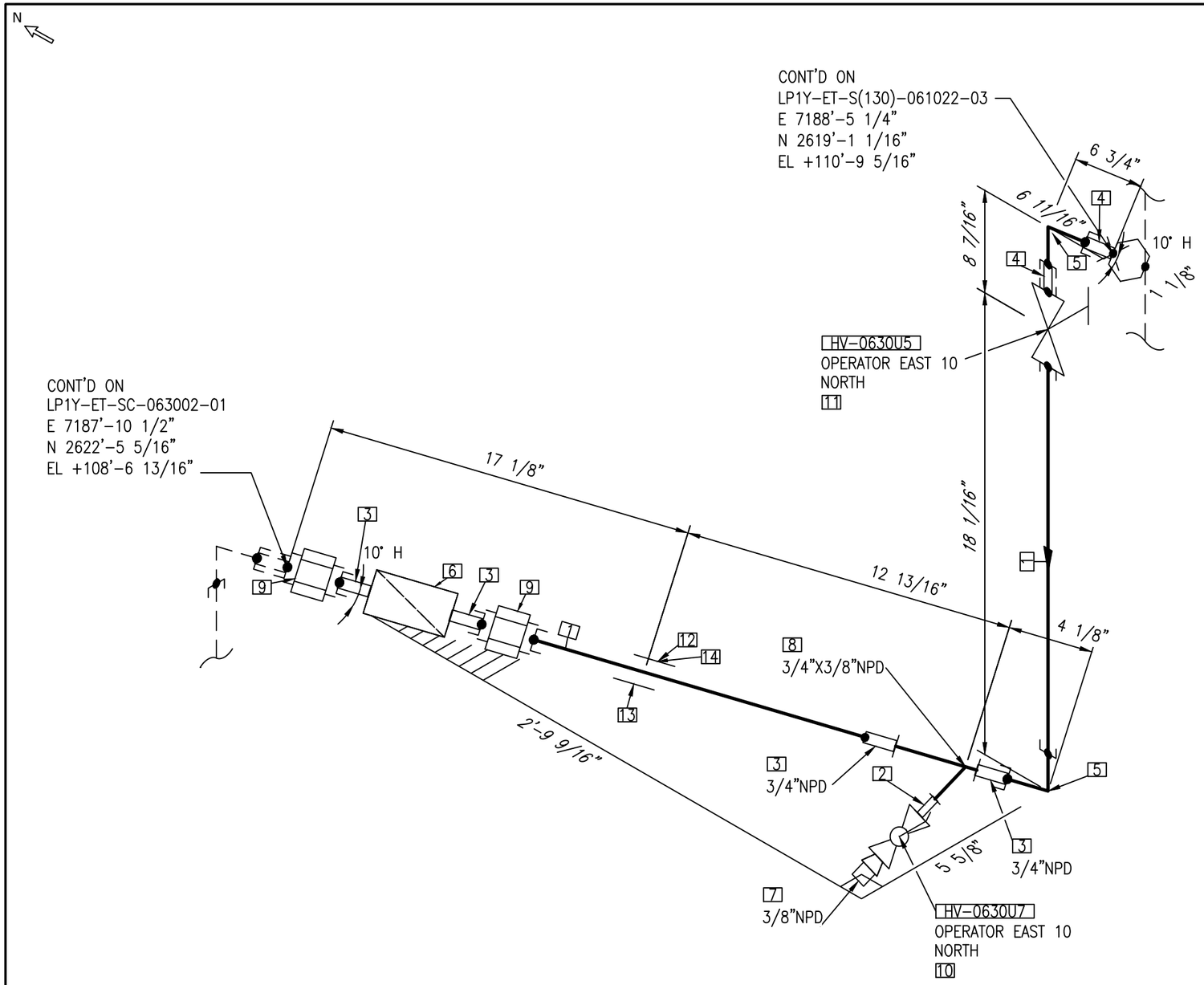

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	2'-7"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
2	1	3/8"	NIPPLE XXS STL A106-B TBE 3" LONG	5539171L
3	4	3/4"	NIPPLE SCH 160 STL A106-B TOE 3" LONG	5539141
4	2	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
5	2	3/4"	ELL 90 DEG 3000# SW STL A105	5478105
6	1	3/4"	ARMSTRONG 2010 NPT INVERTED BUCKET STEAM TRAP	LP1YSTT0630U1
7	1	3/8"	PLUG ROUND HEAD SCRD STL A105	5490665
8	1	3/4"X3/8"	Y-STRAINER, KECKLEY STYLE SB7, CL 600 FNPT	LP1YSTR0630U2
9	2	3/4"	UNION 3000# SW STL A105	5478111
10	1	3/8"	BALL 2000# SCRD STL UNIBODY	5058791L5
11	1	3/4"	GATE 800# SW STL, F6/STEL SEATS	5061780L0
12	1	3/4"	5C (INSULATION CRADLE)	5C-75-15
13	1	3/4"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-75-15
14	1	3/4"	5ABS2 (ANGLE BASE SUPPORT)	5ABS2

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS S(130) (2) (103) 063001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-ET-S(130)-063001-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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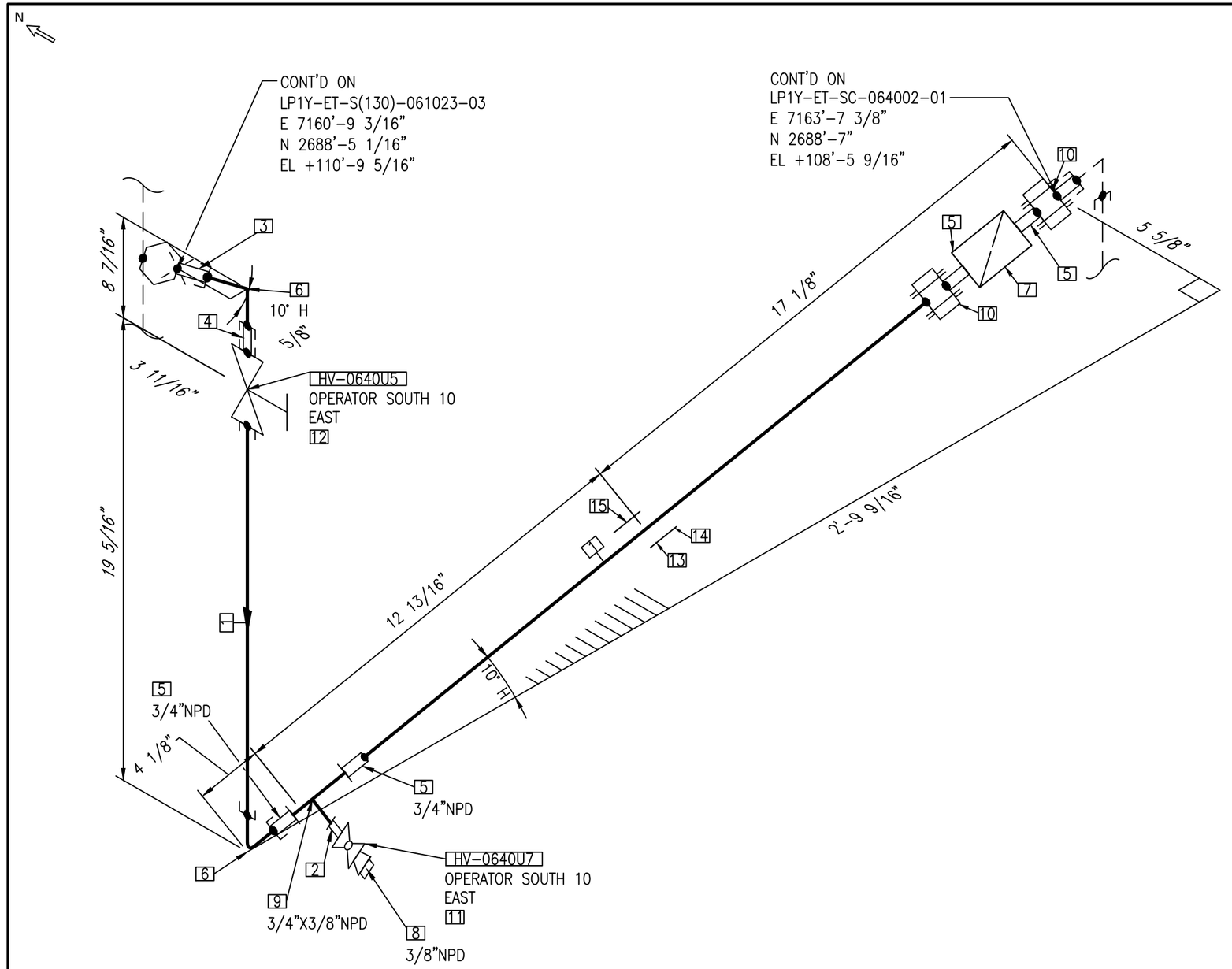

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	2'-8"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
2	1	3/8"	NIPPLE XXS STL A106-B TBE 3" LONG	5539171L
3	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
4	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 6" LONG	5539043L1
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6	2	3/4"	ELL 90 DEG 3000# SW STL A105	5478105
7	1	3/4"	ARMSTRONG 2010 NPT INVERTED BUCKET STEAM TRAP	LP1YSTT0640U1
8	1	3/8"	PLUG ROUND HEAD SCRD STL A105	5490665
9	1	3/4"X3/8"	Y-STRAINER, KECKLEY STYLE SB7, CL 600 FNPT	LP1YSTR0640U1
10	2	3/4"	UNION 3000# SW STL A105	5478111
11	1	3/8"	BALL 2000# SCRD STL UNIBODY	5058791L5
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13	1	3/4"	5C (INSULATION CRADLE)	5C-75-15
14	1	3/4"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-75-15
15	1	3/4"	5FS1 (FIELD SUPPORT)	5FS1S1

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS S(130) (2) (103) 064003

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000064	LP1Y-ET-S(130)-064003-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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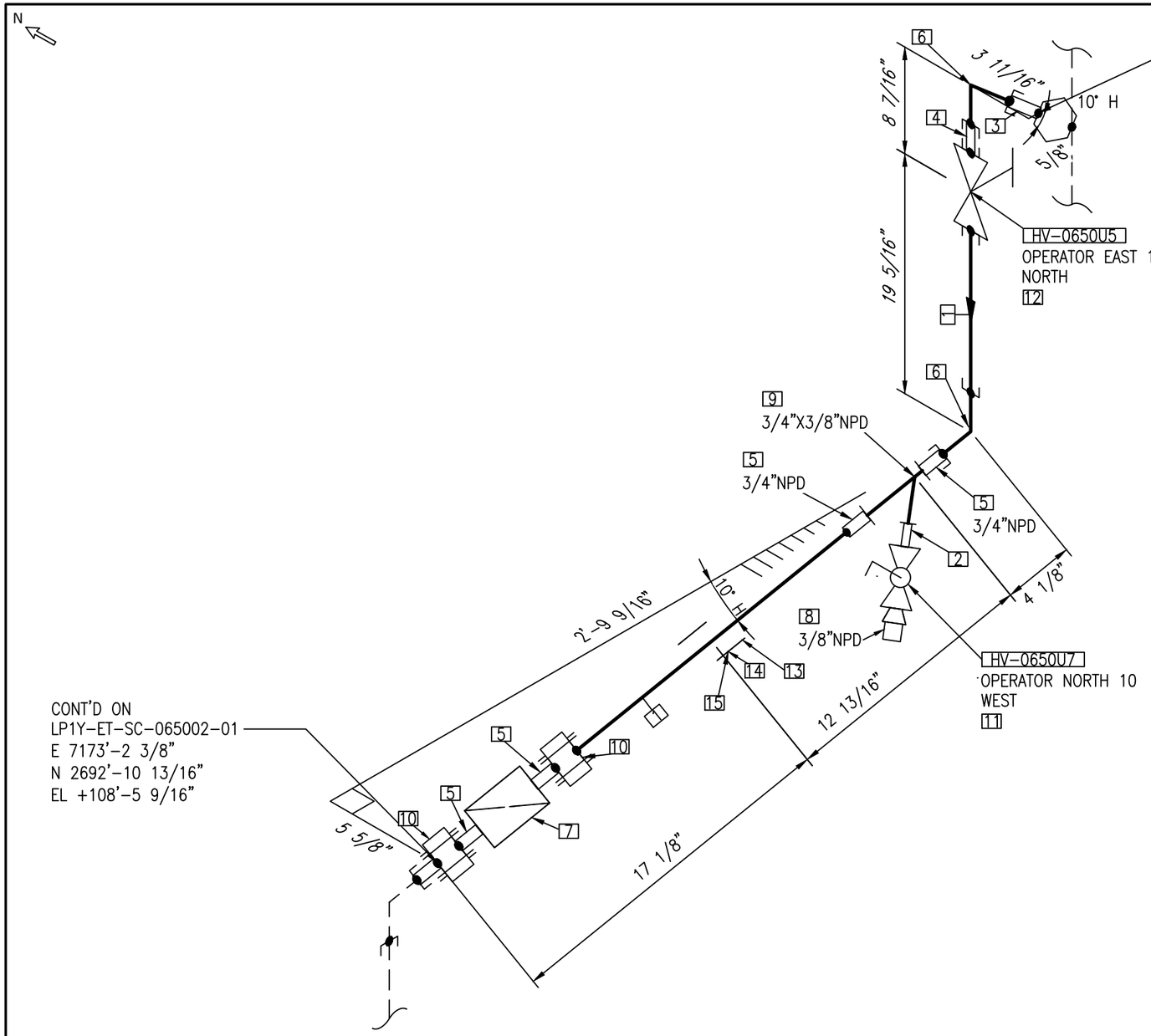

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7	1	3/4"	ARMSTRONG 2010 NPT INVERTED BUCKET STEAM TRAP	LP1YSTT0650U1
8	1	3/8"	PLUG ROUND HEAD SCRD STL A105	5490665
9	1	3/4"X3/8"	Y-STRAINER, KECKLEY STYLE SB7, CL 600 FNPT	LP1YSTR0650U1
10	2	3/4"	UNION 3000# SW STL A105	5478111
11	1	3/8"	BALL 2000# SCRD STL UNIBODY	5058791L5
12	1	3/4"	GATE 800# SW STL, F6/STEL SEATS	5061780L0
13	1	3/4"	5C (INSULATION CRADLE)	5C-75-15
14	1	3/4"	5GUSP (INSULATED U-BOLT GUIDE)	5UGSP-75-15
15	1	3/4"	5FS1 (FIELD SUPPORT)	5FS1S1

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	S(130)
PIPE SCHEDULE:	2
INSULATION CODE:	103
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS S(130) (2) (103) 065003

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000065	LP1Y-ET-S(130)-065003-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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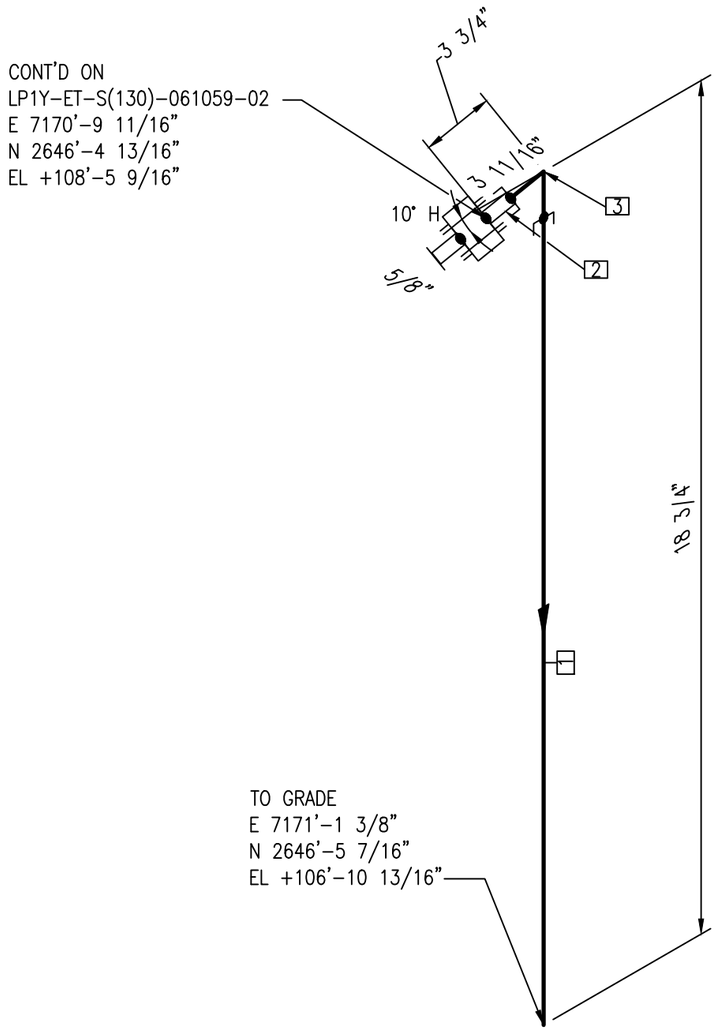
N



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	1'-6"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
2	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
3	1	3/4"	ELL 90 DEG 3000# SW STL A105	5478105

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
						/	
						/	
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	SC
PIPE SCHEDULE:	2
INSULATION CODE:	101
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS SC (2) (101) 061060

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-SC-061060-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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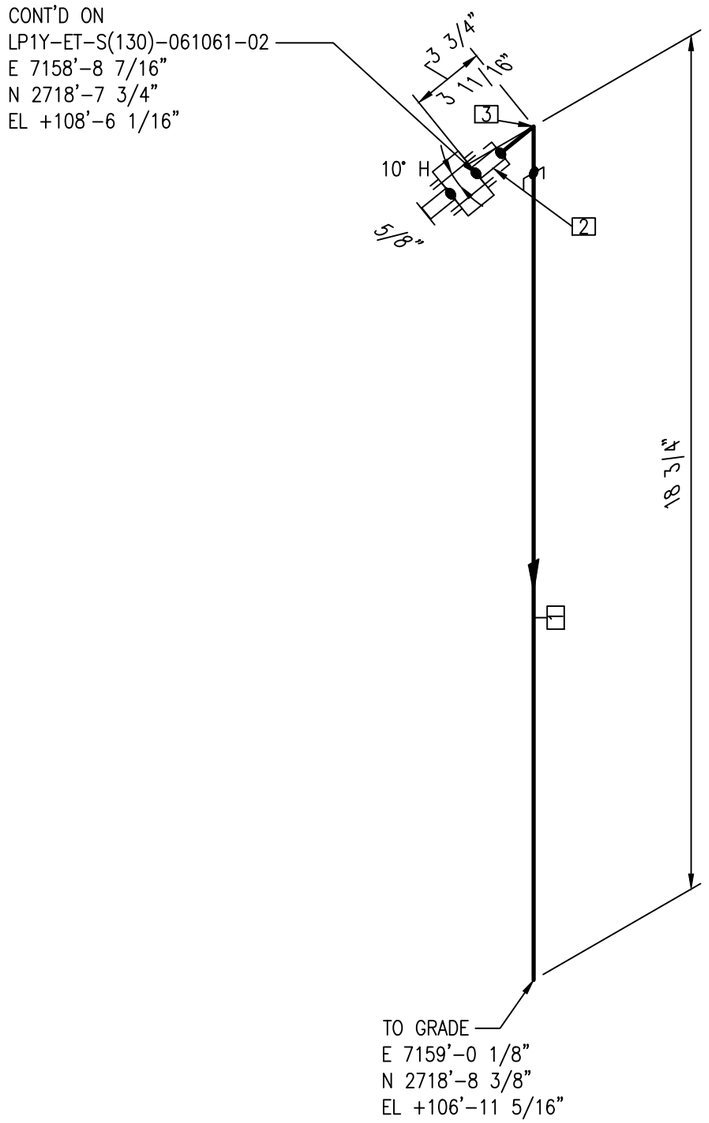
N



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	1'-6"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
2	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
3	1	3/4"	ELL 90 DEG 3000# SW STL A105	5478105

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By dac54131 at Mar 03, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	SC
PIPE SCHEDULE:	2
INSULATION CODE:	101
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS SC (2) (101) 061063

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-ET-SC-061063-01

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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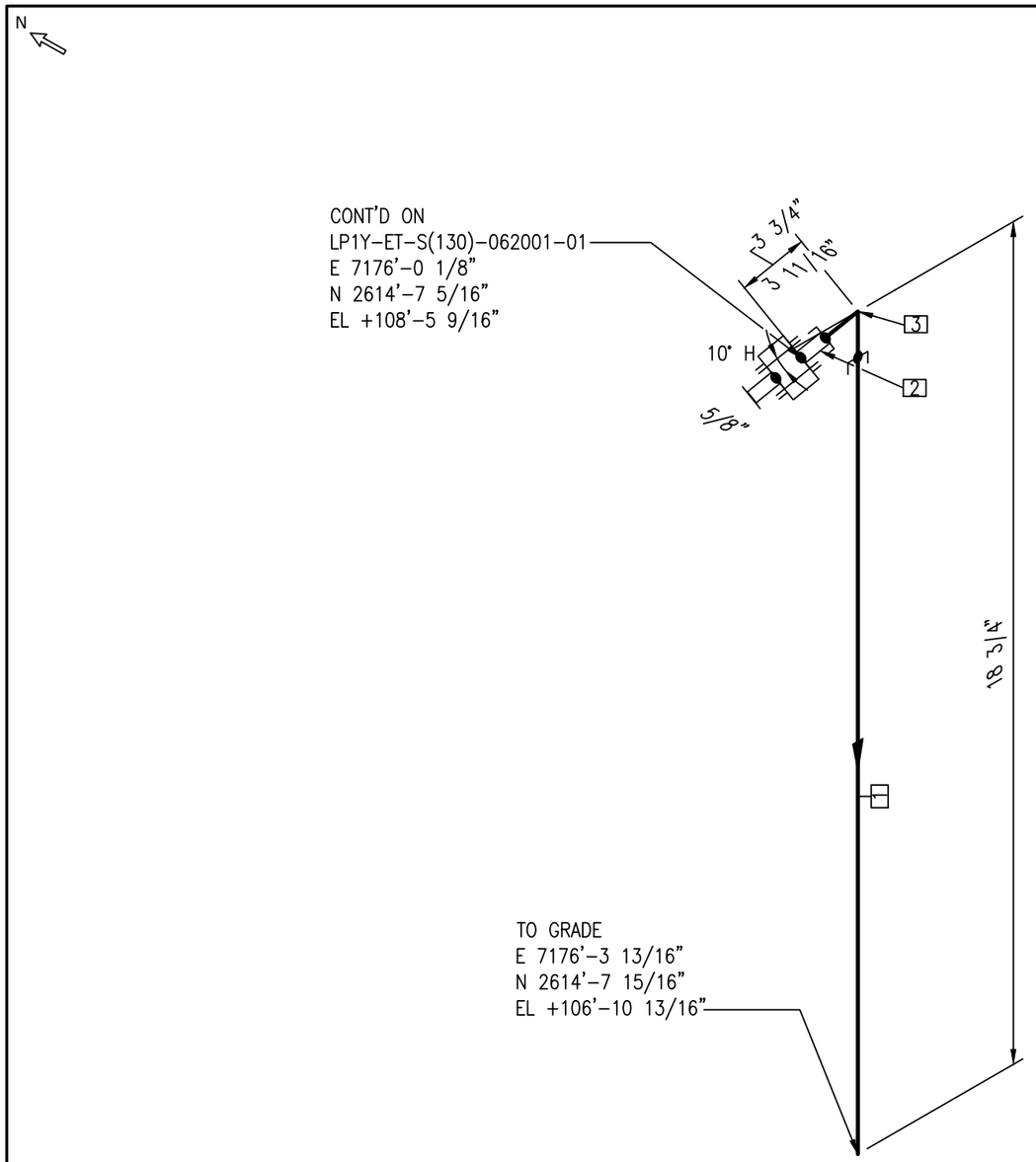

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	1'-6"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
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3	1	3/4"	ELL 90 DEG 3000# SW STL A105	5478105

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By dac54131 at Mar 03, 2026

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0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	SC
PIPE SCHEDULE:	2
INSULATION CODE:	101
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS SC (2) (101) 062002

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000062	LP1Y-ET-SC-062002-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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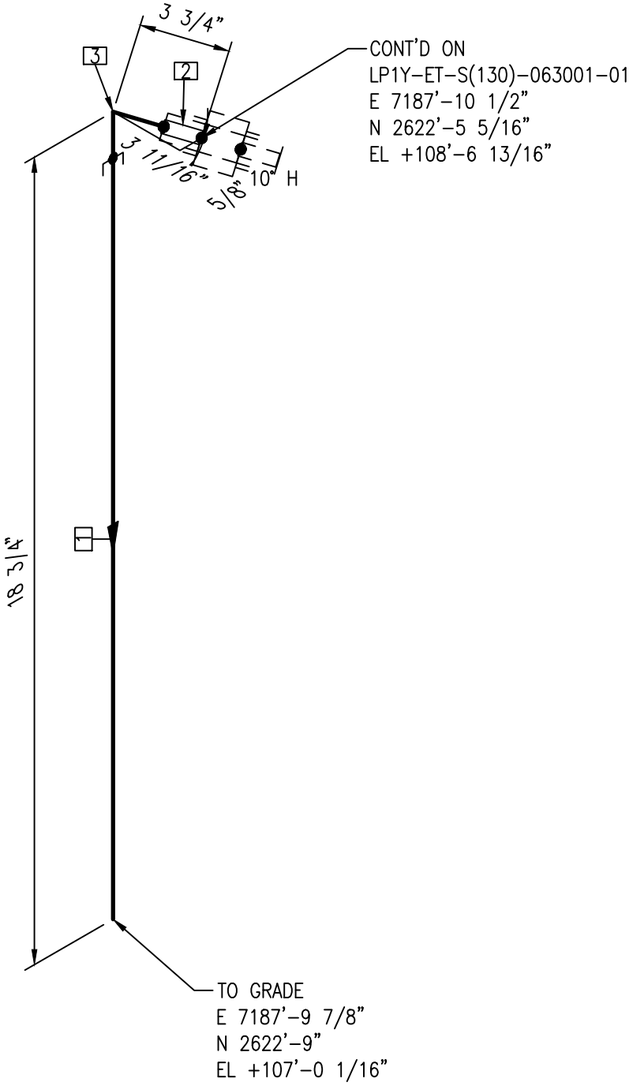
3	1	3/4"	ELL 90 DEG 3000# SW STL A105	5478105



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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	SC
PIPE SCHEDULE:	2
INSULATION CODE:	101
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS SC (2) (101) 063002

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000063	LP1Y-ET-SC-063002-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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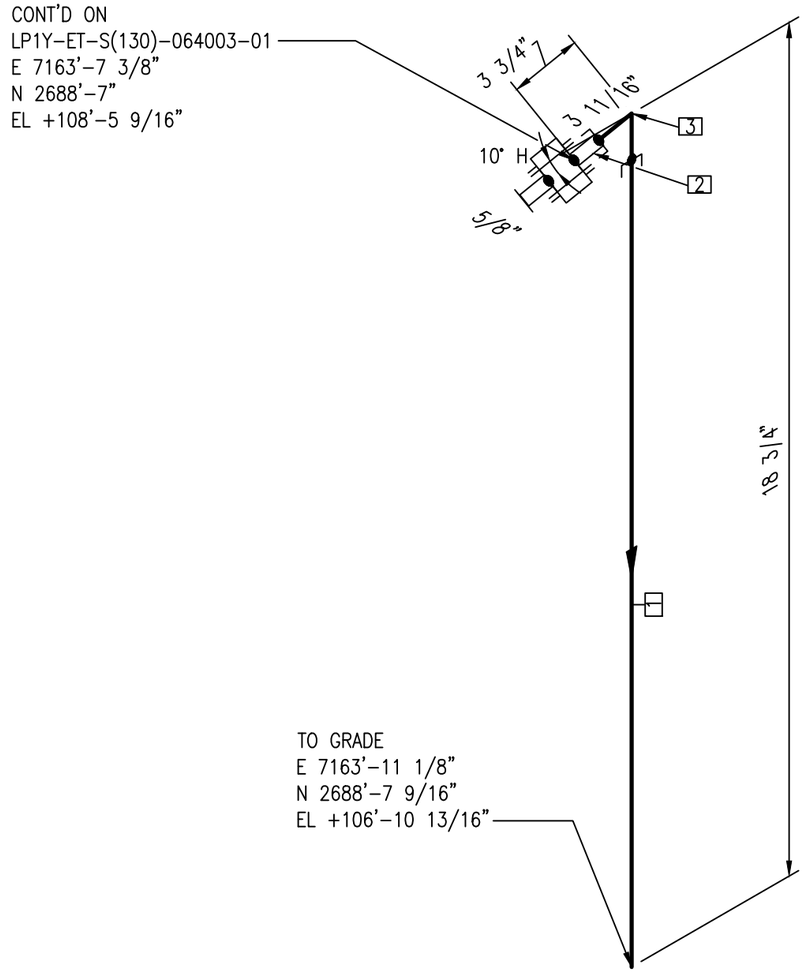

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	1'-6"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
2	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
3	1	3/4"	ELL 90 DEG 3000# SW STL A105	5478105

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By dac54131 at Mar 03, 2026

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						/	
						/	
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	SC
PIPE SCHEDULE:	2
INSULATION CODE:	101
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS SC (2) (101) 064002

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000064	LP1Y-ET-SC-064002-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

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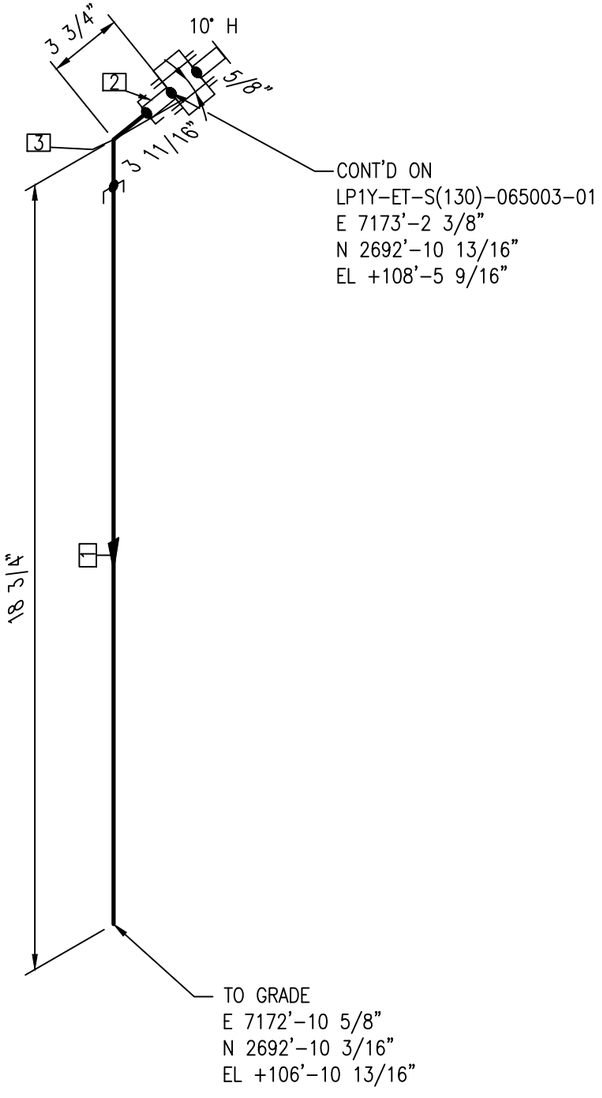

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	1'-6"	3/4"	PIPE SCH 80 ERW STL A53-B	5355898L
2	1	3/4"	NIPPLE SCH 80 STL A53-B PBE 3" LONG	5539041L0
3	1	3/4"	ELL 90 DEG 3000# SW STL A105	5478105

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	24-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

GENERAL NOTE:
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3. WELDS AND/OR COUPLINGS DEFINING RANDOM PIPE SPOOL LENGTHS ARE FOR REFERENCE ONLY. FINAL WELD OR COUPLING LOCATION IS TO BE DETERMINED BY THE CONTRACTOR.
4. THE MECHANICAL CONTRACTOR TO FIELD VERIFY INTERFACE POINTS BETWEEN EQUIPMENT & PLUMBING FIXTURES IN 3rd FIX (3F SERIES CWP) AND YS YARD STICK BUILT (YS SERIES CWP EQUIPMENT CONNECTIONS) ISOMETRICS PRIOR TO FABRICATION.
5. THE MECHANICAL CONTRACTOR IS TO VERIFY ALL DIMENSIONAL DATA WITH ACTUAL FIELD CONDITIONS PRIOR TO INSTALLATION OF THE SYSTEM.
6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	SC
PIPE SCHEDULE:	2
INSULATION CODE:	101
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

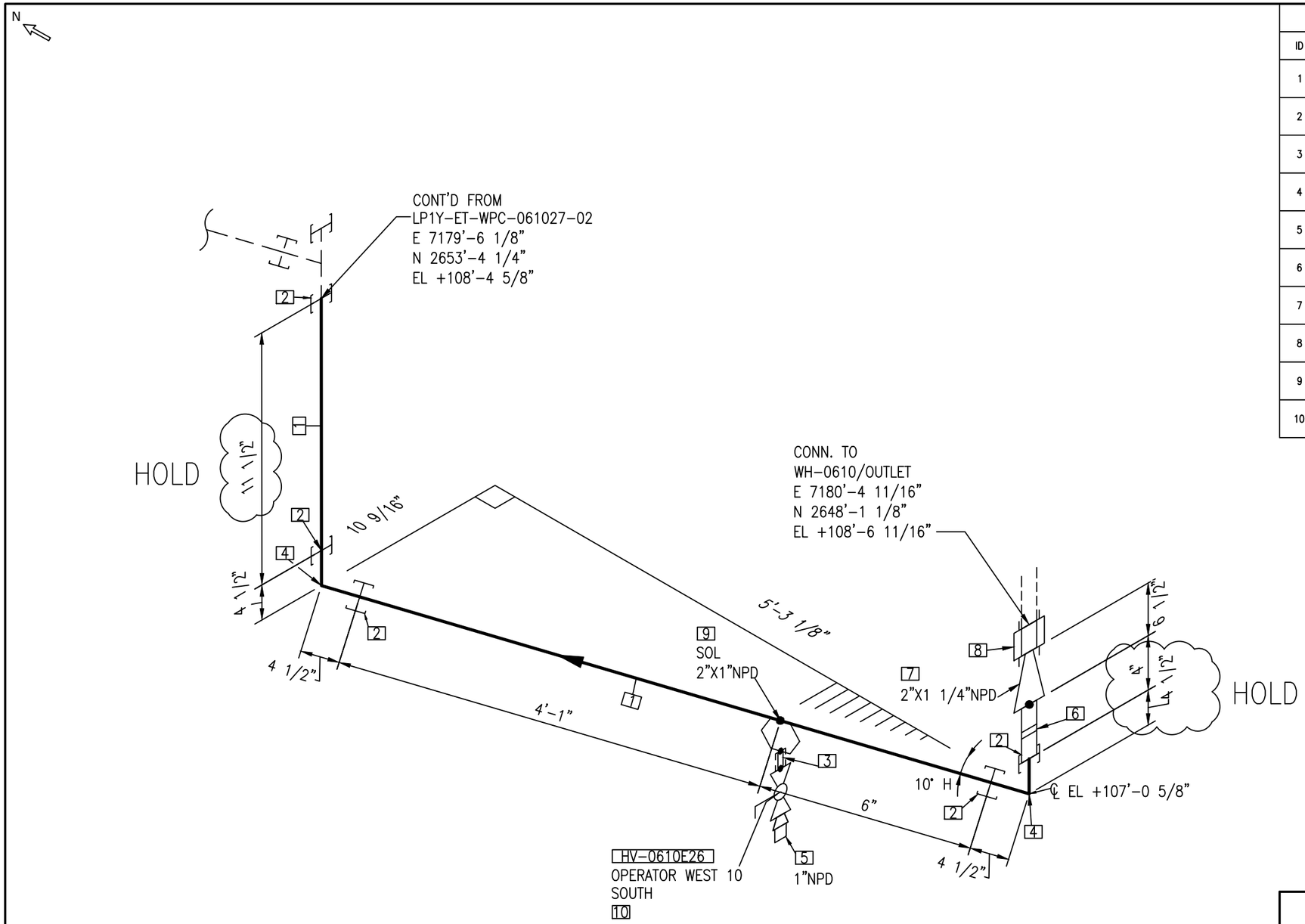
HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3/4" CS SC (2) (101) 065002

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000065	LP1Y-ET-SC-065002-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-FEB-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR. AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	5'-7"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	5	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	1	1"	NIPPLE SCH 10S 316/316L SS PBE 6" LONG	5540493
4	2	2"	ELL 90 DEG GRV 316L SCH 10S NSF	5484951S1
5	1	1"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
6	1	2"	ADAPTER NIPPLE GRV X BE 316L SCH 10S NSF	5540181
7	1	2"X1 1/4"	SWAGE CONC SCH 80S 316/316L SS BLEXTSE	5546622
8	1	1 1/4"	UNION DIELECTRIC 3000# SCRD 316/316L SS NSF 61	5486541L0
9	1	2"X1"	PIPET 10S X 3000# SW 316/316L SS	5537081
10	1	1"	BALL 1440# CWP SW X FNPT 316 SS, 316 TRIM RP EXTD STEM, HNDL OP	5300985L8

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	12-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WPC
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WPC (36) (108) 061027

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-ET-WPC-061027-01

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 12-FEB-2026

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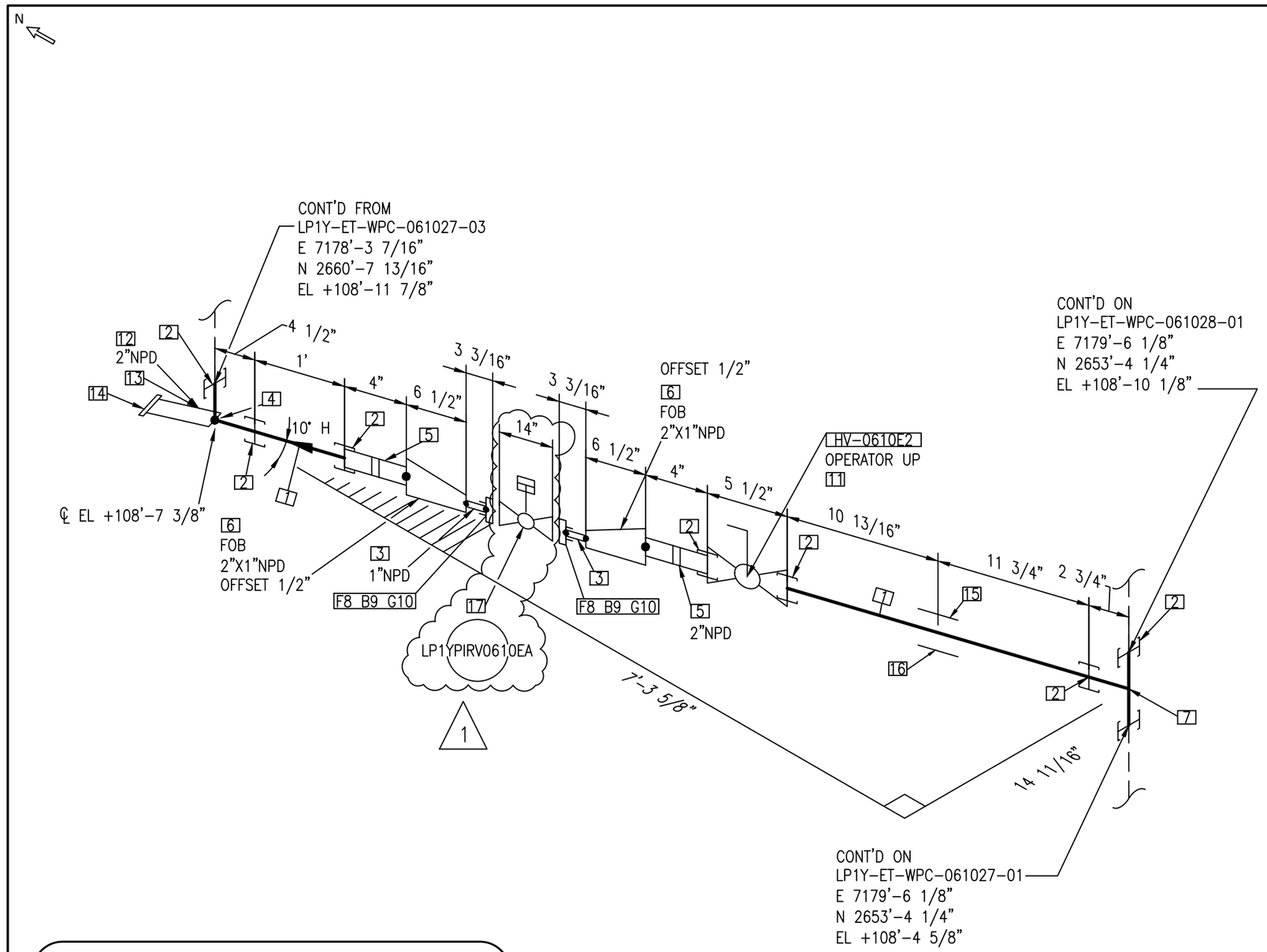

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	2'-11"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	7	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	2	1"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
4	1	2"	ELL 90 DEG GRV 316L SCH 10S NSF	5484951S1
5	2	2"	ADAPTER NIPPLE GRV X BE 316L SCH 10S NSF	5540181
6	2	2"X1"	SWAGE ECC SCH 10S 316/316L SS BBE	5551651
7	1	2"	TEE GRV 316L SCH 10S NSF	5484953S1
8	2	1"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
9	8	1/2"X2 1/4"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
10	2	1"	GASKET 150# SOFT PTFE RING 1/16" THK NSF	5669345
11	1	2"	BALL 1000# CWP GRV SS, RTFE 2-PC RP HNDL OP NSF	511453L3
12	1	2"	5DS1 (DUMMY SUPPORT) L=1'-4"	5DS1-02
13	1	2"	5UG (U-BOLT GUIDE)	5UG-02
14	1	2"	5FS1 (FIELD SUPPORT)	5FS1S1
15	1	2"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-02-01
16	1	2"	5ABS2 (ANGLE BASE SUPPORT)	5ABS2
17	1	1"	PRV CL 150# RF	LP1YPIRV0610EA

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WPC
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW
0	ISSUE FOR CONSTRUCTION	12-FEB-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

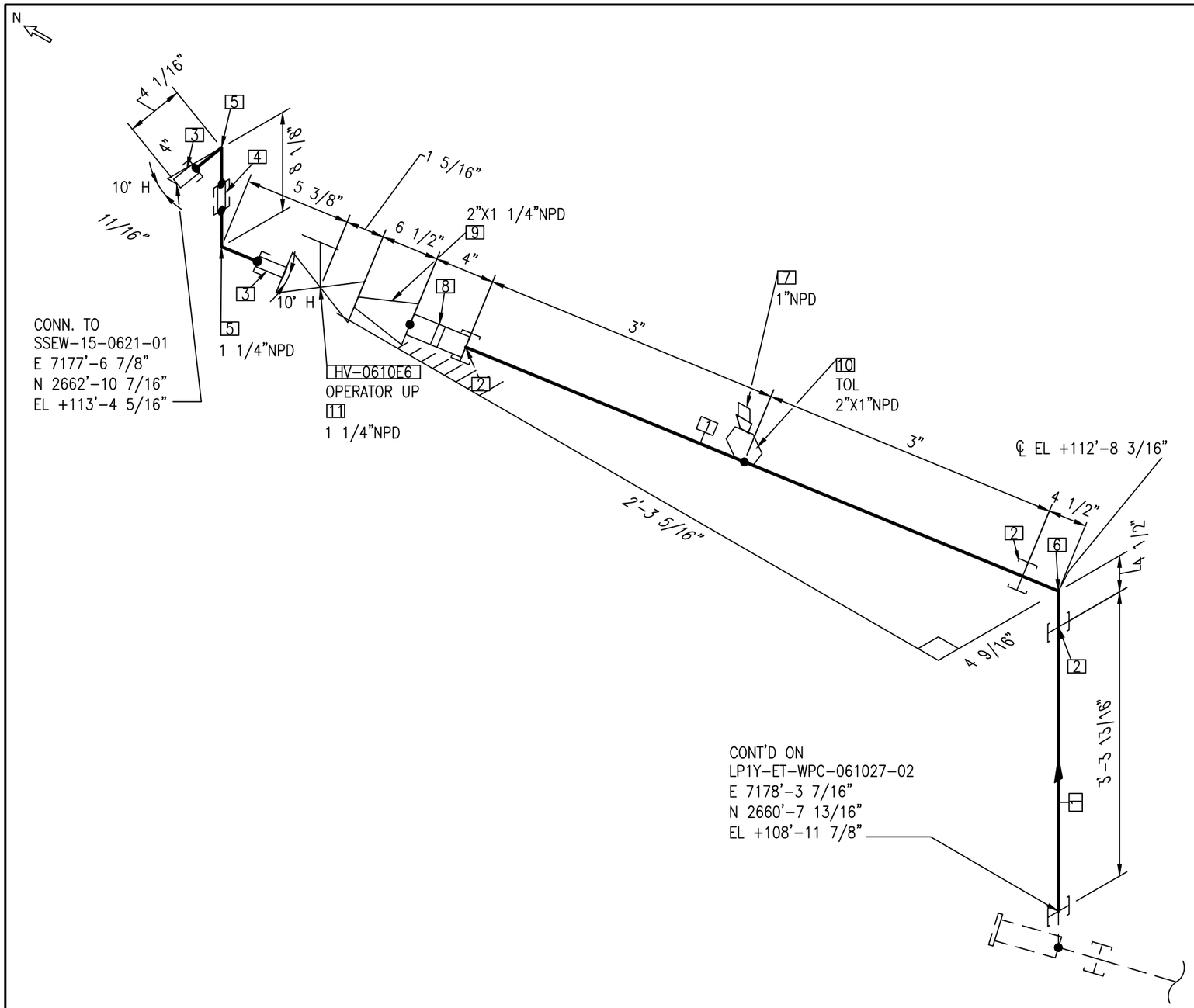
HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WPC (36) (108) 061027

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WPC-061027-02	1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	3'-10"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	3	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	2	1 1/4"	NIPPLE SCH 80S 316/316L SS TOE 3" LONG	5540561
4	1	1 1/4"	NIPPLE SCH 10S 316/316L SS PBE 6" LONG	5540493
5	2	1 1/4"	ELL 90 DEG 3000# SW 316/316L SS	5480138
6	1	2"	ELL 90 DEG GRV 316L SCH 10S NSF	5484951S1
7	1	1"	WATER HAMMER ARRESTOR	LP1YWHA0610E1
8	1	2"	ADAPTER NIPPLE GRV X BE 316L SCH 10S NSF	5540181
9	1	2"X1 1/4"	SWAGE CONC SCH 80S 316/316L SS BLEXTSE	5546622
10	1	2"X1"	PIPET 10S X 3000# SCRD 316/316L SS	5515773
11	1	1 1/4"	GATE 150# SCRD BRONZE LOCKSHIELD	5000510L0

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Mar 03, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WPC
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	12-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WPC (36) (108) 061027

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WPC-061027-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 12-FEB-2026

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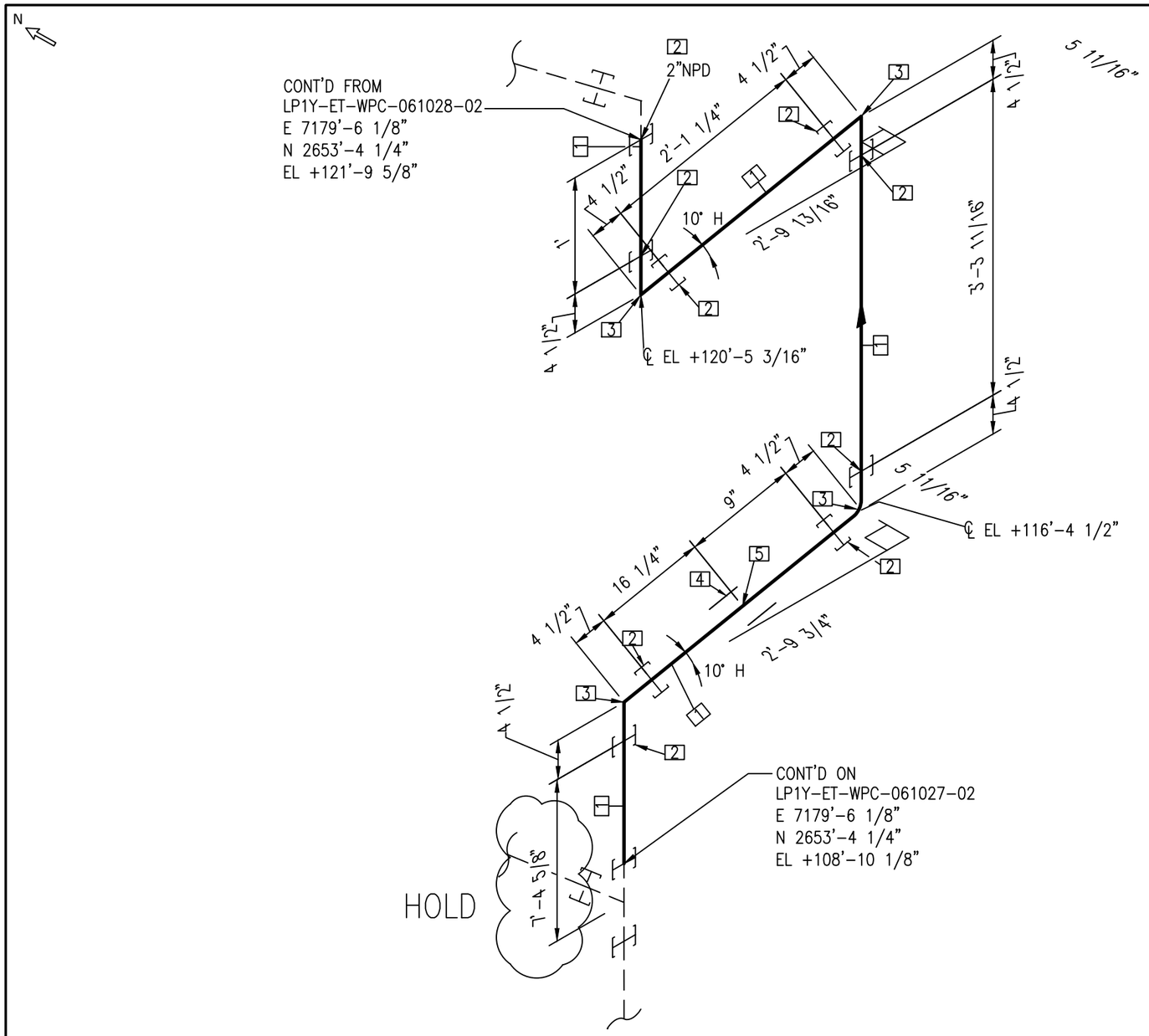

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	15'-9"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	9	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	4	2"	ELL 90 DEG GRV 316L SCH 10S NSF	5484951S1
4	1	2"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-02-01
5	1	2"	5HR4 (HANGER ROD)	5HR4-05

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	12-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WPC
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WPC (36) (108) 061028

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WPC-061028-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 12-FEB-2026

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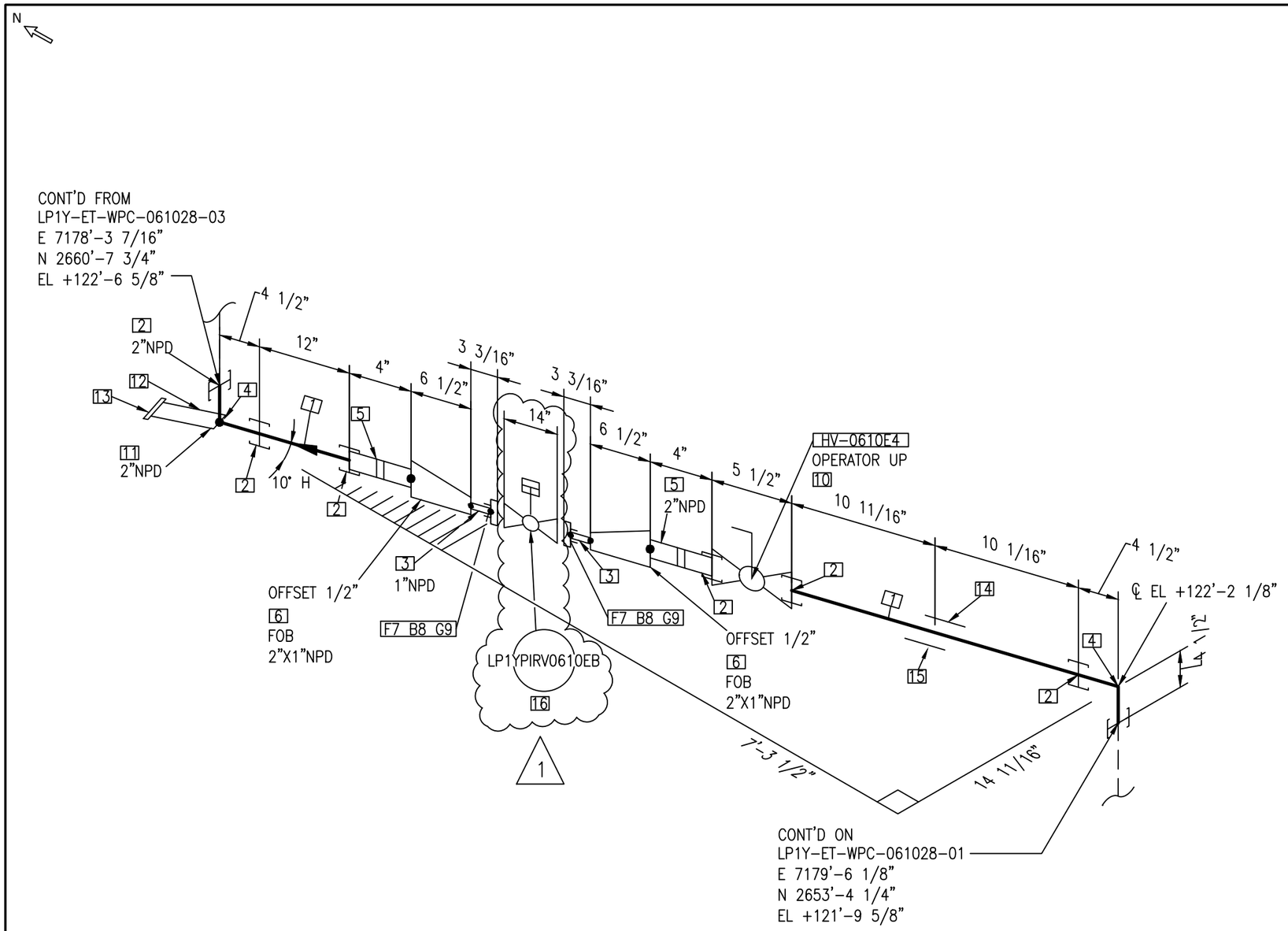

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	2'-9"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	6	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	2	1"	NIPPLE SCH 10S 316/316L SS PBE 3" LONG	5540491
4	2	2"	ELL 90 DEG GRV 316L SCH 10S NSF	5484951S1
5	2	2"	ADAPTER NIPPLE GRV X BE 316L SCH 10S NSF	5540181
6	2	2"X1"	SWAGE ECC SCH 10S 316/316L SS BBE	5551651
7	2	1"	FLG SW 150# RF 316/316L SS 10S BORE	5608148
8	8	1/2"X2 1/4"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
9	2	1"	GASKET 150# SOFT PTFE RING 1/16" THK NSF	5669345
10	1	2"	BALL 1000# CWP GRV SS, RTFE 2-PC RP HNDL OP NSF	511453L3
11	1	2"	5DS1 (DUMMY SUPPORT) L=1'-4"	5DS1-02
12	1	2"	5UG (U-BOLT GUIDE)	5UG-02
13	1	2"	5FS1 (FIELD SUPPORT)	5FS1S1
14	1	2"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-02-01
15	1	2"	5ABS2 (ANGLE BASE SUPPORT)	5ABS2
16	1	1"	PRV CL 150# RF	LP1YPIRV0610EB

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By dac54131 at Apr 10, 2026

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5. THE MECHANICAL CONTRACTOR IS TO VERIFY ALL DIMENSIONAL DATA WITH ACTUAL FIELD CONDITIONS PRIOR TO INSTALLATION OF THE SYSTEM.
6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW
0	ISSUE FOR CONSTRUCTION	12-FEB-2026	DRW	SBR	N/A	JHN / N/A	MDW
REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WPC
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WPC (36) (108) 061028

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-ET-WPC-061028-02

REV
1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026

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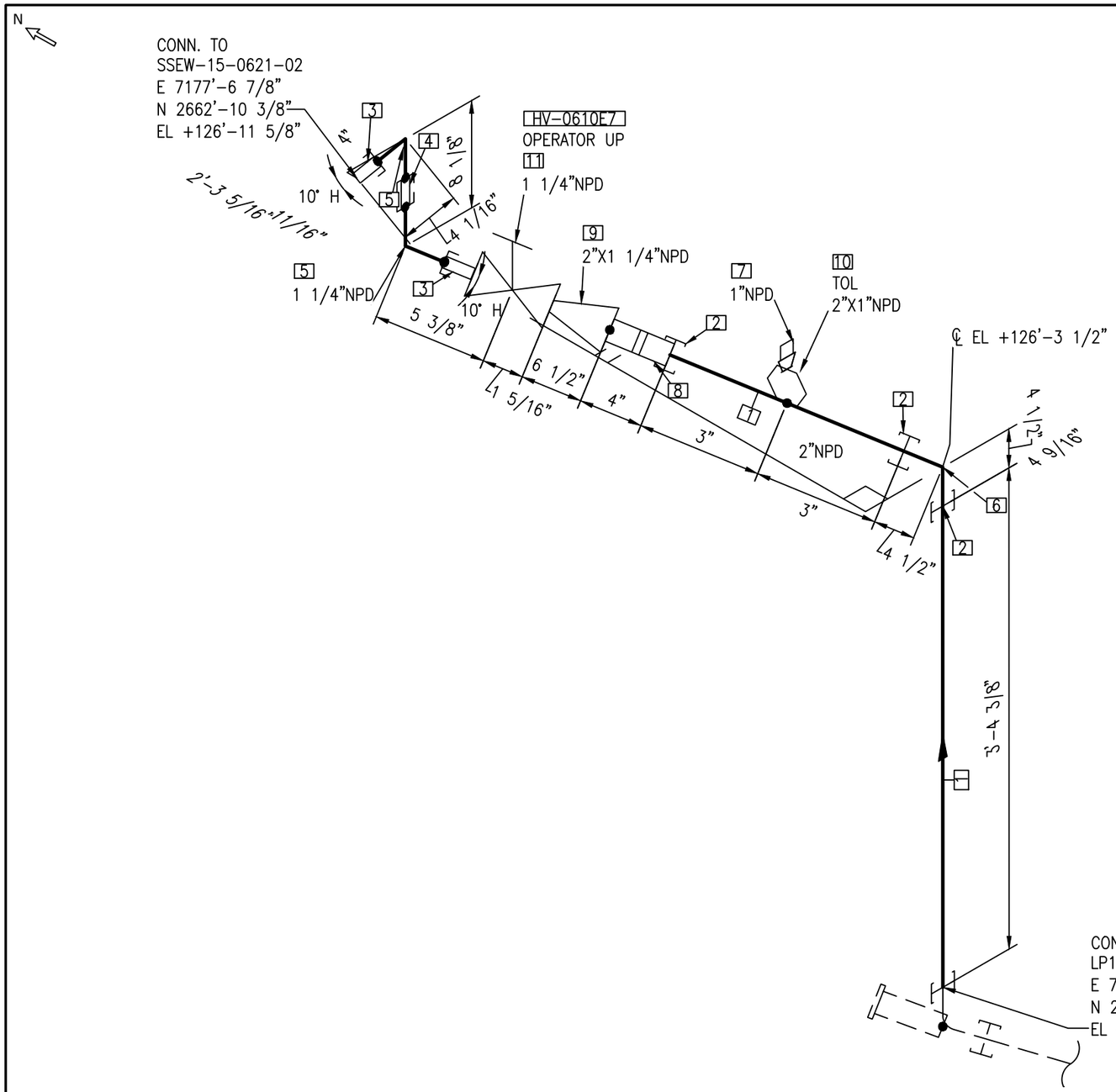

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	3'-11"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	3	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	2	1 1/4"	NIPPLE SCH 80S 316/316L SS TOE 3" LONG	5540561
4	1	1 1/4"	NIPPLE SCH 10S 316/316L SS PBE 6" LONG	5540493
5	2	1 1/4"	ELL 90 DEG 3000# SW 316/316L SS	5480138
6	1	2"	ELL 90 DEG GRV 316L SCH 10S NSF	5484951S1
7	1	1"	WATER HAMMER ARRESTOR	LP1YWHA0610E2
8	1	2"	ADAPTER NIPPLE GRV X BE 316L SCH 10S NSF	5540181
9	1	2"X1 1/4"	SWAGE CONC SCH 80S 316/316L SS BLEXTSE	5546622
10	1	2"X1"	PIPET 10S X 3000# SCRD 316/316L SS	5515773
11	1	1 1/4"	GATE 150# SCRD BRONZE LOCKSHIELD	5000510L0

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
						/	
						/	
0	ISSUE FOR CONSTRUCTION	12-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WPC
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WPC (36) (108) 061028

REFERENCE P&ID NUMBER
LP1Y-DI-00000061

ISOMETRIC DRAWING NUMBER
LP1Y-ET-WPC-061028-03

REV
0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 12-FEB-2026

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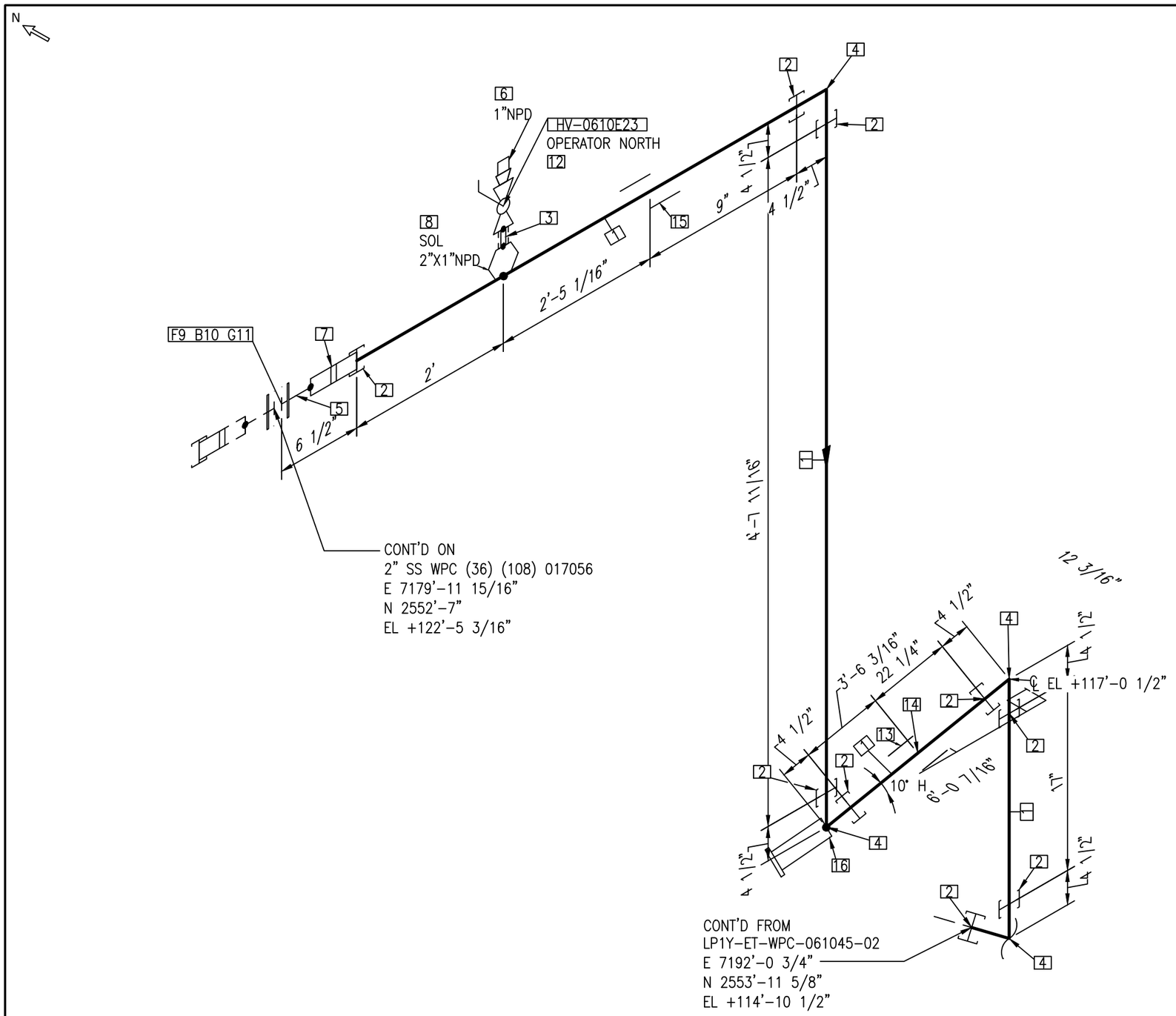

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	16'-8"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	9	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	1	1"	NIPPLE SCH 10S 316/316L SS PBE 6" LONG	5540493
4	4	2"	ELL 90 DEG GRV 316L SCH 10S NSF	5484951S1
5	1	2"	STUB END SCH10S 316/316L A403-W SHORT PATT	5450750L
6	1	1"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
7	1	2"	ADAPTER NIPPLE GRV X BE 316L SCH 10S NSF	5540181
8	1	2"X1"	PIPET 10S X 3000# SW 316/316L SS	5537081
9	1	2"	FLG LAP JOINT 150# 316 SS	5599578L
10	4	5/8"X2 1/2"	STUD-BOLT A193 GR B7 W/A194 GR 2H NUTS	5675648
11	1	2"	GASKET 150# SOFT PTFE RING 1/16" THK NSF	5669345
12	1	1"	BALL 1440# CWP SW X FNPT 316 SS, 316 TRIM RP EXTD STEM, HNDL OP	5300985L8
13	1	2"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-02-01
14	1	2"	5HR4 (HANGER ROD)	5HR4-05
15	1	2"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE, 1" INS)	5ISC-02-01
16	1	2"	5DS1 (DUMMY SUPPORT, L=1' 2")	5DS1-02

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	12-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WPC
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WPC (36) (108) 061045

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WPC-061045-01	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 12-FEB-2026

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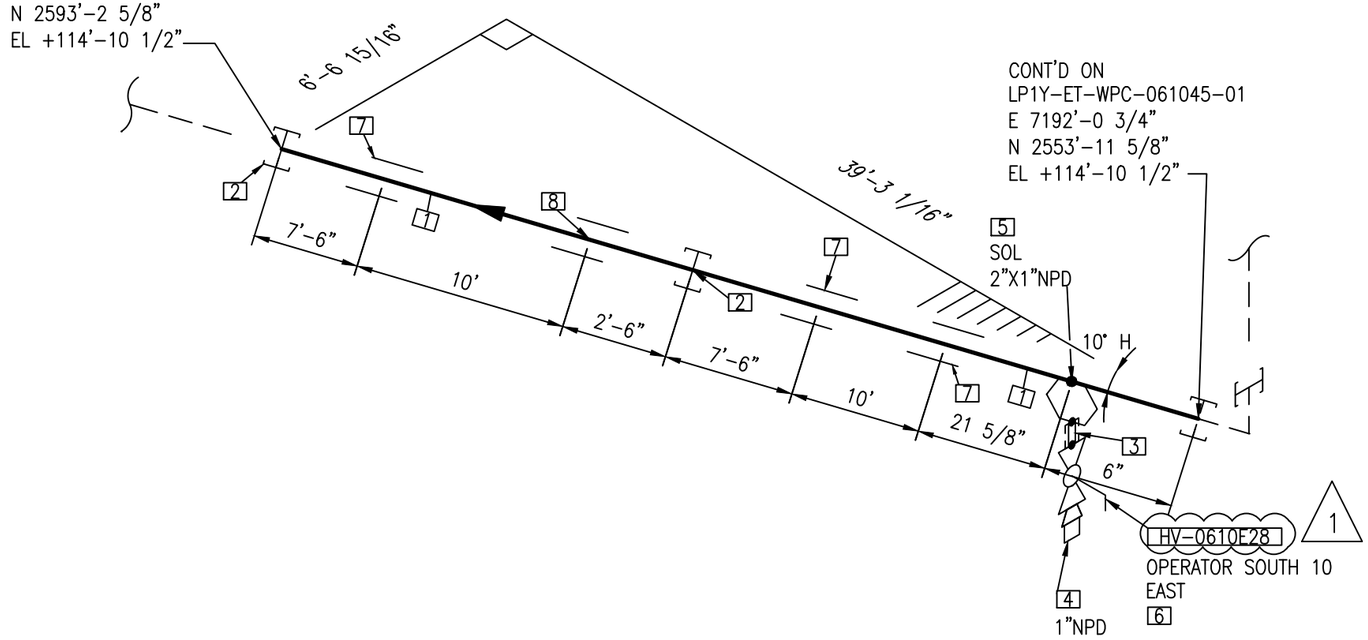
N



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	39'-10"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	2	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	1	1"	NIPPLE SCH 10S 316/316L SS PBE 6" LONG	5540493
4	1	1"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
5	1	2"X1"	PIPET 10S X 3000# SW 316/316L SS	5537081
6	1	1"	BALL 1440# CWP SW X FNPT 316 SS, 316 TRIM RP EXTD STEM, HNDL OP	5300985L8
7	3	2"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE, 1" INS)	5ISC-02-01
8	1	2"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE, 1" INS)	5IGC-02-01

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WPC
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	N/A / N/A	MDW
0	ISSUE FOR CONSTRUCTION	12-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WPC (36) (108) 061045

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WPC-061045-02	1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026

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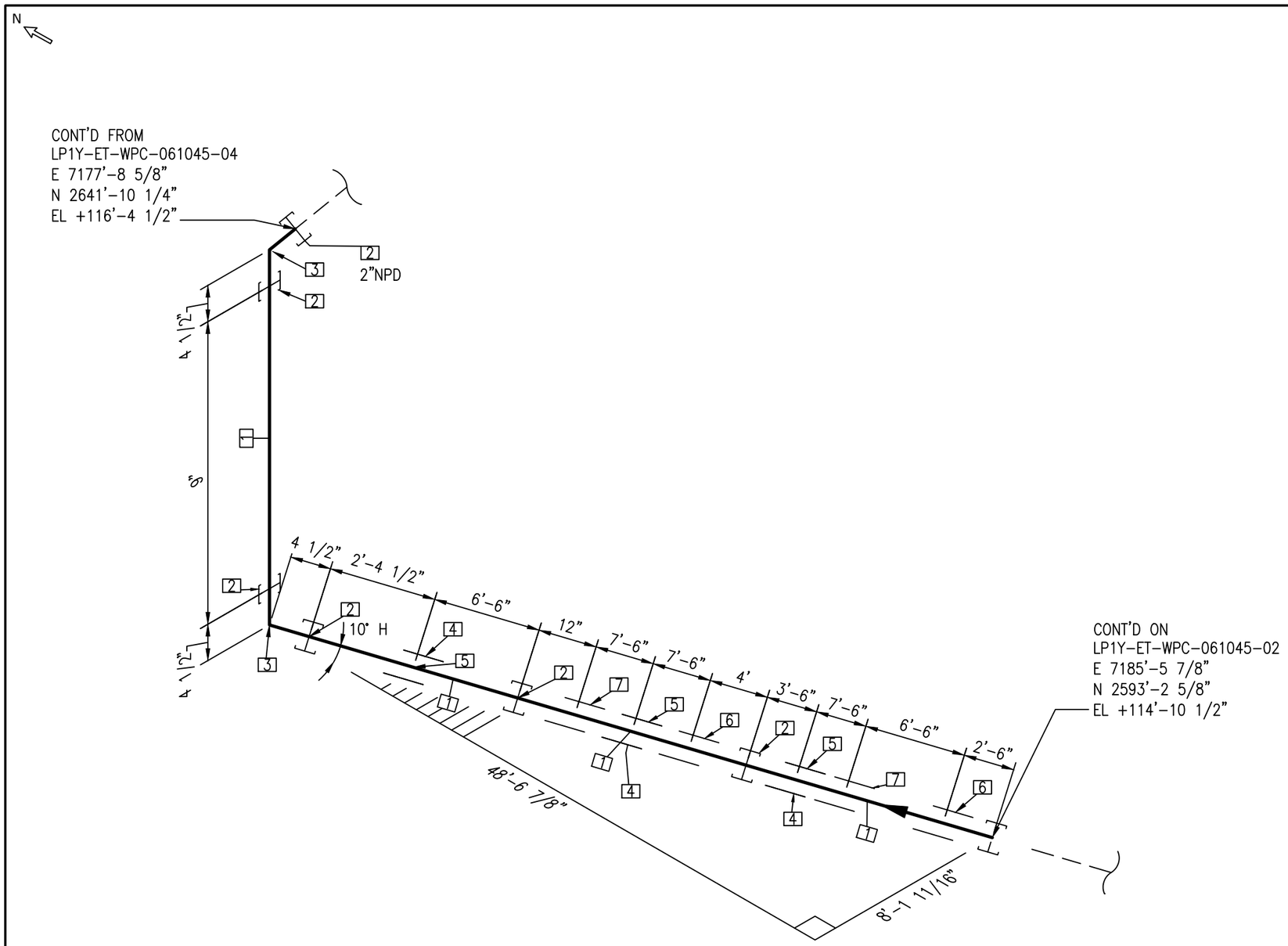

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	49'-8"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	6	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	2	2"	ELL 90 DEG GRV 316L SCH 10S NSF	5484951S1
4	3	2"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-02-01
5	3	2"	5HR4 (HANGER ROD)	5HR4-05
6	2	2"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE, 1" INS)	5ISC-02-01
7	2	2"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE, 1" INS)	5IGC-02-01

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By dac54131 at Mar 03, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WPC
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	12-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WPC (36) (108) 061045

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WPC-061045-03	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 12-FEB-2026

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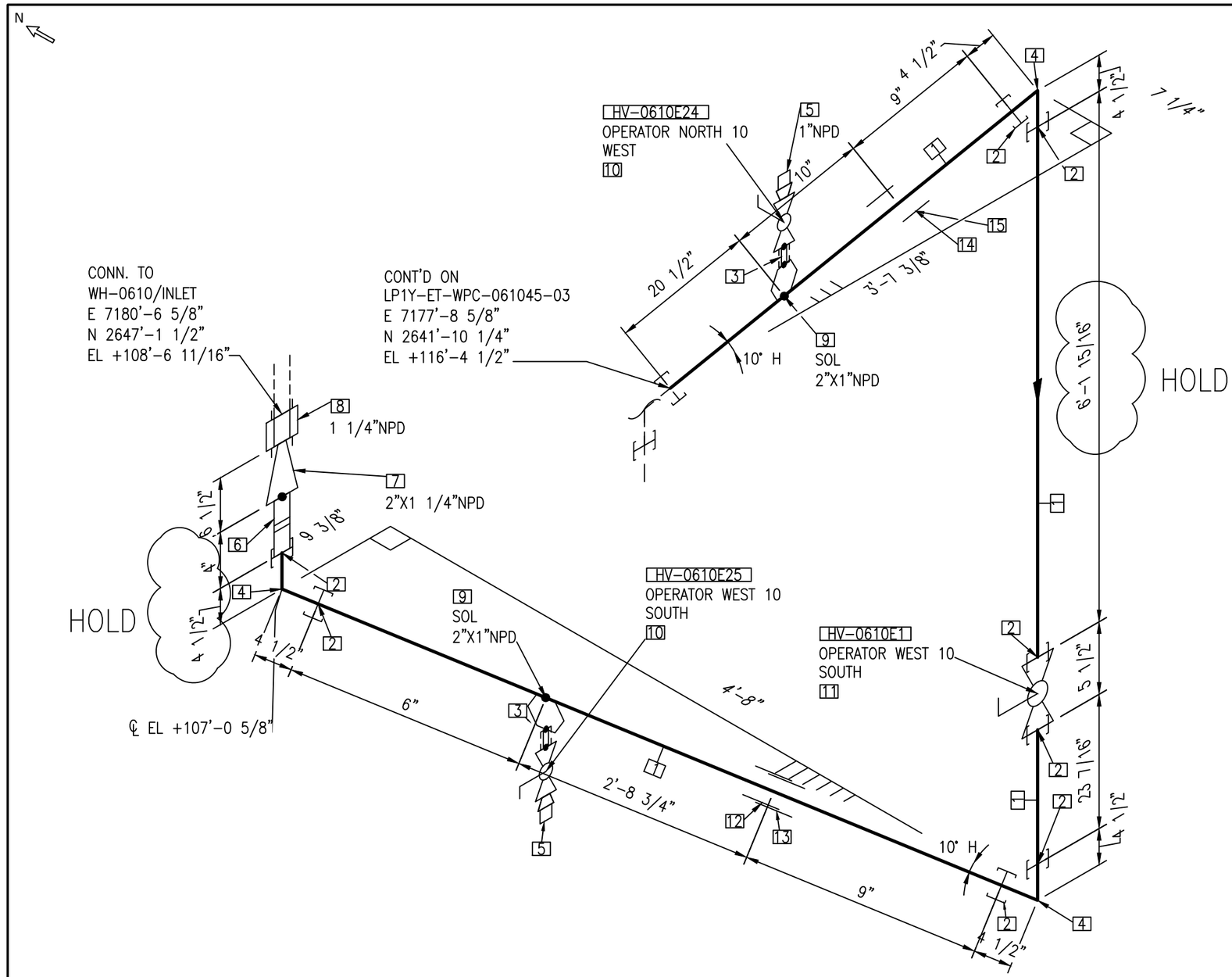

BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	15'-5"	2"	PIPE SCH 10S ERW 316/316L SS A312 NSF	5365189L
2	8	2"	COUPLING RIGID GRV DI GRADE P GASKET PRE-AS NSF	5714906L0
3	2	1"	NIPPLE SCH 10S 316/316L SS PBE 6" LONG	5540493
4	3	2"	ELL 90 DEG GRV 316L SCH 10S NSF	5484951S1
5	2	1"	PLUG ROUND HEAD SCRD 316/316L SS	5498383
6	1	2"	ADAPTER NIPPLE GRV X BE 316L SCH 10S NSF	5540181
7	1	2"X1 1/4"	SWAGE CONC SCH 80S 316/316L SS BLEXTSE	5546622
8	1	1 1/4"	UNION DIELECTRIC 3000# SCRD 316/316L SS NSF 61	5486541L0
9	2	2"X1"	PIPET 10S X 3000# SW 316/316L SS	5537081
10	2	1"	BALL 1440# CWP SW X FNPT 316 SS, 316 TRIM RP EXTD STEM, HNDL OP	5300985L8
11	1	2"	BALL 1000# CWP GRV SS, RTFE 2-PC RP HNDL OP NSF	511453L3
12	1	2"	5CC (COLD SERVICE INSULATION CRADLE)	5CC-02-01
13	1	2"	5ABS2 (ANGLE BASE SUPPORT)	5ABS2
14	1	2"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE, 1" INS)	5ISC-02-01
15	1	2"	5FSS3 (FIELD SUPPORT STEEL)	5FSS3-L-1-3

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5. THE MECHANICAL CONTRACTOR IS TO VERIFY ALL DIMENSIONAL DATA WITH ACTUAL FIELD CONDITIONS PRIOR TO INSTALLATION OF THE SYSTEM.
6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WPC
PIPE SCHEDULE:	36
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	12-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

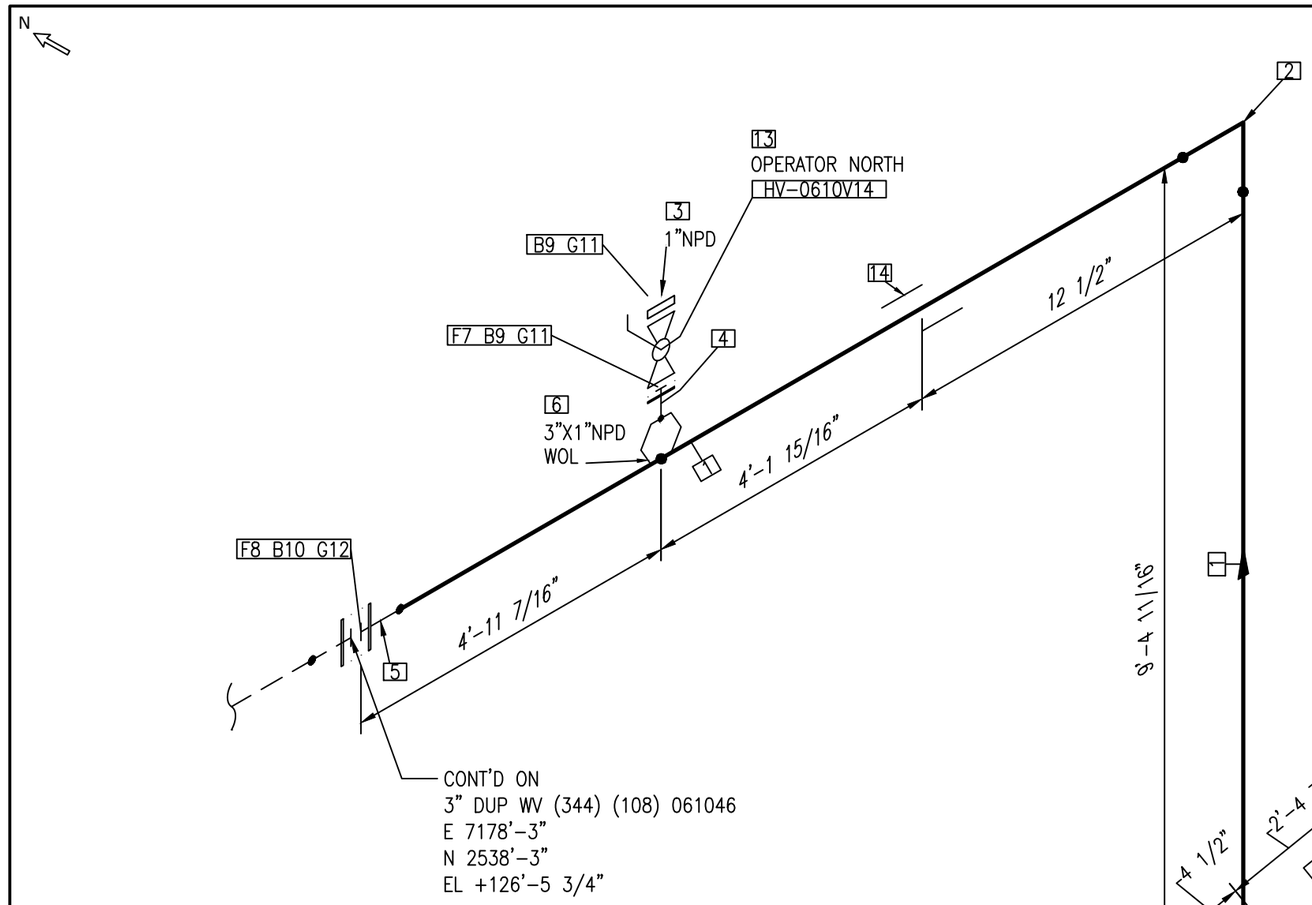
Lilly

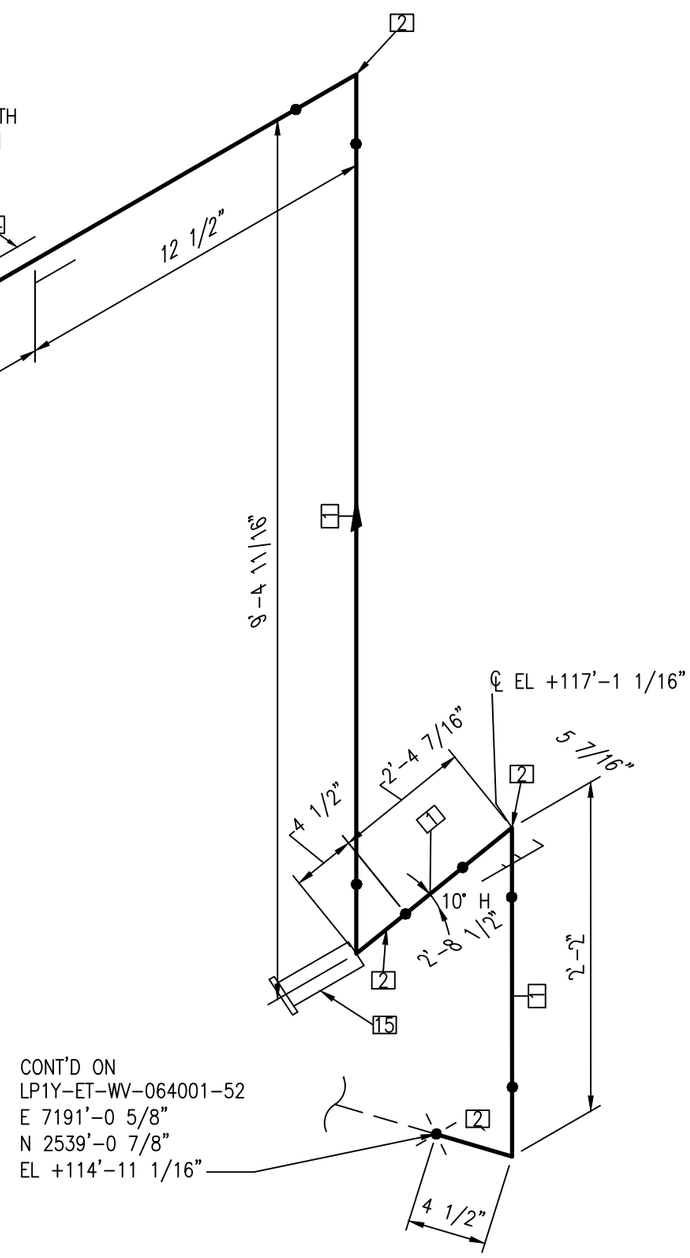
HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 2" SS WPC (36) (108) 061045

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WPC-061045-04	0

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.





BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	21'-8"	3"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	4	3"	ELL 90 DEG LR SCH 10S DUPLEX STL S32205-W	5460743
3	1	1"	FLG BLIND 150# RF DUPLEX STL F60	5616811
4	1	1"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
5	1	3"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
6	1	3"X1"	PIPET SCH 10S DUPLEX STL F60	5512022
7	1	1"	FLG LAP JOINT 150# 304 SS	5599576L
8	1	3"	FLG LAP JOINT 150# 304 SS	5599576L
9	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
10	4	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
11	2	1"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
12	1	3"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
13	1	1"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
14	1	3"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-03-01
15	1	3"	5DS1 (DUMMY SUPPORT) L=1'-1"	5DS1-03

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

GENERAL NOTE:
1. REFER TO THE PIPING LINE LIST FOR PROCESS DATA/CONDITIONS AND OTHER FABRICATION DETAILS SUCH AS TESTING, CLEANING, AND PAINTING REQUIREMENTS.
2. UNLESS OTHERWISE NOTED, ALL WELDS ARE INDICATED AS SHOP WELDS. CONTRACTOR TO DETERMINE FIELD WELDS LOCATIONS AND DOCUMENT ACCORDINGLY ON AS-BUILT RECORDS.
3. WELDS AND/OR COUPLINGS DEFINING RANDOM PIPE SPOOL LENGTHS ARE FOR REFERENCE ONLY. FINAL WELD OR COUPLING LOCATION IS TO BE DETERMINED BY THE CONTRACTOR.
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6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
7. PIPE SUPPORTS INDICATED ON THE DRAWING MAY BE PART OF AN ASSEMBLY ON SEVERAL ISOMETRICS. REFER TO THE 3D MODEL FOR THE COMPLETE PIPE SUPPORT ASSEMBLY.
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WV
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

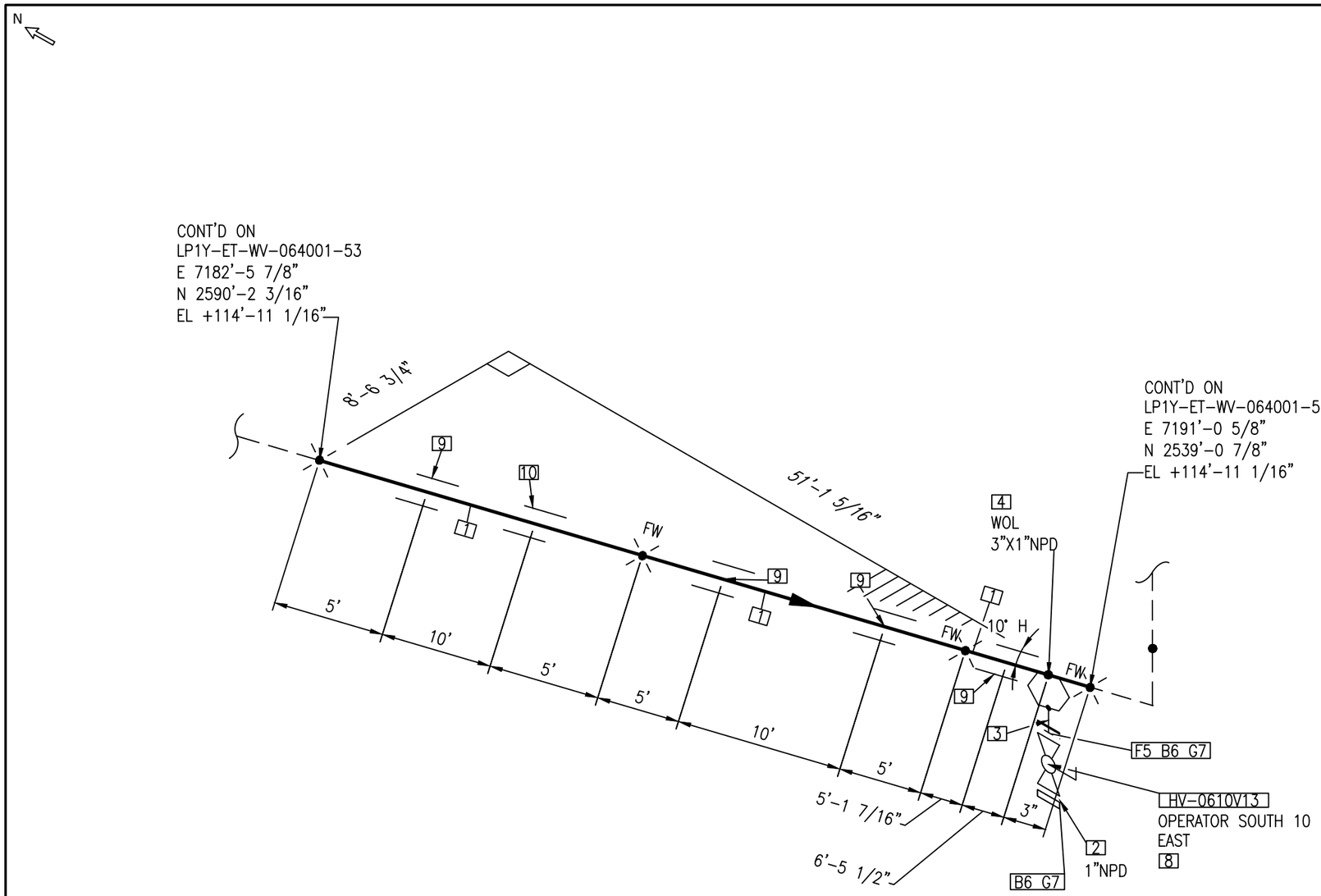
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" DUP WV (344) (108) 064001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WV-064001-51	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 23-FEB-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	51'-10"	3"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	1	1"	FLG BLIND 150# RF DUPLEX STL F60	5616811
3	1	1"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
4	1	3"X1"	PIPET SCH 10S DUPLEX STL F60	5512022
5	1	1"	FLG LAP JOINT 150# 304 SS	5599576L
6	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
7	2	1"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
8	1	1"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
9	4	3"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-03-01
10	1	3"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-03-01

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Mar 03, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

GENERAL NOTE:
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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WV
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" DUP WV (344) (108) 064001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WV-064001-52	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 23-FEB-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.

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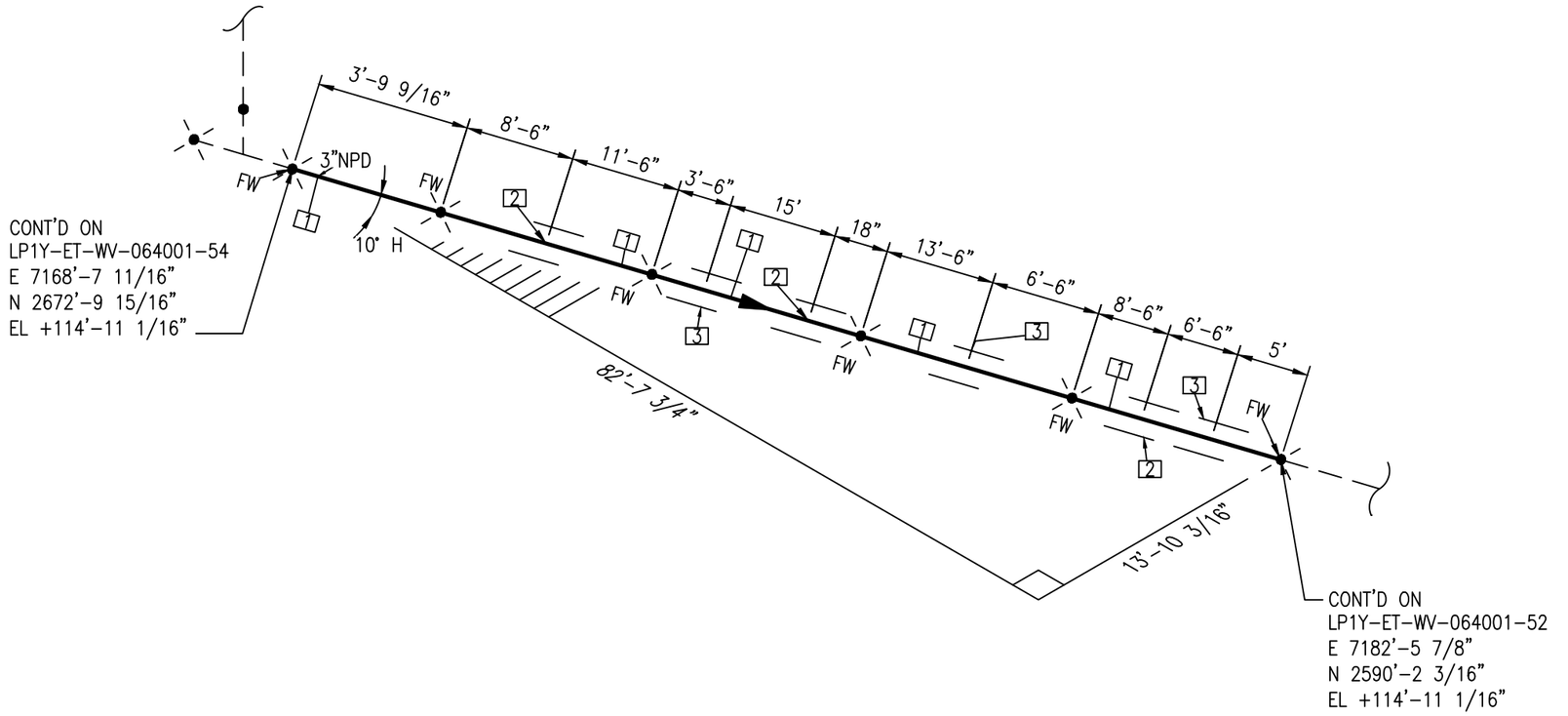

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	83'-10"	3"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	3	3"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-03-01
3	3	3"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-03-01

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Mar 03, 2026

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0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW
REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WV
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" DUP WV (344) (108) 064001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WV-064001-53	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 23-FEB-2026

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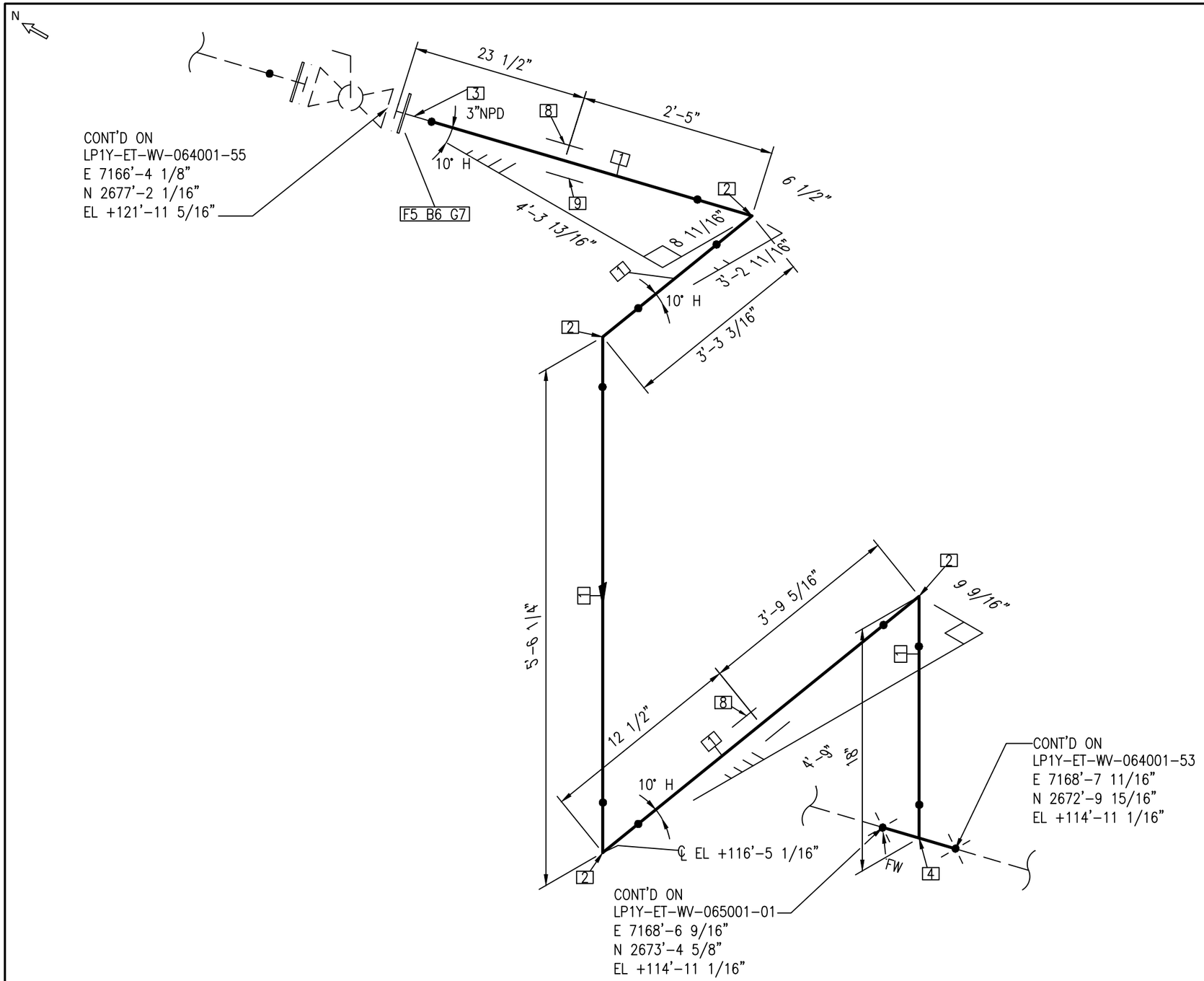

BILL OF MATERIALS

ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	16'	3"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	4	3"	ELL 90 DEG LR SCH 10S DUPLEX STL S32205-W	5460743
3	1	3"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
4	1	3"	TEE SCH 10S DUPLEX STL S32205-W	5460752
5	1	3"	FLG LAP JOINT 150# 304 SS	5599576L
6	4	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
7	1	3"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
8	2	3"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-03-01
9	1	3"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Mar 03, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WV
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" DUP WV (344) (108) 064001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WV-064001-54	0

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 23-FEB-2026

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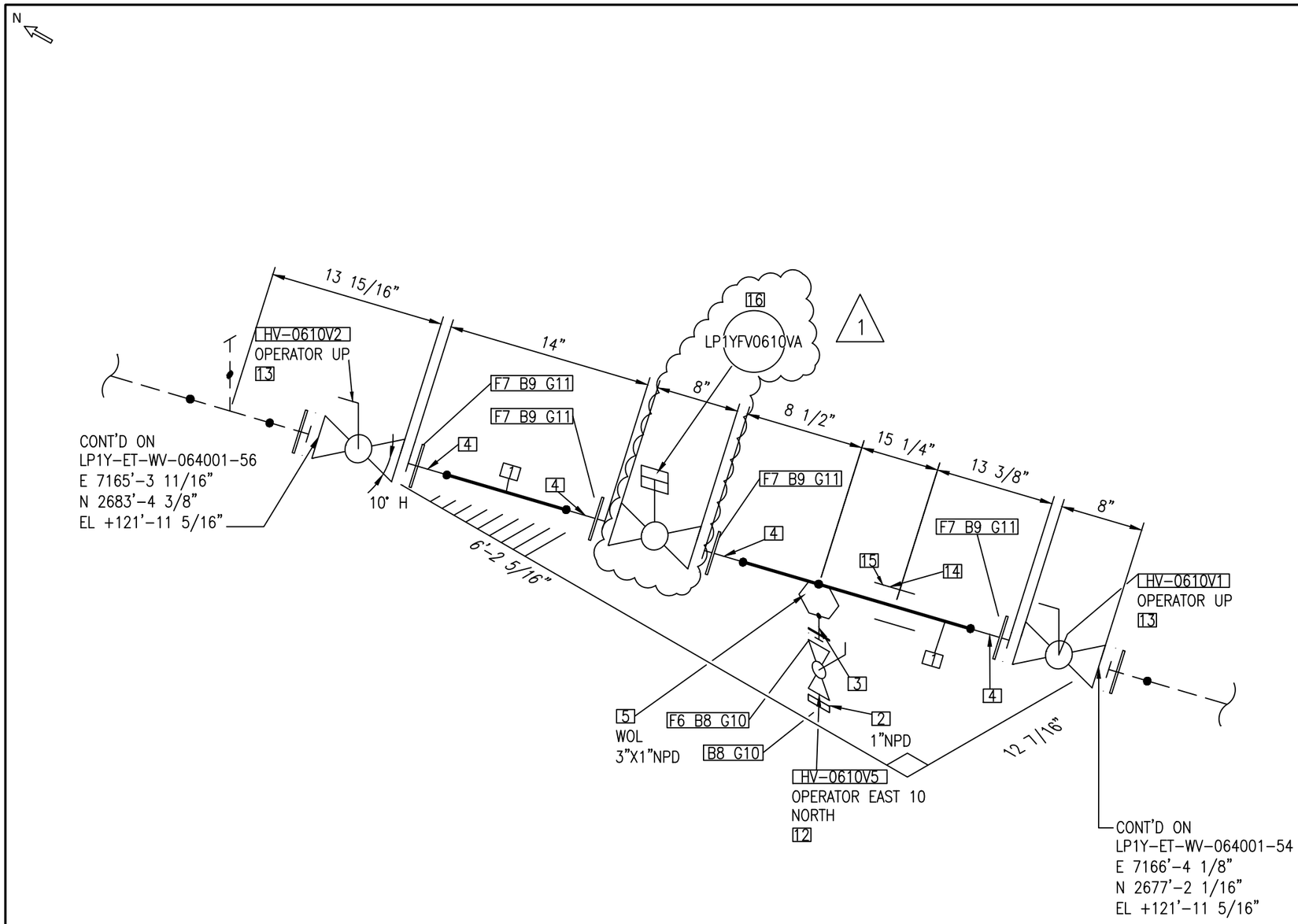

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	3'-6"	3"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	1	1"	FLG BLIND 150# RF DUPLEX STL F60	5616811
3	1	1"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
4	4	3"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
5	1	3"X1"	PIPET SCH 10S DUPLEX STL F60	5512022
6	1	1"	FLG LAP JOINT 150# 304 SS	5599576L
7	4	3"	FLG LAP JOINT 150# 304 SS	5599576L
8	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
9	16	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
10	2	1"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
11	4	3"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
12	1	1"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
13	2	3"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
14	1	3"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-03-01
15	1	3"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
16	1	3"	KNA BALL VALVE ASME CL. 150# RF	LP1YFV0610VA

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	JHN / N/A	MDW

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5. THE MECHANICAL CONTRACTOR IS TO VERIFY ALL DIMENSIONAL DATA WITH ACTUAL FIELD CONDITIONS PRIOR TO INSTALLATION OF THE SYSTEM.
6. PIPE SUPPORTS SHALL BE FURNISHED AND INSTALLED IN ACCORDANCE WITH THIS DRAWING AND SPECIFICATION LP100-250-EMS-15110. MECHANICAL CONTRACTOR SHALL PROVIDE TEMPORARY SUPPORTS AS REQUIRED TO AID PRESSURE TESTING, INSTALLATION ETC.
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8. MECHANICAL CONTRACTOR SHALL NOTE ALL DESIGN CHANGES ON THIS DRAWING AND RETURN IT TO THE CONSTRUCTION WPO FOR AS-BUILT RECORDS.
9. THE BILL OF MATERIAL IS FOR REFERENCE ONLY.
10. SLOPE EQUIVALENTS: 1:27 = 7/16" PER FT, 1:96 = 1/8" PER FT, 1:120 = 1/10" PER FT, 1:192 = 1/16" PER FT, 1:480 = 1/40" PER FT.

NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WV
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

Lilly

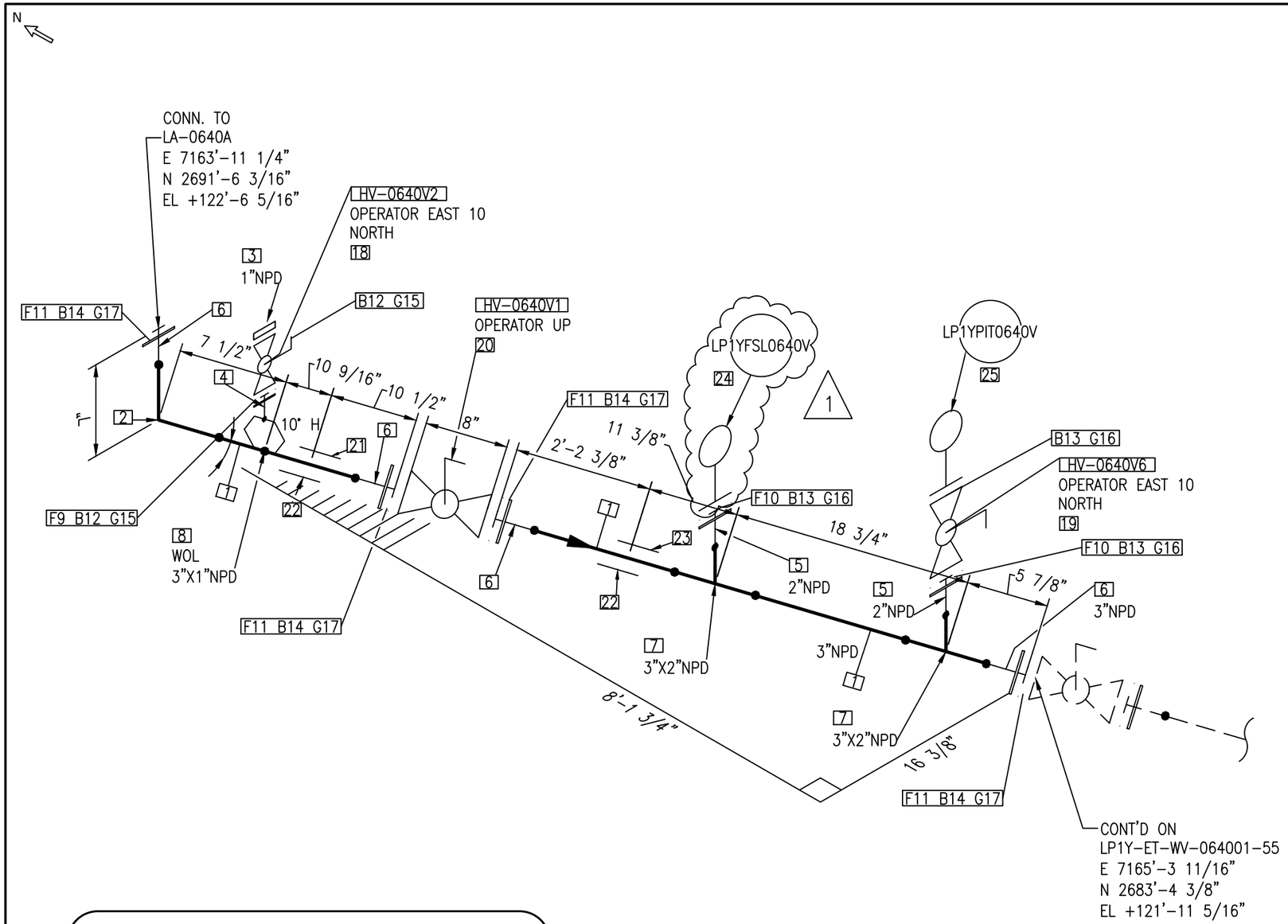
HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" DUP WV (344) (108) 064001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WV-064001-55	1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026

NOTICE: THIS DRAWING HAS NOT BEEN PUBLISHED AND IS THE SOLE PROPERTY OF FLUOR AND IS LENT TO THE BORROWER FOR THEIR CONFIDENTIAL USE ONLY, AND IN CONSIDERATION OF THE LOAN OF THIS DRAWING, THE BORROWER PROMISES AND AGREES TO RETURN IT UPON REQUEST AND AGREES THAT IT WILL NOT BE REPRODUCED, COPIED, LENT, OR OTHERWISE DISPOSED OF DIRECTLY OR INDIRECTLY, NOR USED FOR ANY PURPOSE OTHER THAN THAT FOR WHICH IT IS FURNISHED.



BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	5'-6"	3"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	1	3"	ELL 90 DEG LR SCH 10S DUPLEX STL S32205-W	5460743
3	1	1"	FLG BLIND 150# RF DUPLEX STL F60	5616811
4	1	1"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
5	2	2"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
6	4	3"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
7	2	3"X2"	TEE RED SCH 10S DUPLEX STL S32205-W	5460751
8	1	3"X1"	PIPET SCH 10S DUPLEX STL F60	5512022
9	1	1"	FLG LAP JOINT 150# 304 SS	5599576L
10	2	2"	FLG LAP JOINT 150# 304 SS	5599576L
11	4	3"	FLG LAP JOINT 150# 304 SS	5599576L
12	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
13	12	5/8"X2 1/2"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
14	16	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
15	2	1"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
16	3	2"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
17	4	3"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
18	1	1"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
19	1	2"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
20	1	3"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
21	1	3"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-03-01
22	2	3"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
23	1	3"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-03-01
24	1	2"	FSL	LP1YFSL0640V
25	1	2"	PRESSURE TRANSMITTER	LP1YPIT0640V

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WV
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" DUP WV (344) (108) 064001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WV-064001-56	1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026

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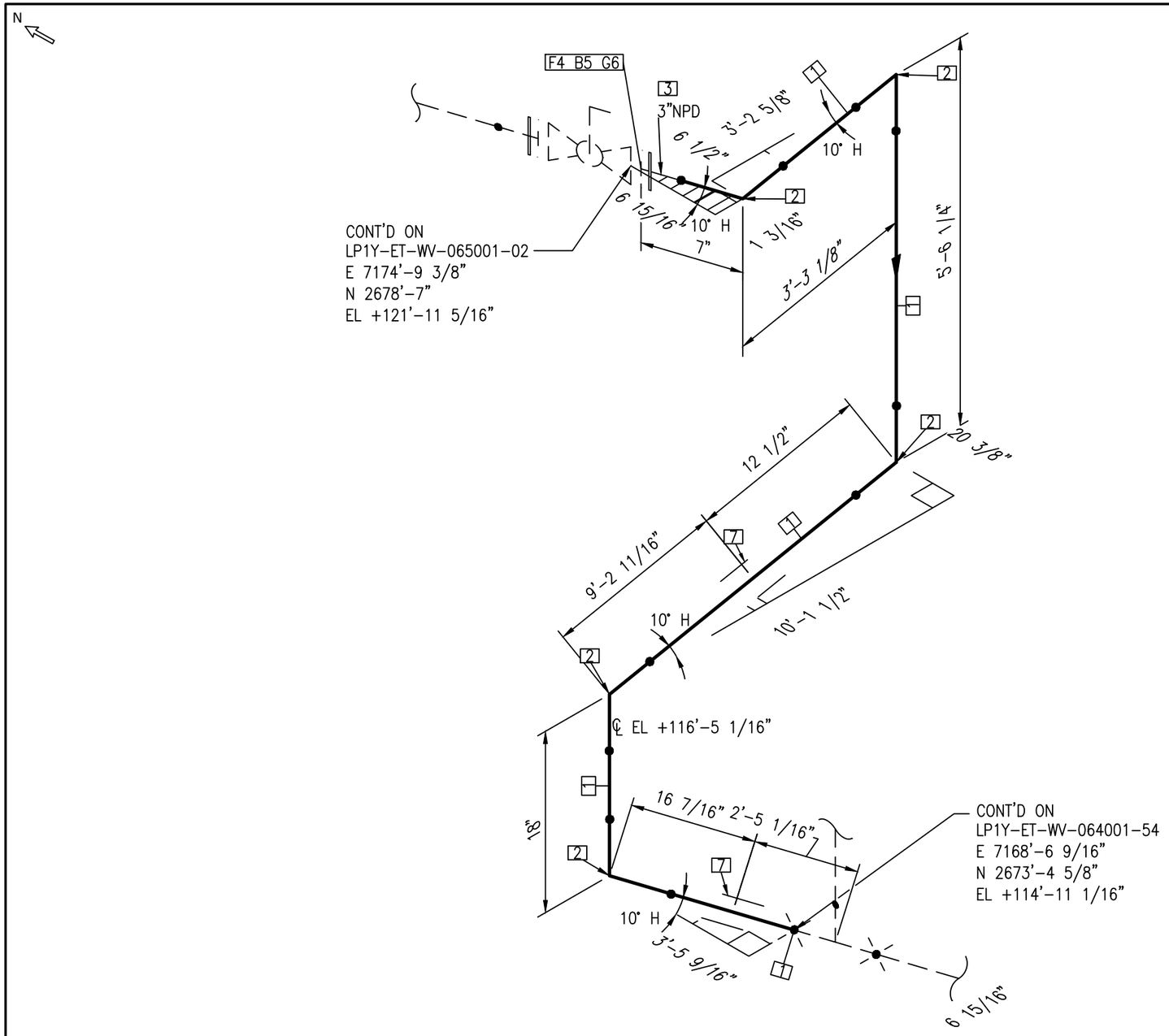

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	20'-9"	3"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	5	3"	ELL 90 DEG LR SCH 10S DUPLEX STL S32205-W	5460743
3	1	3"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
4	1	3"	FLG LAP JOINT 150# 304 SS	5599576L
5	4	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
6	1	3"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
7	2	3"	5ISC (COLD SERVICE INSULATION CRADLE AND SHOE)	5ISC-03-01

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Mar 03, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WV
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	N/A / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" DUP WV (344) (108) 065001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WV-065001-01	0

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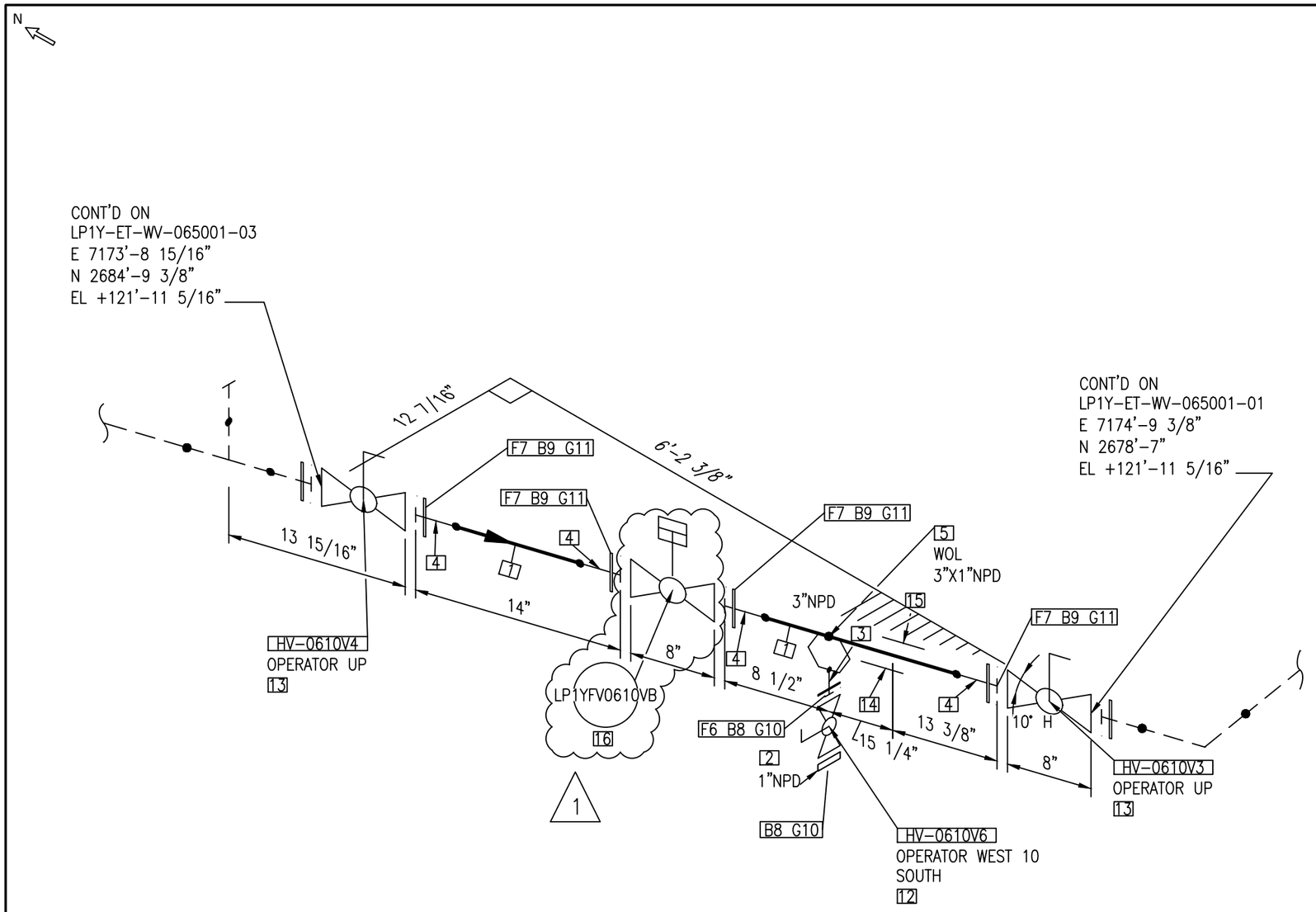

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
1	3'-6"	3"	PIPE SCH 10S EFW DUPLEX STL S32205-W	5366201
2	1	1"	FLG BLIND 150# RF DUPLEX STL F60	5616811
3	1	1"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
4	4	3"	STUB-END SCH10S DUPLEX S32205-W SHORT PATT	5460762L
5	1	3"X1"	PIPET SCH 10S DUPLEX STL F60	5512022
6	1	1"	FLG LAP JOINT 150# 304 SS	5599576L
7	4	3"	FLG LAP JOINT 150# 304 SS	5599576L
8	8	1/2"X2 1/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
9	16	5/8"X2 3/4"	STUD-BOLT A193 GR B8M W/A194 GR 8M NUTS W/ WASHERS	5676576L0
10	2	1"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
11	4	3"	GASKET 150# PTFE RING 1/16" THK LOW STRESS	5669359L
12	1	1"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
13	2	3"	BALL CL 150 RF DI, PFA LINED FP EXTD STEM, HNDL OP	5300941L2
14	1	3"	5IGC (COLD SERVICE INSULATION CRADLE WITH SHOE AND GUIDE)	5IGC-03-01
15	1	3"	5FSS5 (FIELD BASE SUPPORT)	5FSS5-L-1
16	1	3"	KNA BALL VALVE ASME CL. 150# RF	LP1YFV0610VB

A1- DATA COMPLETE - NO COMMENTS - DO NOT RESUBMIT

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WV
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

HANSON
EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" DUP WV (344) (108) 065001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WV-065001-02	1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026

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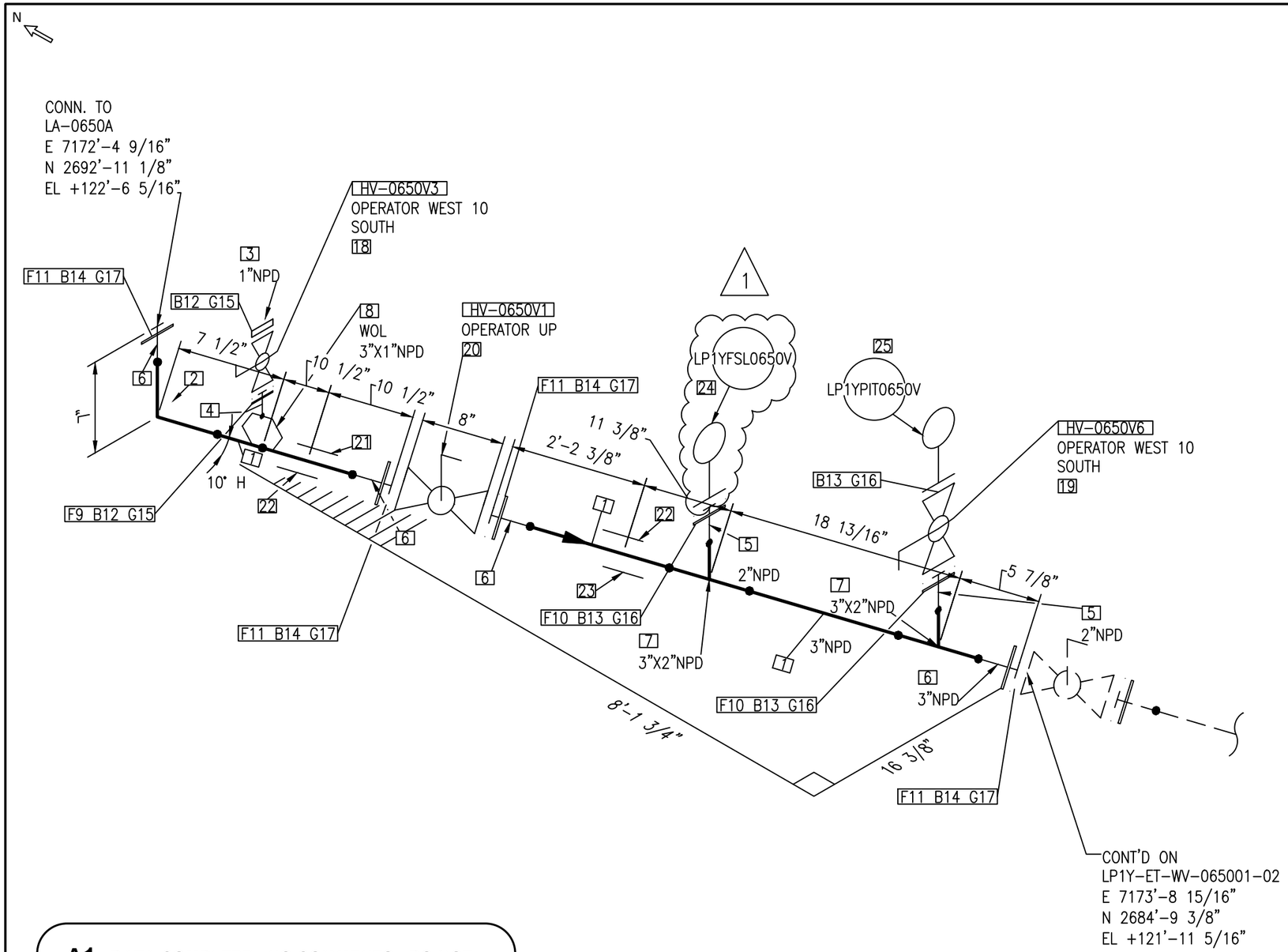

BILL OF MATERIALS				
ID	QTY	ND	DESCRIPTION	CMDTY CODE
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25	1	2"	PRESSURE TRANSMITTER	LP1YPIT0650V

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By dac54131 at Apr 10, 2026

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NODE:	TFS
BUILDING NO:	Y
CWP NO:	N/A
MODULE NO:	N/A
SERVICE CODE:	WV
PIPE SCHEDULE:	344
INSULATION CODE:	108
TRACING TYPE:	ET
ELASTOMER:	-
HYGIENIC: NO	IMPACT: NO

REV. NO.	DESCRIPTION	DATE	DRWN	CHK	STR	INSTR / PRCS	QC
1	ISSUE FOR CONSTRUCTION	24-MAR-2026	DRW	SBR	N/A	JHN / N/A	MDW
0	ISSUE FOR CONSTRUCTION	23-FEB-2026	DRW	SBR	N/A	JHN / N/A	MDW

Lilly

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EFDEE Engineering PC
PROJECT NO. B1LG

LINE NUMBER
LP1Y 3" DUP WV (344) (108) 065001

REFERENCE P&ID NUMBER	ISOMETRIC DRAWING NUMBER	REV
LP1Y-DI-00000061	LP1Y-ET-WV-065001-03	1

AutoCAD Plant 3D PROJECT: LP10Y-TFS-M0-000-000-00-HNS-P3D EXTRACTION DATE: 24-MAR-2026